

PROFESSOR:

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Class 1: Non-cooperative bargaining

Motivation

Bargaining is the problem of dividing a surplus that can be realized only if the parties reach an agreement. This is a pervasive economic problem: firms bargain with workers, buyers bargain with sellers, and countries bargain over trade, debt, or policy. The central economic question is simple but deep: if agreement creates value, how is that value divided?

A fully realistic model of bargaining would typically require a very rich extensive form, with proposals, counterproposals, deadlines, and strategic delays. That level of detail is often too cumbersome if our objective is to isolate the basic forces behind agreement. For that reason, it is useful to begin with very simple models that highlight two issues:

- (i) whether agreement is reached at all, and
- (ii) how the surplus is divided across players.

A Simultaneous-Move Benchmark: Divide a Dollar

We begin with a stark benchmark in which two players simultaneously demand shares of a unit surplus.

Environment

There are two players, indexed by $i \in \{1, 2\}$. Each player simultaneously chooses a demand

$$d_i \in [0, 1].$$

If the demands are jointly feasible, that is, if

$$d_1 + d_2 \leq 1,$$

then agreement is reached and each player receives her demand. If instead

$$d_1 + d_2 > 1,$$

then the demands are incompatible, bargaining breaks down, and both players receive zero.¹

Thus player i 's payoff is

$$u_i(d_1, d_2) = \begin{cases} d_i & \text{if } d_1 + d_2 \leq 1, \\ 0 & \text{if } d_1 + d_2 > 1. \end{cases}$$

Best responses

Fix player j 's demand d_j . Player i then chooses d_i to maximize his payoff. If $d_j < 1$, then player i wants to ask for as much as possible while still preserving agreement. Therefore the unique best response is

$$BR_i(d_j) = \{1 - d_j\}.$$

If $d_j = 1$, then any choice by player i yields payoff 0: choosing $d_i = 0$ leads to agreement with payoff 0, and any $d_i > 0$ leads to disagreement, also with payoff 0. Hence

$$BR_i(1) = [0, 1].$$

Therefore the best-response correspondence is

$$BR_i(d_j) = \begin{cases} \{1 - d_j\} & \text{if } d_j \in [0, 1), \\ [0, 1] & \text{if } d_j = 1. \end{cases}$$

Nash equilibria

From the best-response correspondence, if $d_1 < 1$ and $d_2 < 1$, equilibrium requires

$$d_1 = 1 - d_2 \quad \text{and} \quad d_2 = 1 - d_1,$$

where both imply the sum

$$d_1 + d_2 = 1.$$

In addition, the profile $(1, 1)$ is also a Nash equilibrium. To verify this, we must check both players' best-response conditions. Since

$$BR_i(1) = [0, 1],$$

each player's demand of 1 is a best response to the other player's demand of 1. In particular,

$$1 \in BR_1(1) \quad \text{and} \quad 1 \in BR_2(1).$$

Hence neither player has a profitable deviation from $(1, 1)$, so $(1, 1)$ is a Nash equilibrium.

¹ *Intuition for future me:* The model is intentionally extreme: agreement requires exact compatibility of demands, and disagreement destroys the entire surplus. This lets us focus only on the strategic tension between asking for more and keeping agreement feasible.

Remark. Recall that a Nash equilibrium, in this context, is a pair (d_1, d_2) such that each player's demand is a best response to the other's demand.

It is important not to overstate this observation. The fact that every demand in $[0, 1]$ is a best response to 1 does *not* imply that every profile of the form $(1, d)$ or $(d, 1)$ is a Nash equilibrium. If $d < 1$, then the player facing d has the unique best response $1 - d$, which yields a strictly positive payoff, whereas demanding 1 yields payoff 0. Therefore, only $(1, 1)$ is an equilibrium among profiles in which one player demands 1.

Hence the set of Nash equilibria is

$$\{(d_1, d_2) \in [0, 1]^2 : d_1 + d_2 = 1\} \cup \{(1, 1)\}.$$

Remark. This game has a continuum of efficient Nash equilibria, namely all pairs on the line $d_1 + d_2 = 1$, and one inefficient Nash equilibrium, $(1, 1)$, in which bargaining breaks down and both players get zero.

A Sequential Benchmark: The Ultimatum Game

We now introduce a simple extensive-form bargaining game in which one player proposes a division and the other decides whether to accept it.

Environment

Player 1 makes a take-it-or-leave-it offer

$$x \in [0, 1],$$

where x is Player 1's share of the unit surplus. Player 2 then chooses

$$y \in \{\text{Yes}, \text{No}\}.$$

If Player 2 accepts, the proposed allocation is implemented and payoffs are

$$u(x, \text{Yes}) = (x, 1 - x).$$

If Player 2 rejects, bargaining breaks down and both players get zero:

$$u(x, \text{No}) = (0, 0).$$

Subgame perfect equilibrium

The equilibrium is obtained by backward induction.

Consider first Player 2's decision after an offer $x < 1$. If she accepts, she obtains

$$1 - x > 0.$$

If she rejects, she obtains

$$0.$$

Therefore, for every $x < 1$, Player 2 strictly prefers to accept.

Now consider the subgame following the offer $x = 1$. If Player 2 accepts, she gets 0; if she rejects, she also gets 0. Thus at $x = 1$ she is indifferent between accepting and rejecting.

Turn now to Player 1's initial choice. Since every offer $x < 1$ is accepted, Player 1's payoff from offering $x < 1$ is simply x . Therefore she would like to make x as large as possible. This pushes her to the boundary $x = 1$.

However, for $x = 1$ to actually be optimal, Player 2 must accept that offer. If instead Player 2 rejected at $x = 1$, then Player 1 would prefer to deviate to some $x < 1$ arbitrarily close to 1, which would be accepted and would give her a strictly positive payoff.² Hence in any subgame perfect equilibrium, Player 2 must accept the offer $x = 1$.

We conclude:

Proposition 0.0.1. *The ultimatum game has a unique subgame perfect equilibrium. In that equilibrium,*

- (i) *Player 1 offers $x = 1$;*
- (ii) *Player 2 accepts every offer, including $x = 1$.*

The equilibrium payoff is therefore

$$(1, 0).$$

Remark. The logic is stark. Because rejection gives Player 2 no future strategic advantage and no outside option, acceptance is optimal for any offer that gives her even an arbitrarily small positive amount. Anticipating this, Player 1 extracts essentially the entire surplus.

Two-Round Bargaining

The ultimatum game delivers a very sharp prediction because rejection leads directly to disagreement and a payoff of zero. A natural next step is to allow bargaining to continue for one more round. This introduces a new force: rejecting an offer may be valuable because it gives the responder a chance to become the proposer in the future.

To capture the cost of delay, suppose that payoffs obtained in round 2 are discounted by a factor

$$\delta \in (0, 1).$$

Thus agreement is still possible after a rejection, but delay shrinks the value of the surplus.

² If Player 2 rejected $x = 1$, then Player 1 would face the problem

$$\max_{x < 1} x,$$

subject to acceptance. The supremum is 1, but it is not attained. Hence Player 1 would have no best response. Therefore a strategy profile in which Player 2 rejects $x = 1$ cannot be subgame perfect.

Key economic idea

In dynamic bargaining, a rejection is not merely a refusal to trade today. It is also a choice to wait for a future bargaining position. The responder's threat point is therefore a *continuation value*, not just zero.

Environment

There are two rounds.

Round 1. Player 1 offers

$$x_1 \in [0, 1],$$

where x_1 is Player 1's share of the surplus. Player 2 then chooses

$$y_1 \in \{\text{Yes}, \text{No}\}.$$

If Player 2 accepts, payoffs are

$$(x_1, 1 - x_1).$$

If Player 2 rejects, the game moves to round 2.

Round 2. Player 2 now makes an offer

$$x_2 \in [0, 1],$$

where x_2 is Player 2's share of the surplus. Player 1 then chooses

$$y_2 \in \{\text{Yes}, \text{No}\}.$$

If Player 1 accepts, payoffs are

$$\delta(1 - x_2, x_2).$$

If Player 1 rejects, payoffs are

$$(0, 0).$$

The proposer in each round chooses her own share. Thus the identity of the future proposer matters directly for the value of rejecting today.

Step 1: Solve the second round

We begin from the last round and use backward induction. Fix any round-2 offer $x_2 \in [0, 1]$. If Player 1 accepts, his payoff is

$$\delta(1 - x_2).$$

If he rejects, his payoff is

$$0.$$

Since $x_2 \in [0, 1]$, we have $1 - x_2 \geq 0$, so

$$\delta(1 - x_2) \geq 0.$$

Therefore:

- if $x_2 < 1$, then Player 1 strictly prefers to accept;
- if $x_2 = 1$, then Player 1 is indifferent between accepting and rejecting.

So Player 1's round-2 best response is

$$BR_1^{(2)}(x_2) = \begin{cases} \{\text{Yes}\} & \text{if } x_2 < 1, \\ \{\text{Yes}, \text{No}\} & \text{if } x_2 = 1. \end{cases}$$

Now consider Player 2's problem in round 2. If Player 1 accepts, Player 2 gets

$$\delta x_2.$$

This is increasing in x_2 , so Player 2 wants to set x_2 as high as possible subject to acceptance. This implies that in any subgame perfect equilibrium Player 2 must choose

$$x_2^* = 1.$$

At that offer, Player 1 is indifferent between accepting and rejecting because his payoff is zero either way. However, in a subgame perfect equilibrium Player 1 must accept $x_2 = 1$. If instead he rejected, then Player 2 would profitably deviate to an offer slightly below 1, which would be accepted and would give her a payoff arbitrarily close to δ . Hence the unique equilibrium outcome of round 2 is:

$$x_2^* = 1, \quad y_2^* = \text{Yes},$$

with continuation payoff

$$(0, \delta).$$

Remark. The second round is just an ultimatum game with a discounted pie. Since Player 2 is the proposer in that round, she captures the entire discounted surplus.

Step 2: Solve the first round

Now return to round 1. If Player 2 accepts Player 1's offer x_1 , her payoff is

$$1 - x_1.$$

If she rejects, the game moves to round 2, where her equilibrium continuation payoff is

$$\delta.$$

Therefore Player 2 compares

$$1 - x_1 \quad \text{with} \quad \delta.$$

Her optimal response is:

$$BR_2^{(1)}(x_1) = \begin{cases} \{\text{Yes}\} & \text{if } 1 - x_1 > \delta, \\ \{\text{Yes, No}\} & \text{if } 1 - x_1 = \delta, \\ \{\text{No}\} & \text{if } 1 - x_1 < \delta. \end{cases}$$

Equivalently,

$$BR_2^{(1)}(x_1) = \begin{cases} \{\text{Yes}\} & \text{if } x_1 < 1 - \delta, \\ \{\text{Yes, No}\} & \text{if } x_1 = 1 - \delta, \\ \{\text{No}\} & \text{if } x_1 > 1 - \delta. \end{cases}$$

This is the crucial threshold condition: **Player 2 accepts if and only if the offer gives her at least as much as waiting for round 2.**

Now consider Player 1's choice in round 1. He wants to maximize x_1 , but only offers that Player 2 accepts can be implemented. Thus Player 1 solves

$$\max_{x_1 \in [0,1]} x_1 \quad \text{subject to} \quad 1 - x_1 \geq \delta.$$

Equivalently,

$$x_1 \leq 1 - \delta.$$

Hence Player 1's optimal offer is

$$x_1^* = 1 - \delta.$$

There is a subtle but important point here. At $x_1 = 1 - \delta$, Player 2 is indifferent between accepting and rejecting. In a subgame perfect equilibrium she must accept this offer. If she rejected it, then Player 1 would have no best response in round 1: every accepted offer would require $x_1 < 1 - \delta$, and Player 1 could always do slightly better by increasing x_1 a bit more while still remaining below the threshold. Thus no maximizing offer would exist. Therefore equilibrium requires acceptance at the boundary.

Acceptance at the boundary

At $x_1 = 1 - \delta$, Player 2 is indifferent:

$$1 - x_1 = \delta.$$

If she were to reject at this exact offer, then Player 1 could approach $1 - \delta$ from below but could never attain his supremum payoff. Hence there would be no optimal first-round offer. To sustain a subgame perfect equilibrium, the boundary offer must be accepted.

Subgame perfect equilibrium

We can now summarize the equilibrium.

Proposition 0.0.2. *The two-round bargaining game has a unique subgame perfect equilibrium. It is given by:*

(i) *In round 1, Player 1 offers*

$$x_1^* = 1 - \delta.$$

(ii) *Player 2 accepts in round 1 if and only if*

$$x_1 \leq 1 - \delta.$$

(iii) *If round 2 is reached, Player 2 offers*

$$x_2^* = 1.$$

(iv) *In round 2, Player 1 accepts every offer.*

The equilibrium outcome is immediate agreement in round 1, with payoffs

$$(1 - \delta, \delta).$$

Economic intuition

This equilibrium is best understood through continuation values and bargaining power.

In round 1, Player 2 is not a passive responder. Rejecting today gives her something valuable tomorrow: the right to make the final offer in round 2. Since in round 2 the proposer captures the entire discounted surplus, Player 2 can secure payoff δ by rejecting and waiting. Therefore she will never accept less than δ in round 1.

Player 1 anticipates this. He therefore offers Player 2 exactly her continuation value and keeps the residual surplus for himself. That is why the equilibrium offer is

$$(1 - \delta, \delta).$$

This is the same basic logic that appears throughout bargaining theory: the proposer makes the responder just indifferent between accepting now and continuing the game.

Remark. The responder's equilibrium payoff is pinned down by her outside option *within the game*. Here that outside option is the value of becoming tomorrow's proposer.

Comparative statics

The parameter δ measures how valuable future bargaining opportunities are. In this two-round game, a higher δ does not affect both players symmetrically. What matters is that, after a rejection in round 1, Player 2 becomes the proposer in round 2. Thus her continuation value in round 1 is

$$\delta,$$

since by waiting she can make the final offer herself and capture the entire discounted surplus.

Player 1 must therefore leave Player 2 at least this continuation value in order to obtain acceptance. Since the equilibrium offer is

$$x_1^* = 1 - \delta,$$

Player 1's share falls one-for-one with δ , while Player 2's share rises to

$$1 - x_1^* = \delta.$$

If δ is small, delay is very costly, so Player 2's continuation value is low. Rejecting is then unattractive, and Player 1 can keep almost the whole surplus:

$$x_1^* \rightarrow 1, \quad 1 - x_1^* \rightarrow 0 \quad \text{as } \delta \rightarrow 0.$$

If δ is large, delay is not very costly, so rejecting becomes attractive for Player 2 because waiting allows her to become the final proposer. Hence Player 1 must concede more in round 1:

$$x_1^* \rightarrow 0, \quad 1 - x_1^* \rightarrow 1 \quad \text{as } \delta \rightarrow 1.$$

Finite Alternating-Offer Bargaining

The two-round bargaining game makes clear that a responder evaluates an offer relative to a continuation value: rejecting today may be optimal because it changes tomorrow's bargaining position. A finite alternating-offer model extends this logic to an arbitrary number of rounds and shows how equilibrium offers are pinned down recursively by the value of future bargaining opportunities.

The key insight is simple: in each round, the proposer gives the responder exactly enough to make her indifferent between accepting now and rejecting to continue bargaining.

Logic and reservation values

The finite-horizon bargaining problem is solved by backward induction. The last round is an ultimatum game. Once that last-round continuation value is known, it determines the responder's reservation value in the previous round, which in turn determines the equilibrium offer there, and so on recursively back to round 1.

Environment

There are T rounds, indexed by

$$t = 1, \dots, T.$$

If t is odd, Player 1 is the proposer. If t is even, Player 2 is the proposer.

In round t , the proposer chooses

$$x_t \in [0, 1],$$

where x_t is the proposer's share of the surplus in that round. The responder then chooses

$$y_t \in \{\text{Yes}, \text{No}\}.$$

If the responder accepts in round t , agreement is reached immediately and payoffs are:

$$\delta^{t-1}x_t \quad \text{for the proposer,} \quad \delta^{t-1}(1 - x_t) \quad \text{for the responder,}$$

where $\delta \in (0, 1)$ is the common discount factor.

If the responder rejects and $t < T$, bargaining continues to round $t + 1$. If the responder rejects in the final round T , then bargaining breaks down and payoffs are³

$$(0, 0).$$

Continuation values and backward induction

Let

$$V_t = (V_t^1, V_t^2)$$

denote the equilibrium payoff vector from the beginning of round t onward.

The dynamic structure is recursive: once V_{t+1} is known, the equilibrium in round t can be computed directly.

³ *Intuition for future me:* The surplus is available in every round, but delay shrinks its value by the factor δ^{t-1} . Thus the game is not only about who proposes, but also about when agreement is reached.

Odd rounds. Suppose t is odd, so Player 1 proposes and Player 2 responds.

If Player 2 accepts offer x_t , her payoff is

$$\delta^{t-1}(1 - x_t).$$

If she rejects, her continuation payoff is

$$V_{t+1}^2.$$

Hence Player 2 accepts if and only if

$$\delta^{t-1}(1 - x_t) \geq V_{t+1}^2.$$

Player 1 wants to maximize his own payoff, so he chooses the largest offer that is still accepted. Therefore the equilibrium offer in an odd round satisfies

$$\delta^{t-1}(1 - x_t^*) = V_{t+1}^2,$$

that is,

$$x_t^* = 1 - \frac{V_{t+1}^2}{\delta^{t-1}}.$$

The resulting equilibrium payoff vector in round t is

$$V_t = \left(\delta^{t-1} - V_{t+1}^2, V_{t+1}^2 \right).$$

Even rounds. Suppose instead that t is even, so Player 2 proposes and Player 1 responds.

If Player 1 accepts offer x_t , his payoff is

$$\delta^{t-1}(1 - x_t).$$

If he rejects, his continuation payoff is

$$V_{t+1}^1.$$

Therefore Player 1 accepts if and only if

$$\delta^{t-1}(1 - x_t) \geq V_{t+1}^1.$$

Again, the proposer chooses the largest acceptable share for herself, so the equilibrium offer satisfies

$$\delta^{t-1}(1 - x_t^*) = V_{t+1}^1,$$

hence

$$x_t^* = 1 - \frac{V_{t+1}^1}{\delta^{t-1}}.$$

The resulting equilibrium payoff vector in round t is

$$V_t = \left(V_{t+1}^1, \delta^{t-1} - V_{t+1}^1 \right).$$

Remark. At every history, the responder receives exactly her continuation value. The proposer extracts the entire remaining surplus.

Terminal round

The recursion starts from the final round.

If T is odd, then Player 1 is the proposer in round T . Since rejection leads to $(0, 0)$, round T is just an ultimatum game. Therefore Player 1 offers himself the whole discounted surplus:

$$x_T^* = 1,$$

and Player 2 accepts. Thus

$$V_T = (\delta^{T-1}, 0).$$

If T is even, the same logic gives

$$V_T = (0, \delta^{T-1}).$$

A worked example: $T = 3$

It is useful to work through the three-round case explicitly.

Round 3. Since round 3 is odd, Player 1 proposes. This is the last round, so rejection yields zero. Hence Player 1 offers

$$x_3^* = 1,$$

Player 2 accepts, and the continuation payoff from round 3 is

$$V_3 = (\delta^2, 0).$$

Round 2. Now Player 2 proposes. If Player 1 rejects in round 2, he gets his continuation payoff from round 3, namely

$$V_3^1 = \delta^2.$$

If he accepts round-2 offer x_2 , he gets

$$\delta(1 - x_2).$$

Thus Player 1 accepts if and only if

$$\delta(1 - x_2) \geq \delta^2.$$

Player 2 therefore chooses the largest x_2 that satisfies this constraint:

$$\delta(1 - x_2) = \delta^2 \implies x_2^* = 1 - \delta.$$

Hence the equilibrium payoff from round 2 is

$$V_2 = (\delta^2, \delta - \delta^2).$$

Round 1. Player 1 now proposes. If Player 2 rejects in round 1, she gets her continuation payoff from round 2:

$$V_2^2 = \delta - \delta^2.$$

If she accepts round-1 offer x_1 , she gets

$$1 - x_1.$$

So Player 2 accepts if and only if

$$1 - x_1 \geq \delta - \delta^2.$$

Player 1 therefore chooses

$$x_1^* = 1 - (\delta - \delta^2) = 1 - \delta + \delta^2.$$

The equilibrium payoff from the start of the game is then

$$V_1 = (1 - \delta + \delta^2, \delta - \delta^2).$$

What the T

With three rounds, Player 1's first-round payoff is

$$1 - \delta + \delta^2.$$

The signs alternate because bargaining power alternates across rounds: waiting one round hurts the current proposer, but waiting two rounds can return proposal power to that same player.

Normalized continuation values

The recursion becomes cleaner if we divide round- t payoffs by the size of the available surplus in round t , namely δ^{t-1} .

Define

$$v_t^i \equiv \frac{V_t^i}{\delta^{t-1}},$$

so that v_t^i is player i 's equilibrium share of the round- t surplus.

Then the terminal condition is

$$v_T = \begin{cases} (1, 0) & \text{if } T \text{ is odd,} \\ (0, 1) & \text{if } T \text{ is even.} \end{cases}$$

The recursive formulas become:

If t is odd:

$$v_t^2 = \delta v_{t+1}^2, \quad v_t^1 = 1 - \delta v_{t+1}^2.$$

If t is even:

$$v_t^1 = \delta v_{t+1}^1, \quad v_t^2 = 1 - \delta v_{t+1}^1.$$

These equations summarize the bargaining logic in one line: the responder gets her discounted continuation value, and the proposer gets the residual.

Equilibrium outcome

The recursion implies that agreement occurs immediately in every subgame. The proposer at round t makes an offer that leaves the responder exactly indifferent between accepting now and rejecting to continue.

Proposition 0.0.3. *The finite alternating-offer bargaining game has a unique subgame perfect equilibrium. In every round $t < T$:*

- (i) *the proposer offers the responder exactly her continuation value;*
- (ii) *the responder accepts all offers that give at least her continuation value and rejects all offers that give less;*
- (iii) *on the equilibrium path, agreement occurs immediately in that round.*

In the final round, the proposer offers himself the whole remaining discounted surplus, and the responder accepts.

Remark. The equilibrium is unique because, at each stage, the responder's continuation value is uniquely determined by the next subgame, and the proposer has a unique optimal way to satisfy the responder's participation constraint with equality.

Remark (Why uniqueness rules out nondegenerate mixing at the acceptance threshold). The uniqueness argument can be sharpened as follows.

Fix any subgame in this class of bargaining games. Suppose the responder's continuation value from rejecting is uniquely determined by the next subgame, and let x^* denote the unique offer that makes the responder exactly indifferent between accepting and rejecting.

Then in any subgame perfect equilibrium the responder must accept x^* with probability 1.

To see why, suppose instead that at the threshold offer x^* the responder accepts with probability $p < 1$. Then the proposer's expected payoff from offering x^* is only

$$p x^*.$$

But because the offer space is continuous, for any sufficiently small $\varepsilon > 0$ the proposer can instead offer

$$x^* - \varepsilon,$$

which gives the responder strictly more than her continuation value and is therefore accepted with probability 1. The proposer's payoff from this deviation is

$$x^* - \varepsilon.$$

Choosing $\varepsilon < (1 - p)x^*$ yields

$$x^* - \varepsilon > px^*.$$

Hence offering x^* cannot be optimal unless it is accepted with probability 1.

Therefore any mixed response at the acceptance threshold is incompatible with subgame perfection, except for the degenerate mixture that assigns probability 1 to acceptance. Once the responder's continuation value is pinned down, the proposer has a unique optimal offer: the largest offer that is still accepted. This is the deeper reason why the equilibrium is unique in this class of models.

Closed-form payoffs at $t = 1$

Before writing the closed-form expressions, it is useful to see where they come from. They are obtained by iterating backward the recursive continuation-value equations.

If t is odd, Player 1 is the proposer, so

$$V_t = (\delta^{t-1} - V_{t+1}^2, V_{t+1}^2).$$

If t is even, Player 2 is the proposer, so

$$V_t = (V_{t+1}^1, \delta^{t-1} - V_{t+1}^1).$$

Starting from $t = 1$, this gives

$$V_1^1 = 1 - V_2^2.$$

But

$$V_2^2 = \delta - V_3^1,$$

so

$$V_1^1 = 1 - \delta + V_3^1.$$

Continuing recursively,

$$V_3^1 = \delta^2 - V_4^2, \quad V_4^2 = \delta^3 - V_5^1,$$

and so on. Therefore

$$V_1^1 = 1 - \delta + \delta^2 - \delta^3 + \dots + (-\delta)^{T-1}.$$

Similarly,

$$V_1^2 = \delta - \delta^2 + \delta^3 - \dots + (-1)^{T-2}\delta^{T-1}.$$

These are finite geometric series, so they can be written in closed form as

$$V_1^1 = 1 - \delta + \delta^2 - \delta^3 + \dots + (-\delta)^{T-1} = \frac{1 - (-\delta)^T}{1 + \delta},$$

and

$$V_1^2 = \delta - \delta^2 + \delta^3 - \dots + (-1)^{T-2}\delta^{T-1} = \frac{\delta(1 - (-\delta)^{T-1})}{1 + \delta}.$$

Since agreement occurs immediately in equilibrium, these are also the period-1 payoffs generated by the round-1 equilibrium offer.⁴

Interpretation

Several lessons emerge from the finite-horizon model.

First, the proposer always has a first-mover advantage within a given round: she leaves the responder just indifferent and keeps the residual surplus.

Second, the responder is not weak simply because she moves second. Her payoff is disciplined by her continuation value, which depends on future proposal opportunities.

Third, the horizon matters. As the deadline becomes more distant, the responder's continuation value generally rises because rejecting preserves more future bargaining opportunities.

Finally, the model illustrates a general principle of dynamic games: equilibrium outcomes are determined recursively by continuation payoffs.

⁴ *Intuition for future me:* The alternating signs are not an algebraic accident. They come from the recursive structure of the game. Every time we move back one round, proposal power switches players. That is why one additional round lowers the current proposer's payoff, while one more round after that can raise it again by returning proposal power to the original player.

Bridge to infinite-horizon bargaining

The finite-horizon model is the natural bridge to Rubinstein bargaining. As T becomes large, the first-round payoffs approach

$$\left(\frac{1}{1 + \delta}, \frac{\delta}{1 + \delta} \right),$$

which are precisely the equilibrium shares in the infinite alternating-offer game with a common discount factor.

Infinite-Horizon Alternating-Offer Bargaining

We now turn to the infinite-horizon version of the alternating-offer bargaining game. The strategic logic is similar to the finite-horizon case, but the method of solution changes. Since there is no final round from which to start backward induction, we instead look for a *stationary* equilibrium.

Definition. A strategy profile is *stationary* if players' actions depend only on their current role in the game (proposer or responder), and not on calendar time or on the particular history of past rejections.

Remark. In a stationary equilibrium of this game, whenever a player is the proposer she makes the same offer, and whenever a player is the responder she uses the same acceptance rule. The appeal of stationarity is that, after a rejection, the continuation game looks exactly like the original one, except for discounting.

Existence of a stationary equilibrium

We look for a stationary equilibrium in which, in every round, the proposer keeps a fraction x of the pie and offers the responder the remaining share $1 - x$.

The key idea is that, in such a stationary candidate, the responder must be indifferent at the threshold offer. If she accepts, she gets⁵

$$1 - x.$$

If she rejects, then in the next round she becomes the proposer and obtains

$$\delta x.$$

Therefore indifference requires

$$1 - x = \delta x.$$

Solving,

$$x^* = \frac{1}{1 + \delta}, \quad 1 - x^* = \frac{\delta}{1 + \delta}.$$

Remark. We do *not* impose interiority a priori. We start with a generic $x \in [0, 1]$, and the indifference condition delivers

$$x^* = \frac{1}{1 + \delta} \in (0, 1)$$

because $\delta \in (0, 1)$.

This calculation suggests the following stationary strategy profile:

⁵ *Intuition for future me:* The responder compares accepting today with becoming tomorrow's proposer. Stationarity implies that tomorrow's proposer payoff is the same object x , just discounted by δ .

- (i) whenever a player is the proposer, she offers herself the share

$$x^* = \frac{1}{1 + \delta}$$

and offers the responder

$$1 - x^* = \frac{\delta}{1 + \delta};$$

- (ii) whenever a player is the responder to an offer that gives the proposer share x , she accepts if

$$x \leq x^*,$$

and rejects if

$$x > x^*.$$

Notice that at $x = x^*$ the responder is indifferent. We specify acceptance at the boundary. This is the natural analogue of the acceptance-at-indifference convention used in the ultimatum game and in the finite-horizon model: without acceptance at the threshold, the proposer would not have a best response, since she could approach x^* from below without ever attaining it.

Proposition 0.0.4. *There exists a stationary subgame perfect equilibrium of the infinite-horizon alternating-offer bargaining game in which each proposer offers*

$$\left(\frac{1}{1 + \delta}, \frac{\delta}{1 + \delta} \right)$$

between herself and the responder, and the responder accepts an offer if and only if the proposer's share is at most

$$x^* = \frac{1}{1 + \delta}.$$

Proof. Consider the stationary strategy profile described above.

First, the responder's behavior is sequentially rational. If she receives an offer with proposer share $x < x^*$, then her current payoff from accepting is

$$1 - x > 1 - x^* = \frac{\delta}{1 + \delta} = \delta x^*,$$

which is strictly greater than the continuation payoff from rejecting. Hence she strictly prefers to accept.

If instead $x > x^*$, then

$$1 - x < 1 - x^* = \frac{\delta}{1 + \delta} = \delta x^*,$$

so she strictly prefers to reject and wait until next round, when she becomes the proposer.

If $x = x^*$, then she is indifferent:

$$1 - x^* = \delta x^*.$$

So acceptance at the boundary is sequentially rational.

Second, the proposer's behavior is optimal given the responder's acceptance rule. If the proposer offers some $x < x^*$, the offer is accepted, but her payoff is strictly less than x^* . If she offers some $x > x^*$, the responder rejects, and the proposer becomes the responder next period. Her continuation payoff is then

$$\delta(1 - x^*) = \delta \frac{\delta}{1 + \delta} = \frac{\delta^2}{1 + \delta} < \frac{1}{1 + \delta} = x^*.$$

Thus any offer above x^* yields strictly less than x^* , and any offer below x^* also yields strictly less than x^* . Therefore offering exactly x^* is optimal.

Since the same argument applies after every history, the candidate stationary strategy profile is sequentially rational in every subgame. Hence it is a stationary subgame perfect equilibrium. $\square$

Main intuition

The proposer always pushes the offer up to the responder's reservation level. In the infinite-horizon stationary environment, that reservation level is not zero: it is the discounted value of becoming tomorrow's proposer. This is why the proposer keeps

$$\frac{1}{1 + \delta}$$

and the responder gets

$$\frac{\delta}{1 + \delta}.$$

Uniqueness Proof

Class 2: Repeated Games

Why repeated games matter

Many strategic interactions are not one-shot. Firms compete repeatedly in the same market, workers and employers interact over time, and countries repeatedly negotiate, cooperate, and retaliate. In such environments, an action taken today may affect what other players do tomorrow. This is the central force behind repeated games.

A repeated game is built from a *stage game*: in each period, players play the same underlying strategic situation, observe what happened, and then move on to the next period. What makes repeated games interesting is that strategies can condition on past play. Thus a player's action today can influence future continuation play.

Big picture

The main question in repeated games is whether future rewards and punishments can change behavior today.

Stage games, histories, and subgames

Let the same stage game be played in periods

$$t = 1, 2, \dots, T$$

in the finite-horizon case, or in periods

$$t = 1, 2, \dots$$

in the infinite-horizon case.

A *history* at the beginning of period t is the list of all past action profiles:

$$h^t = (a^1, \dots, a^{t-1}).$$

A strategy in a repeated game specifies an action after every possible history.

Remark. In repeated games, the appropriate equilibrium concept is usually *subgame perfect equilibrium* (SPE), because every history defines a continuation game. Sequential rationality must hold after every possible history, not just along the equilibrium path.⁶

⁶ *Intuition for future me:* A repeated game is not just “the same game many times.” It is one large dynamic game in which future play can depend on past actions.

Finitely Repeated Games

Prisoners’ Dilemma played twice

Consider the one-shot Prisoners’ Dilemma with action set $\{C, D\}$ for each player and payoff matrix

	C	D
C	$(3, 3)$	$(0, 4)$
D	$(4, 0)$	$(1, 1)$

where C stands for *cooperate* and D for *defect*.

The stage game has a unique Nash equilibrium, namely (D, D) . Indeed, for each player, D strictly dominates C .

Now suppose this game is played for two periods.

Why defection must occur in period 2

At the beginning of period 2, whatever happened in period 1 is already fixed. Thus every period-2 continuation game is simply the one-shot Prisoners’ Dilemma.

Since the stage game has unique Nash equilibrium (D, D) , every period-2 subgame must prescribe

$$(D, D).$$

Remark. This is the crucial first step. The logic of backward induction starts from the last period because no future incentives remain there.

Why defection must also occur in period 1

Once period 2 is fixed at (D, D) after every possible history, the continuation payoff from period 2 is the same regardless of what a player does in period 1. Therefore, in period 1 each player compares only current period payoffs.

If Player 2 plays C , Player 1 gets

$$u_1(C, C) = 3 \quad \text{from } C, \quad u_1(D, C) = 4 \quad \text{from } D.$$

If Player 2 plays D , Player 1 gets

$$u_1(C, D) = 0 \quad \text{from } C, \quad u_1(D, D) = 1 \quad \text{from } D.$$

So D is still strictly better in period 1. By symmetry, both players choose D .

Proposition 0.0.5. *In the twice-repeated Prisoners' Dilemma, the unique subgame perfect equilibrium prescribes (D, D) in period 2 after every history, and (D, D) in period 1 as well.*

Proof. In every period-2 subgame, the continuation game is the one-shot Prisoners' Dilemma, whose unique Nash equilibrium is (D, D) . Hence period 2 must feature (D, D) after every history.

Given this, period-2 continuation payoffs are independent of period-1 actions. Therefore each player solves the same optimization problem in period 1 as in the one-shot game. Since D strictly dominates C , period 1 must also feature (D, D) . $\square$

What this example is really teaching

The point is not merely that players defect twice. The point is that a unique equilibrium in the last period leaves no room to reward or punish first-period behavior. Once the future is fixed, current incentives collapse back to the one-shot game.

General lesson from the finite-horizon benchmark

The previous argument extends far beyond the Prisoners' Dilemma itself.

Proposition 0.0.6. *Suppose a stage game has a unique Nash equilibrium action profile a^N . If the stage game is repeated a finite number of times with perfect monitoring, then the unique subgame perfect equilibrium outcome is to play a^N in every period after every history.⁷*

Idea of the argument. In the last period, every continuation game is just the stage game, so a^N must be played. Once the last period is fixed independently of previous actions, the same logic applies to the previous period. Iterating this argument backward yields the result. $\square$

Remark. This is sometimes called the *unraveling logic* of finitely repeated games.

⁷ This equilibrium is also *non-contingent*, in the sense that the action profile prescribed in each period does not depend on the history of past play. We will define non-contingent strategy profiles formally later in the notes.

Why the previous logic is not the whole story

At first sight, one might conclude that finite repetition always kills interesting strategic behavior. That conclusion is too strong.

What the previous argument really used was not only finite horizon, but also the fact that the stage game had a *unique* Nash equilibrium. If the stage game has multiple Nash equilibria, then different continuation equilibria can be assigned to different histories, and these can be used as rewards or punishments.

An augmented Prisoners' Dilemma

Consider now the following stage game, obtained by adding a third action E :

	C	D	E
C	(3,3)	(0,4)	(-5,-5)
D	(4,0)	(1,1)	(-5,-5)
E	(-5,-5)	(-5,-5)	(-4,-4)

This game has two pure-strategy Nash equilibria:

$$(D, D) \quad \text{and} \quad (E, E).$$

Indeed:

- if the opponent plays D , the best response is D ;
- if the opponent plays E , the best response is E .

A subgame perfect equilibrium with first-period cooperation

Now consider the two-period repeated version of this augmented game, and define the following strategy profile:

1. In period 1, both players play C .
2. In period 2, if period-1 play was (C, C) , both players play D .
3. In period 2, after any other period-1 history, both players play E .

That is, for each player $i = 1, 2$,

$$s_i(h) = \begin{cases} C, & \text{if } h = \emptyset, \\ D, & \text{if } h = (C, C), \\ E, & \text{if } h \in H^2 \setminus \{(C, C)\}. \end{cases}$$

We now verify that this is a subgame perfect equilibrium.

Step 1: period-2 subgames. After every possible history, the period-2 prescription is either (D, D) or (E, E) . Since both are Nash equilibria of the stage game, sequential rationality holds in every period-2 subgame.

Step 2: incentives in period 1. Suppose Player 2 follows the proposed strategy and plays C in period 1.

If Player 1 also plays C , his total payoff is

$$u_1(C, C) + u_1(D, D) = 3 + 1 = 4.$$

If instead Player 1 deviates to D , then period-1 outcome is (D, C) , so period-2 play is (E, E) . His total payoff becomes

$$u_1(D, C) + u_1(E, E) = 4 + (-4) = 0.$$

If instead Player 1 deviates to E , then period-1 outcome is (E, C) , so period-2 play is again (E, E) . His total payoff becomes

$$u_1(E, C) + u_1(E, E) = (-5) + (-4) = -9.$$

Therefore Player 1 does not want to deviate from C in period 1. By symmetry, neither does Player 2.

Proposition 0.0.7. *In the twice-repeated augmented Prisoners' Dilemma, the following is a subgame perfect equilibrium:*

1. play (C, C) in period 1;
2. play (D, D) in period 2 after (C, C) ;
3. play (E, E) in period 2 after any other period-1 history.

Proof. The period-2 prescriptions are Nash equilibria of the stage game after every history, so all period-2 subgames are sequentially rational. In period 1, given the continuation strategies, each player's payoff from obeying is strictly larger than from any unilateral deviation, as shown above. Hence the strategy profile is a subgame perfect equilibrium. $\square$

Why this counterexample matters

Finite horizon alone does not rule out cooperative behavior in early periods. What ruled it out in the repeated Prisoners' Dilemma was the combination of finite horizon and a unique stage-game equilibrium.

What this is preparing us for

The two examples above are the conceptual gateway to infinitely repeated games.

In the standard Prisoners' Dilemma, finite repetition does not help because the last period destroys all future incentives. But in an infinite-horizon game there is no final period. This breaks the backward-induction chain.

That is the core reason why cooperation may be sustainable in infinitely repeated games even when the stage game has a unique Nash equilibrium.

Preview: why infinity changes everything

Suppose the one-shot Prisoners' Dilemma is repeated forever, and players discount the future with factor

$$\delta \in (0, 1).$$

A natural candidate strategy is a *grim trigger*:

Play C as long as no deviation has occurred; after any deviation, play D forever.

Under this strategy, cooperating forever gives a player the discounted payoff

$$3 + 3\delta + 3\delta^2 + \dots = \frac{3}{1 - \delta}.$$

A one-shot deviation yields

$$4 + \delta (1 + \delta + \delta^2 + \dots) = 4 + \frac{\delta}{1 - \delta}.$$

So cooperation is sustainable if and only if

$$\frac{3}{1 - \delta} \geq 4 + \frac{\delta}{1 - \delta},$$

that is,

$$\delta \geq \frac{1}{3}.$$

Remark. This is the sharp contrast with the finite-horizon case. When the horizon is infinite, the threat of future punishment may be strong enough to discipline current behavior.⁸

⁸ *Intuition for future me:* In a finite game, punishments eventually run out. In an infinite game, they need not. That is why the infinite horizon changes the entire equilibrium logic.

*Formal Setup of a Repeated Game**Motivation*

A repeated game is built by taking a static game—the *stage game*—and playing it over time. The key new object is not the action chosen in a given period, but the rule that specifies what each player will do after every possible history of past play.

This is what makes repeated games fundamentally dynamic: current actions matter not only because of their current payoff consequences, but also because they affect future continuation play.

Central idea

In a repeated game, incentives are history-dependent. A player's action today may change what everyone does tomorrow.

The stage game

Let the stage game be

$$\langle N, A, \pi \rangle,$$

where:

- $N = \{1, \dots, n\}$ is the set of players;
- $A = A_1 \times \dots \times A_n$ is the set of action profiles;
- $\pi : A \rightarrow \mathbb{R}^n$ is the stage-payoff function.

Thus, if the action profile $a = (a_1, \dots, a_n) \in A$ is played in some period, then player i 's stage payoff is

$$\pi_i(a).$$

Remark. The repeated game does not change the primitive one-period incentives encoded in π . What repetition changes is the set of strategies available to the players, by enlarging it from one-period action choices to history-dependent plans of behavior.

Time horizon and discounting

The stage game is repeated over periods

$$t = 1, \dots, T,$$

where $T \leq \infty$.

Players discount future payoffs using discount factor $\delta > 0$. When $T = \infty$, we typically assume

$$\delta \in (0, 1),$$

so that discounted utility is well defined.

Remark. A higher discount factor means that future payoffs matter more relative to current payoffs. This is why patient players are easier to discipline with future rewards and punishments.

Histories

A history at the beginning of period t is the sequence of all past action profiles:

$$h^t = (a^1, \dots, a^{t-1}), \quad h^1 = \emptyset, \quad \text{where } a^t = (a_1^t, \dots, a_n^t) \in A.$$

The set of possible histories at the beginning of period t is therefore

$$\mathcal{H}^t = A^{t-1} = \underbrace{A \times A \times \dots \times A}_{t-1 \text{ times}}, \quad \mathcal{H}^1 = A^0 = \{\emptyset\}.$$

The set of all finite histories is

$$\mathcal{H} = \bigcup_{t=1}^T \mathcal{H}^t.$$

Remark. Because past action profiles are observed, all players share the same public history h^t at the start of period t . For this reason, it is enough to index strategies by histories rather than by a more elaborate collection of information sets.

A note on information

It is important to distinguish between *complete information* and *perfect information*.

Complete information means that all players know the stage game: who the players are, what actions are available, and what the payoff function is.

Perfect information is a statement about timing and observation: whenever a player moves, she observes all payoff-relevant past moves.

In the repeated games we study here, past action profiles are publicly observed, so the history h^t is common to all players at the start of period t . However, if actions within each period are chosen simultaneously, then players do *not* observe their opponents' current-period actions before acting. Thus these repeated games have public monitoring of past play, but are not necessarily games of perfect information in the usual extensive-form sense.

Why histories are enough

At the start of each period, the only payoff-relevant uncertainty about the past is resolved by the observed history h^t . Hence strategies can be written as functions of histories.

Strategies

A standard behavioral strategy for player i is a map

$$\sigma_i : \mathcal{H} \rightarrow \Delta(A_i),$$

where $\Delta(A_i)$ denotes the set of probability distributions over A_i .

Thus, after every history h^t , player i chooses a (possibly mixed) action according to the distribution $\sigma_i(h^t)$.

A strategy profile is then

$$\sigma = (\sigma_1, \dots, \sigma_n).$$

Sometimes it is convenient to summarize behavior by a single map

$$\sigma : \mathcal{H} \rightarrow \Delta(A),$$

which assigns, after every history, a distribution over action profiles. This notation is useful when one wants to describe directly the induced law of play after each history.

Remark. The notation $\sigma : \mathcal{H} \rightarrow \Delta(A)$ should be read as a shorthand for the joint behavior induced by the players' individual strategies. In equilibrium analysis, however, it is often conceptually cleaner to think in terms of the individual strategy maps σ_i .

Discounted utility

Given a strategy profile σ , let $\Pr(h^t \mid \sigma)$ denote the probability of history h^t under σ .

Then player i 's total discounted utility is

$$\tilde{u}_i(\sigma) = \sum_{t=1}^T \delta^{t-1} \sum_{h^t \in \mathcal{H}^t} \pi_i(\sigma(h^t)) \Pr(h^t \mid \sigma).$$

When $\sigma(h^t)$ is interpreted as a distribution over action profiles, the term

$$\pi_i(\sigma(h^t))$$

means the expected stage payoff induced by that distribution.

Normalized average utility

It is often convenient to normalize discounted utility so that repeated-game payoffs are measured in the same units as stage-game payoffs. Define

$$u_i(\sigma) = \frac{1 - \delta}{1 - \delta^T} \sum_{t=1}^T \delta^{t-1} \sum_{h^t \in \mathcal{H}^t} \pi_i(\sigma(h^t)) \Pr(h^t \mid \sigma).$$

If $T = \infty$, this becomes

$$u_i(\sigma) = (1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} \sum_{h^t \in \mathcal{H}^t} \pi_i(\sigma(h^t)) \Pr(h^t \mid \sigma).$$

Remark. This normalization is convenient because if player i receives the same stage payoff c in every period, then her normalized repeated-game payoff is simply c .

Why this formulation matters

This formal setup makes clear what an equilibrium in a repeated game must do. It must specify behavior after *every* possible history. Hence equilibrium analysis is inseparable from continuation values: after any history h^t , the players face a continuation game, and what can be sustained today depends on what continuation play will follow tomorrow.

Class 3: Non-Contingent Strategies and Equilibrium in Finite Repeated Games

Non-contingent strategy profiles

A strategy profile σ is said to be *non-contingent* if, for every period t and every pair of histories $h^t, \hat{h}^t \in \mathcal{H}^t$,

$$\sigma(h^t) = \sigma(\hat{h}^t).$$

Thus, the prescribed play in period t depends only on the date t , not on the realized history leading to that date.⁹

Equivalently, there exists a sequence

$$\alpha^1, \alpha^2, \dots, \alpha^T \in \Delta(A)$$

such that

$$\sigma(h^t) = \alpha^t \quad \text{for every } h^t \in \mathcal{H}^t.$$

Remark. Non-contingent does *not* mean constant over time. It only means history-independent within each period. Hence one may have $\alpha^1 \neq \alpha^2$ and still have a non-contingent strategy profile.

A useful contrast is the following. In a two-period repeated game, the rule

play D in period 1 and D in period 2 regardless of period-1 play

is non-contingent. By contrast, the rule

play D in period 2 after (C, C) , and E otherwise

is contingent, because period-2 play depends on the realized history.

Proposition 0.0.8 (Non-contingent equilibria). *Suppose $T < \infty$. If σ is a non-contingent strategy profile and $\sigma(h^t)$ is a Nash equilibrium of the stage game for every $t = 1, \dots, T$, then σ is a subgame perfect equilibrium of the repeated game.*

⁹ *Intuition for future me:* A non-contingent strategy profile is a calendar plan: what is played may vary across dates, but not across histories within a given date. In particular, current behavior cannot be used to trigger future rewards or punishments.

Verbal proof and intuition. Fix any subgame beginning after some history h^t . Since σ is non-contingent, the continuation play from period t onward depends only on calendar time, not on the realized history. Therefore, a deviation in the current period cannot change future prescribed play. In that sense, what happens today does not affect what is played tomorrow.

Hence, in each period, a player cannot gain by deviating in order to induce a more favorable continuation, because the continuation is already fixed. The only possible gain from deviating would come from the current-period stage payoff itself. But by assumption, the prescribed action profile in each period is a Nash equilibrium of the stage game, so no player can profit from a unilateral deviation in that period.

This argument applies in every subgame. Therefore sequential rationality holds after every history, and σ is a subgame perfect equilibrium. $\square$

Why the proof is so simple

Once future play is history-independent, repeated-game incentives collapse into a sequence of independent static problems. If each date prescribes a stage-game Nash equilibrium, then there is no profitable deviation at any date and after any history.

Proposition 0.0.9 (Unique stage-game equilibrium). *Suppose $T < \infty$. If α^* is the unique Nash equilibrium of the stage game, then the unique subgame perfect equilibrium in the repeated game is the non-contingent strategy profile*

$$\sigma^*(h) = \alpha^* \quad \text{for all } h \in \mathcal{H}.$$

Proof. Existence follows immediately from the previous proposition: the strategy profile that prescribes α^* after every history is non-contingent, and its period-by-period prescriptions are Nash equilibria of the stage game.

To prove uniqueness, let σ be any subgame perfect equilibrium of the repeated game. Consider any subgame beginning in the last period T . Since there is no future after period T , the prescribed action in that subgame must be a Nash equilibrium of the stage game. By uniqueness of the stage-game equilibrium, it must be α^* . Hence period- T play is α^* after every history.

Now move to period $T - 1$. Since period- T play is already fixed at α^* regardless of what happens in period $T - 1$, continuation payoffs from period T onward are independent of current actions. Therefore, in every period- $(T - 1)$ subgame, the prescribed action must again be a Nash equilibrium of the stage game. By uniqueness, it must be α^* .

Iterating this argument backward through periods $T - 2, T - 3, \dots, 1$, we conclude that after every history and in every period, the only

possible equilibrium prescription is α^* . Hence the unique subgame perfect equilibrium is the non-contingent profile $\sigma^*(h) = \alpha^*$ for all $h \in \mathcal{H}$. $\square$

Remark. The uniqueness conclusion relies crucially on uniqueness of the stage-game Nash equilibrium. If the stage game has multiple Nash equilibria, then finite repeated games may admit contingent subgame perfect equilibria in which different continuation equilibria are used as rewards or punishments after different histories.

More interesting equilibria in finitely repeated games

The previous impossibility result for finite repetition relied heavily on the stage game having a unique Nash equilibrium. Once the stage game has multiple Nash equilibria, continuation play may depend on past behavior, and this opens the door to more interesting subgame perfect equilibria.

Main idea

With multiple stage-game equilibria, different continuation equilibria can be assigned to different histories. This allows the players to use future equilibrium play as a reward for obedience and as a punishment for deviation.

Consider the following stage game:

	A	B	C
D	(5, 4)	(4, 0)	(4, 2)
E	(4, 1)	(2, 9)	(8, 6)
F	(4, 2)	(6, 5)	(9, 6)

We set $\delta = 1$, so players simply sum payoffs across periods.

Step 1: stage-game Nash equilibria

We first identify the pure-strategy Nash equilibria of the stage game.

For Player 1:

- if Player 2 plays A, D yields 5, E yields 4, F yields 4,
so the best response is D;
- if Player 2 plays B, D yields 4, E yields 2, F yields 6,
so the best response is F;
- if Player 2 plays C, D yields 4, E yields 8, F yields 9,
so the best response is F.

For Player 2:

- if Player 1 plays D , A yields 4, B yields 0, C yields 2,
so the best response is A ;
- if Player 1 plays E , A yields 1, B yields 9, C yields 6,
so the best response is B ;
- if Player 1 plays F , A yields 2, B yields 5, C yields 6,
so the best response is C .

Hence the stage game has two pure-strategy Nash equilibria:

$$(D, A) \quad \text{and} \quad (F, C).$$

Remark. These two equilibria play different roles in what follows. The profile (F, C) is the “good” continuation equilibrium, while (D, A) is the “bad” continuation equilibrium. The latter will be used as a credible punishment.

Step 2: can we support (E, C) in period 1 when $T = 2$?

We now ask whether the non-equilibrium stage outcome (E, C) can be sustained in period 1 of a two-period repeated game.¹⁰

A natural candidate strategy profile is:

$$\begin{aligned} \sigma(\emptyset) &= (E, C), \\ \sigma(E, C) &= (F, C), \\ \sigma(h^2) &= (D, A) \quad \text{for every } h^2 \neq (E, C). \end{aligned}$$

¹⁰ *Intuition for future me:* The idea is simple: if period 1 goes as planned, players are rewarded with the good equilibrium (F, C) in period 2. If someone deviates in period 1, then period 2 switches to the worse equilibrium (D, A) .

Sequential rationality in period 2. After every possible period-1 history, the prescribed continuation is either (F, C) or (D, A) . Since both are Nash equilibria of the stage game, every period-2 subgame is sequentially rational.

Period-1 incentive constraints. We now verify whether either player wants to deviate from (E, C) in period 1.

For Player 1, obeying yields

$$\pi_1(E, C) + \pi_1(F, C) = 8 + 9 = 17.$$

If Player 1 deviates to F in period 1, the continuation switches to (D, A) , so the payoff becomes

$$\pi_1(F, C) + \pi_1(D, A) = 9 + 5 = 14.$$

Hence Player 1 does not want to deviate.

For Player 2, obeying yields

$$\pi_2(E, C) + \pi_2(F, C) = 6 + 6 = 12.$$

If Player 2 deviates to B in period 1, the continuation switches to (D, A) , so the payoff becomes

$$\pi_2(E, B) + \pi_2(D, A) = 9 + 4 = 13.$$

Hence Player 2 *does* want to deviate.

Therefore (E, C) cannot be supported in period 1 when $T = 2$.

Remark. The one-period punishment is not strong enough. Player 2 gains $9 - 6 = 3$ from deviating immediately, but loses only $6 - 4 = 2$ in the continuation. The punishment does not outweigh the temptation.¹¹

Step 3: can we support (E, C) in period 1 when $T = 3$?

Now suppose the game lasts for three periods. We try again, but this time with two future punishment periods available.¹²

Consider the following strategy profile:

$$\sigma(\emptyset) = (E, C),$$

$$\sigma(h^2) = \begin{cases} (F, C) & \text{if } h^2 = (E, C), \\ (D, A) & \text{if } h^2 \neq (E, C), \end{cases}$$

and in period 3,

$$\sigma(h^3) = \begin{cases} (F, C) & \text{if the period-1 outcome was } (E, C), \\ (D, A) & \text{otherwise.} \end{cases}$$

Sequential rationality in periods 2 and 3. In every continuation subgame, the prescribed play is again either (F, C) or (D, A) , both of which are Nash equilibria of the stage game. Hence continuation play is sequentially rational after every history.

Period-1 incentive constraints. For Player 1, obeying yields

$$\pi_1(E, C) + \pi_1(F, C) + \pi_1(F, C) = 8 + 9 + 9 = 26.$$

If Player 1 deviates to F in period 1, the continuation becomes (D, A) in periods 2 and 3, so the payoff becomes

$$\pi_1(F, C) + \pi_1(D, A) + \pi_1(D, A) = 9 + 5 + 5 = 19.$$

Hence Player 1 does not want to deviate.

¹¹ This is the basic incentive-comparison logic that later leads to the idea of *minmax* or worst credible continuation payoffs. Here we are simply seeing it in a finite-horizon example: a punishment works only if the continuation loss from deviating is large enough to outweigh the current gain. We return to this more formally in the discussion of Nash threats and minmax punishments.

¹² *Intuition for future me:* If the desired first-period outcome is achieved, players are rewarded with two periods of the good equilibrium (F, C) . If not, they are punished with two periods of the bad equilibrium (D, A) . The extra period of punishment may now be enough to deter deviations.

For Player 2, obeying yields

$$\pi_2(E, C) + \pi_2(F, C) + \pi_2(F, C) = 6 + 6 + 6 = 18.$$

If Player 2 deviates to B in period 1, the continuation becomes (D, A) in periods 2 and 3, so the payoff becomes

$$\pi_2(E, B) + \pi_2(D, A) + \pi_2(D, A) = 9 + 4 + 4 = 17.$$

Hence Player 2 does not want to deviate either.

Therefore (E, C) can be supported in period 1 when $T = 3$.

Proposition 0.0.10. *In this finitely repeated game with $\delta = 1$, the outcome (E, C) cannot be supported in period 1 when $T = 2$, but it can be supported in period 1 when $T = 3$.¹³*

Idea of the argument. When $T = 2$, only one future punishment period is available, and this is not enough to deter Player 2's deviation. When $T = 3$, the punishment can last for two periods, which increases the continuation loss from deviating and makes obedience incentive compatible for both players. $\square$

¹³ Any subgame perfect equilibrium that supports (E, C) in period 1 must be contingent. Since (E, C) is not sustained by one-period incentives alone, it can be supported only through continuation play that depends on the history.

What this example is really teaching

The point is not that players want to choose a bad outcome. The point is that they may want to sustain an outcome today that is not a stage-game equilibrium, and they can do so by making future equilibrium play contingent on current behavior. The bad equilibrium is useful not on the equilibrium path, but as a credible off-path punishment.

Why is this useful?

At first glance, it may seem strange to ask whether players would ever want to sustain a stage outcome that is not immediately optimal. But this is exactly what repeated interaction often does in practice.

- In tacit collusion, firms may want to sustain high prices even though each firm would like to undercut in the short run.
- In relational contracts, parties may sustain effort, trust, or favorable treatment today because deviation would damage the future relationship.
- In international cooperation, countries may comply with an agreement today because deviation would trigger future retaliation.

Remark. So the economic value of the example is not that players enjoy punishments. Rather, repeated interaction allows them to use future punishments to support current behavior that would otherwise unravel.

General lesson

This example shows that in finitely repeated games, the number of remaining periods matters because it determines how much continuation value can be used to discipline current deviations.¹⁴

Remark. Still, one should not overstate the conclusion. It is not true in general that “any” first-period action profile can be supported once the horizon is long enough. What is true is that a longer horizon expands the set of outcomes that can be supported by history-dependent equilibrium strategies, provided the stage game offers enough credible continuation equilibria.

¹⁴ *Intuition for future me:* More remaining periods means more room for rewards and punishments. The larger is the continuation value at stake, the larger is the set of first-period outcomes that can potentially be sustained.

Infinitely Repeated Games

Why the infinite horizon changes the logic

In a finitely repeated game, backward induction starts from the last period and progressively eliminates the possibility of using future incentives. In an infinitely repeated game, there is no terminal date. For this reason, continuation values may genuinely discipline current behavior.

Central contrast

In a finite repeated game, punishments eventually run out. In an infinitely repeated game, they need not. This is the fundamental reason why cooperation may be sustainable under infinite repetition.

The infinitely repeated Prisoners’ Dilemma

Consider the Prisoners’ Dilemma stage game

	C	D
C	(3,3)	(0,4)
D	(4,0)	(1,1)

and suppose the game is repeated infinitely many times with common discount factor $\delta \in (0,1)$.

A trivial subgame perfect equilibrium. The strategy profile

$$\sigma(h) = (D,D) \quad \text{for all } h \in \mathcal{H}$$

is a subgame perfect equilibrium. Indeed, after every history the continuation game prescribes the stage-game Nash equilibrium, and D strictly dominates C in the one-shot Prisoners’ Dilemma.

A more interesting candidate: *grim trigger*. Define the following strategy profile:

$$\sigma(h) = \begin{cases} (C, C) & \text{if every action profile in } h \text{ is } (C, C), \\ (D, D) & \text{otherwise.} \end{cases}$$

Thus players cooperate as long as no deviation has ever occurred; once any deviation is observed, they defect forever.

Proposition 0.0.11. *The grim-trigger strategy profile is a subgame perfect equilibrium if and only if*

$$\delta \geq \frac{1}{3}.$$

Idea of the argument. There are only two types of subgames to check.

First, consider any history after which a deviation has already occurred. In such a subgame, the continuation play is (D, D) forever. Since D is a best response to D in the stage game, no player wishes to deviate.

Second, consider a history on the cooperative path, say

$$h^t = ((C, C), \dots, (C, C)),$$

with $t - 1$ periods of past cooperation. If player i cooperates at date t , then cooperation continues forever, yielding

$$\pi_i(C, C) + \sum_{s=t+1}^{\infty} \delta^{s-t} \pi_i(C, C).$$

If instead player i deviates once while the opponent plays C , then the player obtains the one-shot gain $\pi_i(D, C)$ today and triggers permanent defection tomorrow, yielding

$$\pi_i(D, C) + \sum_{s=t+1}^{\infty} \delta^{s-t} \pi_i(D, D).$$

Using

$$\pi_i(C, C) = 3, \quad \pi_i(D, C) = 4, \quad \pi_i(D, D) = 1,$$

the no-deviation condition becomes

$$3 + \sum_{s=t+1}^{\infty} \delta^{s-t} 3 \geq 4 + \sum_{s=t+1}^{\infty} \delta^{s-t} 1.$$

Since

$$\sum_{s=t+1}^{\infty} \delta^{s-t} = \delta + \delta^2 + \dots = \frac{\delta}{1-\delta},$$

this is equivalent to

$$3 + \frac{3\delta}{1-\delta} \geq 4 + \frac{\delta}{1-\delta},$$

or

$$\delta \geq \frac{1}{3}.$$

Hence grim trigger is a subgame perfect equilibrium precisely when players are sufficiently patient. $\square$

Remark. The summation index s above is just a dummy variable indexing future periods. Some presentations write $\hat{t}$, but there is no special meaning attached to the hat; it is simply the period being summed over.

The one-shot deviation principle

At first glance, checking whether a strategy profile is subgame perfect seems difficult because one must rule out arbitrary deviations: a player might deviate today, again tomorrow, and condition future behavior on what others do in response.

The one-shot deviation principle provides a major simplification.

Definition. Fix a strategy profile $\sigma = (\sigma_i, \sigma_{-i})$. A strategy $\hat{\sigma}_i \neq \sigma_i$ is a *one-shot deviation* for player i if there exists a unique history $\hat{h} \in \mathcal{H}$ such that

$$\hat{\sigma}_i(\hat{h}) \neq \sigma_i(\hat{h}), \quad \hat{\sigma}_i(h) = \sigma_i(h) \quad \text{for all } h \neq \hat{h}.$$

It is a *profitable one-shot deviation* if, in the subgame beginning at $\hat{h}$,

$$\tilde{u}_i(\hat{\sigma}_i|_{\hat{h}}, \sigma_{-i}|_{\hat{h}}) > \tilde{u}_i(\sigma|_{\hat{h}}).$$

Remark. The notation $\tilde{u}_i(\cdot|_{\hat{h}})$ refers to continuation utility in the subgame that starts after history $\hat{h}$. Thus the comparison is local to that subgame, not to the game from date 1.

Theorem 0.0.12 (One-shot deviation principle). *In a finitely or infinitely repeated game, a strategy profile is a subgame perfect equilibrium if and only if it admits no profitable one-shot deviations.*

Sketch of the argument. The “only if” direction is immediate. If σ is subgame perfect, then after every history $\hat{h}$ it must be sequentially rational in the continuation game starting at $\hat{h}$. Hence no unilateral deviation—in particular, no one-shot deviation—can be profitable there.

For the converse, suppose σ is not subgame perfect. Then there exists some history $\hat{h}$ and some player i for whom a deviation plan τ_i is profitable in the continuation game starting at $\hat{h}$.

If the horizon is finite, choose a history at which τ_i differs from σ_i as late as possible. Because this is the last point of disagreement, from that point onward the continuation coincides again with σ_i . Therefore the gain from the deviation must already be present in that single local change. Hence there exists a profitable one-shot deviation.

If the horizon is infinite, let $\varepsilon > 0$ denote the gain from the profitable deviation τ_i . Since stage payoffs are bounded and discounted, for T large enough the maximal contribution of the tail after date T is smaller than $\varepsilon/2$:

$$\frac{\delta^T}{1-\delta} \left(\max_{a \in A} \pi_i(a) - \min_{a \in A} \pi_i(a) \right) \leq \frac{\varepsilon}{2}.$$

Hence truncating the deviation after T periods still leaves a profitable deviation. Once the problem is truncated, the argument reduces to the finite-horizon case, which yields a profitable one-shot deviation.

Therefore, whenever a strategy profile is not subgame perfect, there must exist a profitable one-shot deviation. $\square$

Why the principle is so useful

To verify subgame perfection, one does not need to inspect arbitrarily complicated multi-period deviations. It is enough to check whether a player can gain by changing behavior at a single history and then reverting to the original strategy thereafter.

Why the principle matters for grim trigger

The previous proposition about grim trigger checked only a unilateral deviation at one cooperative history. This is not a shortcut or a heuristic. It is justified by the one-shot deviation principle.¹⁵

¹⁵ *Intuition for future me:* If no player can gain from a single local deviation at any history, then no more elaborate deviation plan can help either. This is why the grim-trigger calculation reduces to comparing cooperation today with a one-time deviation today.

Appendix to Class 3: Grim Trigger strategy

Consider the following stage game:

	A	B	C
D	(4, 0)	(6, 9)	(2, 8)
E	(7, 5)	(3, 6)	(2, 2)
F	(3, 0)	(1, 3)	(7, 4)

We study the two-period repeated game with $T = 2$ and ask whether there exists a subgame perfect equilibrium in which (E, A) is played in period 1.

Step 1: Pure Nash equilibria of the stage game

We first compute best responses.

For Player 1:

if Player 2 plays A, $D \mapsto 4$, $E \mapsto 7$, $F \mapsto 3$, so $BR_1(A) = \{E\}$;
 if Player 2 plays B, $D \mapsto 6$, $E \mapsto 3$, $F \mapsto 1$, so $BR_1(B) = \{D\}$;
 if Player 2 plays C, $D \mapsto 2$, $E \mapsto 2$, $F \mapsto 7$, so $BR_1(C) = \{F\}$.

For Player 2:

if Player 1 plays D, $A \mapsto 0$, $B \mapsto 9$, $C \mapsto 8$, so $BR_2(D) = \{B\}$;
 if Player 1 plays E, $A \mapsto 5$, $B \mapsto 6$, $C \mapsto 2$, so $BR_2(E) = \{B\}$;
 if Player 1 plays F, $A \mapsto 0$, $B \mapsto 3$, $C \mapsto 4$, so $BR_2(F) = \{C\}$.

Hence the pure-strategy Nash equilibria of the stage game are

$$(D, B) \quad \text{and} \quad (F, C).$$

Step 2: Can (E, A) be supported in period 1 when $T = 2$?

Yes. The key observation is that period 2 is the last period, so after every period-1 history the continuation must prescribe a stage-game Nash equilibrium. Since the stage game has two different Nash equilibria, we can use one as a reward and the other as a punishment.

Consider the following strategy profile:

$$\sigma(\emptyset) = (E, A),$$

and for period 2,

$$\sigma(h^2) = \begin{cases} (D, B), & \text{if } h^2 = (E, A), \\ (F, C), & \text{if } h^2 \in \{(E, B), (E, C)\}, \\ (D, B), & \text{otherwise.} \end{cases}$$

Remark. This strategy says: play (E, A) in period 1. If Player 2 follows the prescribed action A , then reward the period-1 outcome with the continuation equilibrium (D, B) , which is especially favorable for Player 2. If Player 2 deviates from A , switch in period 2 to (F, C) , which gives Player 2 a lower continuation payoff.

Step 3: Sequential rationality in period 2

After every possible history in period 1, the strategy prescribes either (D, B) or (F, C) in period 2. Since both are Nash equilibria of the stage game, continuation play is sequentially rational in every period-2 subgame.

Step 4: Incentive constraints in period 1

We now check whether either player has a profitable one-shot deviation in period 1.

Player 1. If Player 1 follows the prescription and plays E against A , his total payoff is

$$u_1(E, A) + u_1(D, B) = 7 + 6 = 13.$$

If instead Player 1 deviates in period 1, the best deviation against A is D , since

$$u_1(D, A) = 4 \quad \text{and} \quad u_1(F, A) = 3.$$

Under the proposed strategy, the continuation after such a history is still a stage-game Nash equilibrium, namely (D, B) ¹⁶. Hence the maximal payoff from deviation is

$$u_1(D, A) + u_1(D, B) = 4 + 6 = 10.$$

Therefore Player 1 does not want to deviate.

Player 2. If Player 2 follows the prescription and plays A against E , his total payoff is

$$u_2(E, A) + u_2(D, B) = 5 + 9 = 14.$$

If Player 2 deviates in period 1, the best deviation against E is B , since

$$u_2(E, B) = 6 \quad \text{and} \quad u_2(E, C) = 2.$$

After such a deviation, the continuation switches to (F, C) , so the deviation payoff is

$$u_2(E, B) + u_2(F, C) = 6 + 4 = 10.$$

¹⁶ This continuation is not arbitrary. After the on-path history (E, A) , period-2 play must reward Player 2 for complying in period 1. If we replaced (D, B) by (F, C) after (E, A) , then Player 2 would again prefer to deviate in period 1, since compliance would yield $u_2(E, A) + u_2(F, C) = 5 + 4 = 9$, whereas deviating to B would yield $u_2(E, B) + u_2(F, C) = 6 + 4 = 10$. Thus the continuation after the on-path history must be sufficiently favorable to Player 2, and (D, B) serves precisely that role.

Hence Player 2 does not want to deviate either.

We conclude that no player has a profitable one-shot deviation in period 1. Therefore the proposed strategy profile is a subgame perfect equilibrium.

Proposition 0.0.13. *In the two-period repeated game, there exists a subgame perfect equilibrium in which (E, A) is played in period 1.*

Proof. The period-2 continuation after every history is a stage-game Nash equilibrium, so all continuation subgames are sequentially rational. The period-1 incentive constraints computed above show that neither player has a profitable deviation from (E, A) . Hence the strategy profile is subgame perfect. $\square$

Interpretation

The economic logic is that (E, A) is not itself a stage-game Nash equilibrium, but it can still be sustained in the repeated game because future equilibrium play is made contingent on current behavior. Since there are multiple stage-game Nash equilibria, one can be used as a reward and another as a punishment.

Remark. The right intuition is not to think literally of grim trigger. Rather, the useful analogy is this: in a repeated game, continuation play can depend on the realized history. In a *finite* repeated game, however, the last period must always prescribe a stage-game Nash equilibrium. Therefore, the only credible way to create incentives in period 1 is to assign *different stage-game Nash equilibria* to different period-1 histories. That is exactly what happens here: period 2 always plays a stage-game equilibrium, but which equilibrium is played depends on whether the prescribed first-period action was followed.

Remark (Why this equilibrium is not non-contingent). This example is useful because it clarifies the difference between *non-contingent* and *contingent* subgame perfect equilibria.

Recall that a strategy profile is non-contingent if, at each date t , the prescribed action depends only on the calendar date and not on the history leading to that date. In a two-period repeated game, this means that period-2 play must be the same after every possible period-1 history:

$$\sigma(h^2) = \sigma(\hat{h}^2) \quad \text{for all } h^2, \hat{h}^2.$$

The equilibrium constructed in this exercise is *not* non-contingent. The reason is that period-2 play depends on what happened in period 1: after the on-path history (E, A) , the continuation prescribes one stage-game Nash equilibrium, while after a deviation it prescribes another.

Thus period-2 behavior is history-dependent.

This distinction is economically essential. If period-2 play were non-contingent, then a deviation in period 1 would not affect the continuation. In that case, no reward or punishment could be attached to first-period behavior, so the incentive constraints in period 1 would collapse to a comparison of current-period payoffs only. But (E, A) is not a Nash equilibrium of the stage game, since Player 2 would prefer to deviate from A to B :

$$u_2(E, A) = 5 \quad \text{while} \quad u_2(E, B) = 6.$$

Hence (E, A) could not be sustained by any non-contingent strategy profile.

The lesson is that the equilibrium works precisely because continuation play is contingent on past behavior. Even in a finite repeated game, and even though the last period must always prescribe a stage-game Nash equilibrium, different stage-game equilibria can still be assigned to different histories and thereby used as credible rewards and punishments.

Part (c): No SPE (pure or mixed) can play (D, C) in period 1 when $T = 2$

We now show that in the two-period repeated game there is no subgame perfect equilibrium—whether in pure or mixed strategies—that prescribes (D, C) in period 1.

Step 1: Stage-game Nash equilibria and equilibrium payoffs. From part (a), the pure-strategy Nash equilibria of the stage game are

$$(D, B) \quad \text{and} \quad (F, C),$$

with payoffs

$$(6, 9) \quad \text{and} \quad (7, 4),$$

respectively.

There is also a mixed-strategy Nash equilibrium in which Player 1 mixes between D and F , and Player 2 mixes between B and C . Let Player 1 play D with probability p , and let Player 2 play B with probability q . Indifference requires

$$u_1(D \mid q) = u_1(F \mid q) \quad \text{and} \quad u_2(B \mid p) = u_2(C \mid p).$$

For Player 1,

$$u_1(D \mid q) = 6q + 2(1 - q) = 2 + 4q,$$

$$u_1(F \mid q) = 1q + 7(1 - q) = 7 - 6q,$$

so

$$2 + 4q = 7 - 6q \implies q = \frac{1}{2}.$$

For Player 2,

$$u_2(B | p) = 9p + 3(1 - p) = 3 + 6p,$$

$$u_2(C | p) = 8p + 4(1 - p) = 4 + 4p,$$

so

$$3 + 6p = 4 + 4p \implies p = \frac{1}{2}.$$

Hence the mixed stage-game equilibrium is

$$\left(\frac{1}{2}D + \frac{1}{2}F, \frac{1}{2}B + \frac{1}{2}C \right),$$

with payoff

$$(4, 6).$$

Therefore, the set of stage-game equilibrium payoffs is

$$(6, 9), \quad (7, 4), \quad (4, 6).$$

Step 2: Why (D, C) cannot be supported. Suppose, toward a contradiction, that there exists an SPE σ such that

$$\sigma(\emptyset) = (D, C).$$

Since $T = 2$, period 2 is the last period. Therefore, after every period-1 history, the continuation play in period 2 must be a Nash equilibrium of the stage game, possibly mixed.

Let $V_1(h^2)$ denote Player 1's continuation payoff in period 2 after history h^2 . Because continuation play must be a stage-game Nash equilibrium, Player 1's continuation payoff must satisfy

$$4 \leq V_1(h^2) \leq 7 \quad \text{for every } h^2.$$

Now consider Player 1 in period 1. If he follows the prescribed action D while Player 2 plays C , his current-period payoff is

$$u_1(D, C) = 2.$$

If instead he deviates to F , while Player 2 still plays C , his current-period payoff becomes

$$u_1(F, C) = 7.$$

Thus the deviation yields an immediate gain of 5.

For (D, C) to be sustainable, Player 1 must not profit from this one-shot deviation. Hence we would need

$$u_1(D, C) + V_1(D, C) \geq u_1(F, C) + V_1(F, C),$$

that is,

$$2 + V_1(D, C) \geq 7 + V_1(F, C).$$

But even under the most favorable possible continuation assignment for sustaining obedience, we can do no better than

$$V_1(D, C) = 7 \quad \text{and} \quad V_1(F, C) = 4,$$

since 7 is the largest equilibrium continuation payoff available to Player 1 and 4 is the smallest.

Therefore the strongest possible incentive inequality would be

$$2 + 7 \geq 7 + 4,$$

that is,

$$9 \geq 11,$$

which is false.

This contradiction shows that Player 1 always has a profitable deviation from (D, C) in period 1. Hence no SPE can prescribe (D, C) in the first period when $T = 2$, even allowing mixed continuation equilibria.

Proposition 0.0.14. *In the two-period repeated game, there is no subgame perfect equilibrium—in pure or mixed strategies—that prescribes (D, C) in period 1.*

Remark. The proof is stronger than the pure-strategy argument. It does not merely say that pure continuation equilibria are insufficient. It says that even after enlarging the set of credible continuation payoffs to include the mixed stage-game equilibrium payoff $(4, 6)$, the deviation incentive of Player 1 remains too strong.

Part (d): Finite horizons and the role of mixed continuation equilibria

We now ask whether (D, C) can be supported in period 1 when the game is repeated T times, with $T < \infty$.¹⁷

The key point is that the mixed stage-game equilibrium payoff $(4, 6)$ gives a lower continuation payoff to Player 1 than either pure equilibrium payoff. This makes it a more severe punishment for Player 1 and allows (D, C) to be supported with a shorter finite horizon.

A mixed continuation construction with $T = 3$. Consider a three-period repeated game. Let

$$m = \left(\frac{1}{2}D + \frac{1}{2}F, \frac{1}{2}B + \frac{1}{2}C \right)$$

denote the mixed stage-game Nash equilibrium, with payoff

$$u(m) = (4, 6).$$

¹⁷ A useful general procedure for mixed-strategy SPE problems is the following. First, compute *all* stage-game Nash equilibria, including mixed ones, and record their payoff vectors. Second, use the fact that after any history in the final period, continuation play must be one of those stage-game equilibria. Third, to support a target nonequilibrium outcome in an earlier period, work directly with continuation *equilibrium payoffs*: choose on-path continuation payoffs to reward obedience and off-path continuation payoffs to punish deviations. Only after the incentive inequalities are satisfied does one go back and specify the mixed action profile that implements the relevant equilibrium payoff. In repeated-game problems, it is often much easier to reason first in terms of equilibrium payoff vectors than in terms of mixing probabilities.

We construct the following strategy profile:

$$\sigma(\emptyset) = (D, C).$$

If period 1 play is (D, C) , then the continuation is

$$(D, B) \text{ in period 2, } (F, C) \text{ in period 3.}$$

If Player 1 is the first player to deviate from the prescribed period-1 action, then the continuation is

$$m \text{ in periods 2 and 3.}$$

If Player 2 is the first player to deviate from the prescribed period-1 action, then the continuation is

$$(F, C) \text{ in periods 2 and 3.}$$

After any other off-path history, prescribe any stage-game Nash equilibrium, say m .

Sequential rationality. From period 2 onward, the prescribed continuation play is always a stage-game Nash equilibrium: either (D, B) , (F, C) , or m . Hence every continuation subgame is sequentially rational.

Incentive constraints in period 1. We now check deviations in period 1.

For Player 1, obeying yields

$$u_1(D, C) + u_1(D, B) + u_1(F, C) = 2 + 6 + 7 = 15.$$

The most profitable deviation is to switch from D to F , which yields

$$u_1(F, C) + u_1(m) + u_1(m) = 7 + 4 + 4 = 15.$$

Thus Player 1 does not profit from deviating.

For Player 2, obeying yields

$$u_2(D, C) + u_2(D, B) + u_2(F, C) = 8 + 9 + 4 = 21.$$

The most profitable deviation is to switch from C to B , which yields

$$u_2(D, B) + u_2(F, C) + u_2(F, C) = 9 + 4 + 4 = 17.$$

Thus Player 2 does not profit from deviating either.

Therefore (D, C) can be supported in period 1 when $T = 3$.

Proposition 0.0.15. *There exists a finite-horizon subgame perfect equilibrium with $T = 3$ in which (D, C) is played in period 1.*

Proof. The proposed continuation play after every history is a stage-game Nash equilibrium, so all subgames from period 2 onward are sequentially rational. The incentive inequalities above show that neither player has a profitable deviation in period 1. Hence the strategy profile is a subgame perfect equilibrium. $\square$

Remark. The mixed equilibrium is useful because it delivers the lowest equilibrium continuation payoff to Player 1, namely 4. This is what makes the punishment strong enough already with only two future periods remaining. By contrast, if one restricts attention to the two pure stage-game equilibria only, then the lowest continuation payoff available to Player 1 is 6, which makes punishment weaker. In that sense, mixed continuation equilibria are what allow the short-horizon construction.

Part (e): Infinite horizon

We now consider the infinitely repeated game with discount factor $\delta \in (0, 1)$, and show that if

$$\delta \geq \frac{1}{2},$$

then there exists a subgame perfect equilibrium in which (E, A) is played in every period along the equilibrium path.

Candidate strategy profile. The equilibrium path prescribes

$$(E, A) \text{ in every period.}$$

To sustain this outcome, define the following contingent strategy profile:

- As long as no deviation from (E, A) has ever occurred, play (E, A) .
- If Player 1 is the first player to deviate from the prescribed action E while Player 2 plays A , then from the next period onward play (D, B) forever.
- If Player 2 is the first player to deviate from the prescribed action A while Player 1 plays E , then from the next period onward play (F, C) forever.
- If both players deviate simultaneously at the first off-path history, prescribe any stage-game Nash equilibrium forever, say (D, B) .

Remark. This construction uses (D, B) as the punishment path for Player 1 and (F, C) as the punishment path for Player 2. The reason is that (D, B) gives Player 1 a lower continuation payoff than the cooperative path, while (F, C) gives Player 2 a lower continuation payoff than the cooperative path.

Why the continuation paths are credible. Both (D, B) and (F, C) are Nash equilibria of the stage game. Hence the non-contingent strategy profiles

$$(D, B) \text{ forever} \quad \text{and} \quad (F, C) \text{ forever}$$

are themselves subgame perfect equilibria of the infinitely repeated game. Therefore the punishment paths are credible.

Checking incentives on the equilibrium path. By the one-shot deviation principle, it is enough to check whether either player wants to deviate at a history on the equilibrium path.

Player 1. If Player 1 obeys and plays E while Player 2 plays A , then his continuation payoff is

$$7 + \delta 7 + \delta^2 7 + \dots = \frac{7}{1 - \delta}.$$

If instead Player 1 deviates, the best deviation against A is to play D , which gives current payoff 4. From the next period onward, the punishment path (D, B) is played forever, yielding continuation payoff 6 each period. Thus deviation gives

$$4 + \delta \frac{6}{1 - \delta}.$$

Hence Player 1 does not deviate if

$$\frac{7}{1 - \delta} \geq 4 + \delta \frac{6}{1 - \delta}.$$

Multiplying by $1 - \delta$ gives

$$7 \geq 4(1 - \delta) + 6\delta = 4 + 2\delta.$$

Equivalently,

$$3 \geq 2\delta.$$

This holds for every $\delta \in (0, 1)$. So Player 1 never has a profitable deviation.

Player 2. If Player 2 obeys and plays A while Player 1 plays E , then his continuation payoff is

$$5 + \delta 5 + \delta^2 5 + \dots = \frac{5}{1 - \delta}.$$

If instead Player 2 deviates, the best deviation against E is to play B , which gives current payoff 6. From the next period onward, the punishment path (F, C) is played forever, yielding continuation payoff 4 each period. Thus deviation gives

$$6 + \delta \frac{4}{1 - \delta}.$$

Hence Player 2 does not deviate if

$$\frac{5}{1-\delta} \geq 6 + \delta \frac{4}{1-\delta}.$$

Multiplying by $1 - \delta$ gives

$$5 \geq 6(1 - \delta) + 4\delta = 6 - 2\delta.$$

Equivalently,

$$2\delta \geq 1,$$

that is,

$$\delta \geq \frac{1}{2}.$$

Conclusion. If

$$\delta \geq \frac{1}{2},$$

then neither player has a profitable one-shot deviation on the equilibrium path, and the punishment paths are themselves subgame perfect. Therefore the proposed strategy profile is a subgame perfect equilibrium.

Proposition 0.0.16. *For every $\delta \geq \frac{1}{2}$, there exists a subgame perfect equilibrium of the infinitely repeated game in which (E, A) is played in every period along the equilibrium path.*

Proof. The strategy profile described above prescribes (E, A) forever on path, switches to (D, B) forever after a unilateral deviation by Player 1, and switches to (F, C) forever after a unilateral deviation by Player 2. Since both punishment paths are repetitions of stage-game Nash equilibria, they are credible. The incentive constraints above show that Player 1 never wants to deviate, and Player 2 does not want to deviate if and only if $\delta \geq \frac{1}{2}$. Hence the strategy profile is subgame perfect whenever $\delta \geq \frac{1}{2}$. $\square$

What a complete exam answer should contain

A good answer has three ingredients. First, it must specify the *equilibrium path*: here, (E, A) forever. Second, it must specify the *punishment path after a deviation by Player 1*: here, (D, B) forever. Third, it must specify the *punishment path after a deviation by Player 2*: here, (F, C) forever. Once those paths are clearly stated, the rest is to verify that the punishment paths are credible and that neither player has a profitable one-shot deviation from the equilibrium path.

Class 4: Infinite Repeated Duopoly and Collusion

Collusion in infinitely repeated duopoly

We now move from finite repetition to infinite repetition. The key new force is that, with no last period, future punishments can be used to sustain outcomes that are not stage-game Nash equilibria. Two canonical examples are infinitely repeated Cournot and Bertrand duopoly.

Infinite Cournot Duopoly

In each period, two firms simultaneously choose quantities. The inverse demand curve is

$$P = 12 - Q, \quad \text{where } Q = q_1 + q_2,$$

and production costs are zero. Firms discount future payoffs with factor $\delta < 1$.

Stage-game Nash equilibrium Firm i 's one-period profit is

$$\pi_i(q_i, q_j) = Pq_i = (12 - q_i - q_j)q_i.$$

Taking the derivative with respect to q_i ,

$$\frac{\partial \pi_i}{\partial q_i} = 12 - 2q_i - q_j.$$

Thus the best response of firm i is

$$q_i = 6 - \frac{q_j}{2}.$$

By symmetry, the stage-game Nash equilibrium is

$$q_1^{NE} = q_2^{NE} = 4.$$

The associated one-period profit of each firm is

$$\pi_i(4, 4) = (12 - 4 - 4) \cdot 4 = 16.$$

Collusive benchmark If the two firms behaved like a monopolist, they would solve

$$\max_Q (12 - Q)Q,$$

whose first-order condition is

$$12 - 2Q = 0 \quad \implies \quad Q^m = 6.$$

A natural symmetric collusive allocation splits monopoly output equally:

$$(q_1, q_2) = (3, 3).$$

Under this allocation, each firm obtains

$$\pi_i(3, 3) = (12 - 3 - 3) \cdot 3 = 18.$$

Remark. The symmetric split $(3, 3)$ is only one collusive allocation. More generally, any pair (q_1, q_2) with $q_1 + q_2 = 6$ reproduces monopoly output. We focus on the symmetric split because it is the cleanest benchmark.

A grim-trigger strategy profile Let h^t denote the history before period t . Consider the following strategy profile:

$$\sigma(h^t) = \begin{cases} (3, 3), & \text{if every past outcome in } h^t \text{ was } (3, 3), \\ (4, 4), & \text{otherwise.} \end{cases}$$

Thus the firms collude forever as long as no deviation has occurred, and switch permanently to the stage-game Nash equilibrium after the first deviation.

Remark. At every off-path history, continuation play is perpetual repetition of $(4, 4)$, which is a Nash equilibrium of the stage game. Hence sequential rationality is automatic after those histories. **The only non-trivial issue is whether a firm wants to deviate on the equilibrium path from $(3, 3)$.**

Proposition 0.0.17. *The grim-trigger profile above is a subgame perfect equilibrium if and only if*

$$\delta \geq \frac{9}{17}.$$

Proof. If a firm follows the collusive recommendation forever, its discounted payoff is

$$18 + \delta 18 + \delta^2 18 + \cdots = \frac{18}{1 - \delta}.$$

Now consider a one-shot deviation when the rival plays 3. The deviating firm solves

$$\max_{q_i} (12 - q_i - 3)q_i = (9 - q_i)q_i.$$

The first-order condition is

$$9 - 2q_i = 0 \quad \implies \quad q_i = \frac{9}{2}.$$

So the best deviation from (3, 3) is

$$BR_i(3) = \frac{9}{2},$$

which yields current profit

$$\pi_i\left(\frac{9}{2}, 3\right) = \left(12 - \frac{9}{2} - 3\right) \frac{9}{2} = \frac{81}{4} = 20.25.$$

After that deviation, the strategy profile prescribes (4, 4) forever, so continuation profit is

$$16 + \delta 16 + \delta^2 16 + \dots = \frac{16}{1 - \delta},$$

but discounted from next period onward. Hence the total payoff from deviating is

$$20.25 + \delta \frac{16}{1 - \delta}.$$

The collusive path is incentive compatible if

$$\frac{18}{1 - \delta} \geq 20.25 + \delta \frac{16}{1 - \delta}.$$

Multiplying both sides by $1 - \delta$,

$$18 \geq 20.25 - 4.25\delta,$$

so

$$4.25\delta \geq 2.25,$$

which gives

$$\delta \geq \frac{2.25}{4.25} = \frac{9}{17}.$$

Therefore the grim-trigger strategy profile is a subgame perfect equilibrium exactly when $\delta \geq 9/17$.¹⁸ $\square$

Infinite Bertrand Duopoly

In each period, two firms simultaneously choose prices. There is a unit mass of consumers, each demanding one unit at any price weakly below 1. The lowest price gets the whole market; if prices tie, firms split demand equally. Costs are zero, and firms discount future payoffs with factor $\delta < 1$.

A convenient reduced-form profit function is

$$\pi_i(p_i, p_j) = \begin{cases} p_i, & \text{if } p_i < p_j \text{ and } p_i \leq 1, \\ \frac{p_i}{2}, & \text{if } p_i = p_j \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

¹⁸ *Intuition for future me:* In Cournot, collusion raises each firm's current profit from 16 to 18, but it also creates a temptation to expand output and grab a larger one-period payoff. **The role of repetition is to make this short-run temptation costly by threatening a permanent return to the competitive Cournot equilibrium.**

Stage-game Nash equilibrium The stage-game Nash equilibrium is the standard Bertrand outcome

$$(p_1, p_2) = (0, 0),$$

with profits

$$\pi_i(0, 0) = 0.$$

Collusive benchmark The monopoly price is

$$p^m = 1,$$

since demand remains positive up to price 1. If both firms set 1, they split the market equally, so each obtains

$$\pi_i(1, 1) = \frac{1}{2}.$$

A grim-trigger strategy profile Consider the following strategy profile:

$$\sigma(h^t) = \begin{cases} (1, 1), & \text{if every past outcome in } h^t \text{ was } (1, 1), \\ (0, 0), & \text{otherwise.} \end{cases}$$

Thus the firms charge the monopoly price forever unless a deviation occurs, in which case they revert forever to the Bertrand stage-game equilibrium.

Remark. As in Cournot, every off-path continuation is sequentially rational because perpetual play of $(0, 0)$ simply repeats the stage-game Nash equilibrium. The only issue is whether a firm wants to deviate from $(1, 1)$ along the equilibrium path.

Remark (Why there is no best deviation from $(1, 1)$). This case is slightly subtler than Cournot. If the rival charges 1, then any deviation $p_i < 1$ captures the whole market and yields current profit p_i . Hence the deviating firm would like to choose a price as close as possible to 1, while still remaining strictly below it.

But the set of such deviations is open: there is no price $p_i < 1$ that maximizes profit. There is therefore no *best deviation*; instead there is only a supremum:

$$\sup_{p_i < 1} \pi_i(p_i, 1) = \sup_{p_i < 1} p_i = 1.$$

It is also clear why one does not need to consider deviations with $p_i \leq \frac{1}{2}$. Such a deviation yields current profit at most $\frac{1}{2}$, which is weakly below the collusive current payoff $\pi_i(1, 1) = \frac{1}{2}$, even before any future punishment is taken into account. Thus only deviations with

$$p_i \in \left(\frac{1}{2}, 1\right)$$

can possibly be profitable, and their payoff approaches 1 as $p_i \uparrow 1$.

Proposition 0.0.18. *The grim-trigger profile above is a subgame perfect equilibrium if and only if*

$$\delta \geq \frac{1}{2}.$$

Proof. If a firm follows the collusive recommendation forever, its discounted payoff is

$$\frac{1}{2} + \delta \frac{1}{2} + \delta^2 \frac{1}{2} + \cdots = \frac{1/2}{1-\delta}.$$

If instead it deviates from $(1, 1)$, then for any $p_i < 1$ it captures the whole market in the current period and earns p_i . From the next period onward, the punishment $(0, 0)$ yields zero forever. Thus the highest achievable deviation payoff is not a maximum but the supremum

$$\sup_{p_i < 1} \left\{ p_i + \delta \cdot 0 + \delta^2 \cdot 0 + \cdots \right\} = \sup_{p_i < 1} p_i = 1.$$

Therefore the collusive path is incentive compatible if

$$\frac{1/2}{1-\delta} \geq 1.$$

This inequality is equivalent to

$$\frac{1}{2} \geq 1 - \delta,$$

hence

$$\delta \geq \frac{1}{2}.$$

So the grim-trigger profile is a subgame perfect equilibrium exactly when $\delta \geq 1/2$.¹⁹ $\square$

The Nash-Threat Folk Theorem

We now move from examples to a general characterization. The key question is the following: in an infinitely repeated game, which payoff vectors can be sustained as subgame perfect equilibrium payoffs when players are sufficiently patient?

Let the stage game have player set $N = \{1, \dots, n\}$, action set

$$A = \prod_{i=1}^n A_i,$$

and stage-payoff function

$$\pi : A \rightarrow \mathbb{R}^n.$$

¹⁹ *Intuition for future me:* The difference with Cournot is instructive. In Cournot, a deviating firm has a well-defined best response to the collusive quantity. In Bertrand, undercutting by an arbitrarily small amount is enough to steal the whole market, so the temptation payoff has a supremum but no maximizer. The economics is the same—future punishment disciplines current behavior—but the mathematics of the deviation problem is slightly subtler.

If a strategy profile σ induces a random sequence of action profiles $(a^0, a^1, a^2, \dots)$, its normalized discounted payoff vector is

$$v(\sigma, \delta) = (1 - \delta) \mathbb{E}_\sigma \left[\sum_{t=0}^{\infty} \delta^t \pi(a^t) \right].$$

The normalization by $1 - \delta$ is useful because it makes equilibrium payoffs directly comparable to one-period payoff vectors of the stage game.

Theorem 0.0.19 (Nash-Threat Folk Theorem). *If v is feasible and Nash-individually rational, then there exists $\bar{\delta} < 1$ such that, for every $\delta \in (\bar{\delta}, 1)$, the vector v is the average utility profile of a subgame perfect equilibrium.*

Feasibility and Nash-individual rationality

Definition. A payoff vector $v \in \mathbb{R}^n$ is *feasible* if

$$v \in \text{co}(\{\pi(a) : a \in A\}),$$

that is, if v belongs to the convex hull of the stage-game payoff vectors.

Remark. The convex hull appears because, in a repeated game, players can *time-share* across different action profiles. If one action profile is played for a fraction λ of the time and another for a fraction $1 - \lambda$, then the long-run average payoff is approximately the convex combination of their stage payoffs. Thus feasibility means: *this payoff vector can be generated on average by appropriately mixing or cycling across stage outcomes.*

Definition. A payoff vector $v \in \mathbb{R}^n$ is *Nash-individually rational* if

$$v_i > \underline{v}_i^N \quad \text{for every } i,$$

where

$$\underline{v}_i^N := \min_{\alpha \text{ is a Nash equilibrium of the stage game}} \pi_i(\alpha).$$

Remark. This condition is the right one for a folk theorem based on *Nash threats*. If a deviation occurs, the continuation punishment must itself be sequentially rational, so the natural punishments are stage-game Nash equilibria repeated forever. Hence player i can never be threatened with less than the worst payoff that i receives across stage-game Nash equilibria. Nash-individual rationality therefore says: *the target payoff must give each player strictly more than what that player can be forced down to by a credible Nash punishment.*

Remark. This is slightly different from the more standard folk theorem based on minmax payoffs. There, the relevant lower bound is the minmax value. Here, because punishments are restricted to Nash equilibria, the relevant lower bound is the *worst Nash equilibrium payoff*.

Geometric intuition

To visualize the definitions, it is useful to look at a simple 2×2 example:

	C	D
C	(4,4)	(0,5)
D	(5,0)	(1,1)

This is the standard Prisoner's Dilemma. The unique Nash equilibrium is (D, D) , with payoff vector $(1, 1)$.

Step 1: Feasible payoffs. The feasible set is the convex hull of the four stage-game payoff vectors:

$$(4, 4), \quad (0, 5), \quad (5, 0), \quad (1, 1).$$

Geometrically, this is the shaded region in the left panel below.

Step 2: Nash-individual rationality. Since the unique Nash equilibrium is (D, D) , we have

$$\underline{v}_1^N = \underline{v}_2^N = 1.$$

Therefore Nash-individual rationality means

$$v_1 > 1 \quad \text{and} \quad v_2 > 1.$$

So the relevant set is the portion of the feasible region lying strictly above the horizontal line $v_2 = 1$ and strictly to the right of the vertical line $v_1 = 1$, as shown in the middle panel.

Step 3: If there are several Nash equilibria. When the stage game has multiple Nash equilibria, the relevant thresholds are

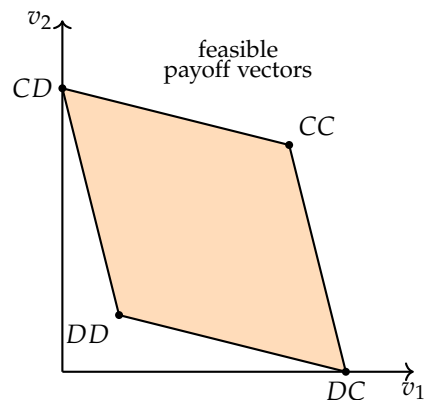
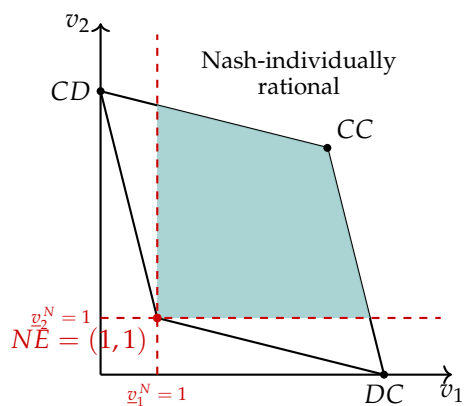
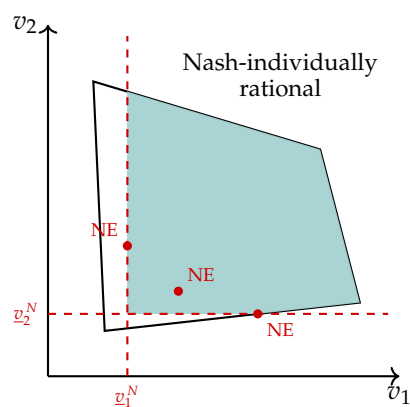
$$\underline{v}_1^N = \min_{\alpha \text{ Nash}} \pi_1(\alpha), \quad \underline{v}_2^N = \min_{\alpha \text{ Nash}} \pi_2(\alpha).$$

Importantly, these minima need not come from the *same* Nash equilibrium. So the Nash-individually rational region is defined coordinate by coordinate: it is the part of the feasible set above $v_2 = \underline{v}_2^N$ and to the right of $v_1 = \underline{v}_1^N$. The right panel below illustrates that idea.

Why these two conditions are the right ones

The theorem has a very intuitive logic.

First, a target payoff must be *feasible*. This is unavoidable: the repeated game cannot create average payoffs out of thin air. Any equilibrium payoff must be built from the stage-game payoffs by averaging over time.

*Feasible set**Unique Nash equilibrium**Several Nash equilibria*

Feasibility and Nash-individual rationality. From top to bottom: the feasible set as the convex hull of stage-game payoff vectors; the Nash-individually rational region when the stage game has a unique Nash equilibrium payoff $(1, 1)$; and the corresponding region when the stage game has several Nash equilibria, so the relevant thresholds are the coordinatewise worst Nash equilibrium payoffs.

Second, a target payoff must be *Nash-individually rational*. If a player deviates, the continuation punishment must still be credible. Under Nash threats, credibility means that after a deviation the continuation play must be some Nash equilibrium of the stage game. Therefore the worst continuation value that can be imposed on player i is $\underline{v}_i^N$. If the target payoff gave player i no more than that amount, then the threat of punishment would have no bite.

Third, when δ is close to 1, future losses matter a lot. A deviation may produce a one-period gain, but it triggers a persistent fall in continuation payoffs. If players are sufficiently patient, the present value of that future loss dominates the short-run temptation. That is the force behind the folk theorem.

Remark (Proof intuition). A standard proof has two ingredients.

Step 1: Construct the target path. Because v is feasible, one can find a finite cycle of stage-action profiles whose average payoff is v , or arbitrarily close to v . The equilibrium path repeatedly implements that cycle.

Step 2: Deter deviations with Nash punishments. If a player deviates, the continuation switches to a Nash equilibrium that is particularly bad for that player. Since $v_i > \underline{v}_i^N$, this punishment is genuinely worse than staying on the equilibrium path. When δ is high enough, the continuation loss outweighs any one-shot gain from deviating.

Bottom line. The Nash-Threat Folk Theorem says that, in an infinitely repeated game with sufficiently patient players, the set of SPE payoff vectors is essentially the set of payoff vectors that satisfy two simple requirements: **feasible** and **above the worst credible Nash punishment for every player**. That is the geometric core of the theorem.

Infinite Prisoner's Dilemma: supporting an alternating path

Consider the infinitely repeated Prisoner's Dilemma with stage game

	C	D
C	(3,3)	(0,4)
D	(4,0)	(1,1)

and discount factor $\delta < 1$.

We now ask whether the repeated game can support the nonstationary path

$$CC, DC, CC, DC, \dots$$

That is, periods alternate between (C, C) and (D, C) .

Proposed strategy profile Consider the following strategy profile:

- On the equilibrium path, players alternate forever between CC and DC , starting from CC .
- If any player deviates at any history, both players switch to DD forever.

Since DD is the stage-game Nash equilibrium, continuation play after any deviation is sequentially rational. Thus the only issue is whether either player wants to deviate along the proposed path.

Continuation values on the path Starting from a period in which the prescribed action profile is CC :

$$\text{Player 1: } 3 + \delta 4 + \delta^2 3 + \delta^3 4 + \dots = \frac{3 + 4\delta}{1 - \delta^2}.$$

$$\text{Player 2: } 3 + \delta 0 + \delta^2 3 + \delta^3 0 + \dots = \frac{3}{1 - \delta^2}.$$

Starting from a period in which the prescribed action profile is DC :

$$\text{Player 1: } 4 + \delta 3 + \delta^2 4 + \delta^3 3 + \dots = \frac{4 + 3\delta}{1 - \delta^2}.$$

$$\text{Player 2: } 0 + \delta 3 + \delta^2 0 + \delta^3 3 + \dots = \frac{3\delta}{1 - \delta^2}.$$

Deviation constraints We now check incentives at each on-path state.

1. *No deviation at CC : player 1.* If player 1 follows the prescribed action C , his continuation payoff is

$$\frac{3 + 4\delta}{1 - \delta^2}.$$

If instead he deviates to D while player 2 plays C , he gets 4 today and then DD forever:

$$4 + \delta \frac{1}{1 - \delta}.$$

Hence we need

$$\frac{3 + 4\delta}{1 - \delta^2} \geq 4 + \delta \frac{1}{1 - \delta}.$$

This is equivalent to

$$3\delta^2 + 3\delta - 1 \geq 0,$$

so

$$\delta \geq \frac{\sqrt{21} - 3}{6}.$$

2. *No deviation at CC: player 2.* If player 2 follows the prescribed action C , his continuation payoff is

$$\frac{3}{1 - \delta^2}.$$

If instead he deviates to D while player 1 plays C , he gets 4 today and then DD forever:

$$4 + \delta \frac{1}{1 - \delta}.$$

Hence we need

$$\frac{3}{1 - \delta^2} \geq 4 + \delta \frac{1}{1 - \delta}.$$

This is equivalent to

$$3\delta^2 - \delta - 1 \geq 0,$$

so

$$\delta \geq \frac{1 + \sqrt{13}}{6}.$$

3. *No deviation at DC: player 1.* If player 1 follows the prescribed action D , his continuation payoff is

$$\frac{4 + 3\delta}{1 - \delta^2}.$$

If instead he deviates to C while player 2 plays C , he gets 3 today and then DD forever:

$$3 + \delta \frac{1}{1 - \delta}.$$

Hence we need

$$\frac{4 + 3\delta}{1 - \delta^2} \geq 3 + \delta \frac{1}{1 - \delta}.$$

This inequality always holds. Indeed, it simplifies to

$$2\delta^2 + 2\delta + 1 \geq 0,$$

which is true for all $\delta \in (0, 1)$.

4. *No deviation at DC: player 2.* If player 2 follows the prescribed action C , his continuation payoff is

$$\frac{3\delta}{1 - \delta^2}.$$

If instead he deviates to D , the current outcome becomes DD , so he gets

$$1 + \delta \frac{1}{1 - \delta} = \frac{1}{1 - \delta}.$$

Hence we need

$$\frac{3\delta}{1 - \delta^2} \geq \frac{1}{1 - \delta}.$$

This is equivalent to

$$3\delta \geq 1 + \delta,$$

that is,

$$\delta \geq \frac{1}{2}.$$

Proposition 0.0.20. *The alternating path*

$$CC, DC, CC, DC, \dots$$

can be supported as a subgame perfect equilibrium by the strategy profile above if and only if

$$\delta \geq \frac{1 + \sqrt{13}}{6}.$$

Proof. The four on-path incentive constraints above are necessary and sufficient. The strongest one is player 2's no-deviation condition at CC, which requires

$$\delta \geq \frac{1 + \sqrt{13}}{6}.$$

All other constraints are weaker. Therefore the proposed strategy profile is an SPE exactly for

$$\delta \geq \frac{1 + \sqrt{13}}{6}. \quad \square$$

Remark. The binding constraint is player 2 at CC. This is economically intuitive. Along the alternating path, player 2 gets

$$3, 0, 3, 0, \dots$$

so at a CC period he may be tempted to deviate immediately, obtain 4 today, and accept the punishment afterwards. Sustaining the arrangement therefore requires players to be sufficiently patient.

Normalized payoff and geometric interpretation The normalized discounted payoff generated by the path

$$CC, DC, CC, DC, \dots$$

starting from CC is

$$v(\delta) = (1 - \delta) \left[(3, 3) + \delta(4, 0) + \delta^2(3, 3) + \delta^3(4, 0) + \dots \right].$$

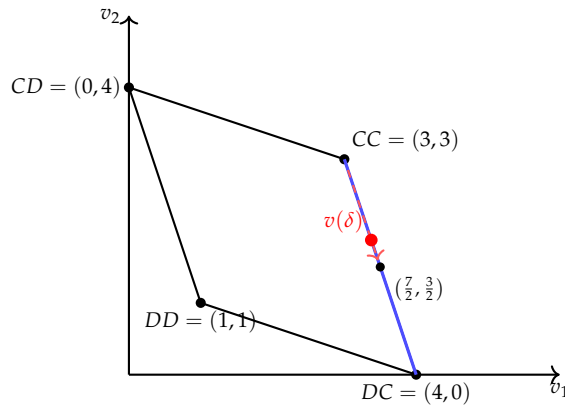
Summing the geometric series,

$$v(\delta) = \left(\frac{3 + 4\delta}{1 + \delta}, \frac{3}{1 + \delta} \right).$$

As $\delta \rightarrow 1$,

$$v(\delta) \rightarrow \left(\frac{7}{2}, \frac{3}{2} \right),$$

which is exactly the midpoint of the segment joining $CC = (3, 3)$ and $DC = (4, 0)$. Thus, for patient players, alternating between CC and DC approximately implements the average payoff of those two stage outcomes.



For $\delta \in [0, 1)$, the alternating path $CC, DC, CC, DC, \dots$, starting from CC , generates a normalized payoff on the segment from CC toward $(\frac{7}{2}, \frac{3}{2})$. As $\delta \rightarrow 1$, the payoff converges to this midpoint.

Remark. This example illustrates the logic of the folk theorem. Repetition allows the players to sustain payoff vectors that are not stage-game Nash equilibrium payoffs, and even to sustain nonstationary patterns of play. By alternating across stage outcomes and using credible punishments after deviations, the repeated game can implement points in the feasible set that lie strictly above the Nash threat point.

Appendix to Class 4: Repeated Stackelberg with Random Leadership

Consider the infinitely repeated duopoly in which, in each period, one firm is randomly selected to be the Stackelberg leader with probability $1/2$, while the other becomes the follower. Demand is

$$p(Q) = A - bQ, \quad Q = q_X + q_Y,$$

and both firms have zero marginal cost.

One-period benchmark. In the stage game, if a firm is the leader, it chooses

$$q^L = \frac{A}{2b},$$

while the follower chooses

$$q^F = \frac{A}{4b}.$$

Hence, in any realization of the stage game,

$$Q = \frac{3A}{4b}, \quad p = \frac{A}{4}.$$

So the leader's profit is

$$\pi^L = \frac{A^2}{8b},$$

and the follower's profit is

$$\pi^F = \frac{A^2}{16b}.$$

Since each firm is leader with probability $1/2$, the expected stage-game equilibrium payoff of each firm is

$$v^N = \frac{1}{2}\pi^L + \frac{1}{2}\pi^F = \frac{3A^2}{32b}.$$

Finite horizon. If the game is repeated for $T < \infty$ periods, backward induction implies that in the last period the unique stage-game equilibrium must be played after every history. Since this removes any possibility of future rewards or punishments, the same logic applies recursively to every earlier period. Therefore, for any finite T , the only subgame perfect equilibrium payoff is

$$\left(\frac{3A^2}{32b}, \frac{3A^2}{32b} \right).$$

Infinite horizon and grim-trigger strategies. Now suppose the game is repeated infinitely many times and firms discount the future with common factor $\delta \in (0, 1)$. We restrict attention to grim-trigger strategies in which, after any deviation, firms play the stage-game equilibrium forever.

The first step is to characterize the feasible set of one-period payoffs. Since total profit in any period is

$$\pi_X + \pi_Y = p(Q)Q = (A - bQ)Q,$$

the maximal joint profit is the monopoly profit

$$\Pi^M = \max_Q (A - bQ)Q = \frac{A^2}{4b}.$$

Therefore, the set of feasible average payoff vectors is

$$F = \left\{ (\pi_X, \pi_Y) \in \mathbb{R}_+^2 : \pi_X + \pi_Y \leq \frac{A^2}{4b} \right\}.$$

Take any target payoff vector $\pi = (\pi_X, \pi_Y) \in F$. Consider a grim-trigger strategy profile such that along the equilibrium path the firms play quantities that generate the stage payoff π in every period, and after any deviation they revert forever to the stage-game equilibrium.

If firm i does not deviate, its discounted value is

$$\frac{\pi_i}{1 - \delta}.$$

If it deviates once, its current gain can never exceed the monopoly profit Π^M , and from the next period onward it receives the punishment payoff v^N . Hence its deviation payoff is bounded above by

$$\Pi^M + \frac{\delta v^N}{1 - \delta}.$$

A sufficient incentive constraint is therefore

$$\frac{\pi_i}{1 - \delta} \geq \Pi^M + \frac{\delta v^N}{1 - \delta},$$

or equivalently,

$$\pi_i \geq (1 - \delta)\Pi^M + \delta v^N.$$

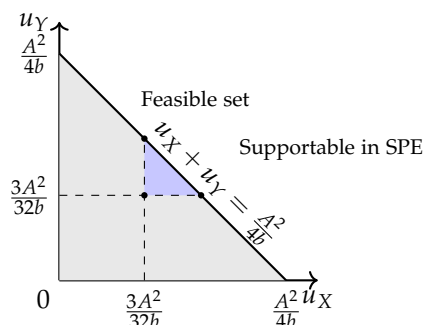
As $\delta \rightarrow 1$, the right-hand side converges to v^N . Thus, in the limit, any feasible payoff vector that gives each firm at least the stage-game equilibrium payoff can be sustained.

It follows that the set of average utilities supportable in subgame perfect equilibrium using these grim-trigger strategies is

$$V^{\text{SPE}} = \left\{ (u_X, u_Y) \in \mathbb{R}_+^2 : u_X + u_Y \leq \frac{A^2}{4b}, u_X \geq \frac{3A^2}{32b}, u_Y \geq \frac{3A^2}{32b} \right\}.$$

Equivalently, this is the truncated feasible triangle whose vertices are

$$\left(\frac{3A^2}{32b}, \frac{3A^2}{32b} \right), \quad \left(\frac{5A^2}{32b}, \frac{3A^2}{32b} \right), \quad \left(\frac{3A^2}{32b}, \frac{5A^2}{32b} \right).$$



Feasible and supportable average payoffs under grim-trigger strategies.

Geometric intuition. The geometry is especially transparent. The large triangle is the feasible set: no payoff pair above the line

$$u_X + u_Y = \frac{A^2}{4b}$$

can be achieved because monopoly profit is the maximal total surplus available in a period. The grim-trigger punishment imposes a lower bound on each firm's continuation value: each firm must receive at least the punishment payoff

$$v^N = \frac{3A^2}{32b}.$$

This removes the lower-left region of the feasible triangle and leaves only the smaller truncated triangle.

Economic intuition. The key point is that the punishment here is not a minmax punishment, but rather permanent reversion to the stage-game equilibrium. Therefore, the individual rationality threshold is not the minmax payoff, but the stage-game equilibrium payoff itself. When δ is close to one, future losses dominate any one-shot gain from deviation, so every feasible payoff above that threshold becomes sustainable. In this sense, part (c) is a restricted folk-theorem result: the sustainable set is the feasible set intersected with the half-spaces

$$u_X \geq v^N, \quad u_Y \geq v^N.$$

Takeaway. For finite T , repetition does not enlarge the equilibrium payoff set because backward induction eliminates the disciplinary role of the future. For infinite horizon, by contrast, the future becomes valuable enough to support a nontrivial set of cooperative outcomes. Under grim-trigger strategies with reversion to the stage-game equilibrium, the supportable set is exactly the feasible set truncated by the requirement that each firm receive at least its static equilibrium payoff.

Class 5: Nash Threat and MinMax Automaton

Nash threats and the folk theorem

We ended the previous class with the folk theorem. A convenient way to understand the proof is through a *Nash-threat automaton*: on the equilibrium path the players repeatedly play a target action profile α^* , and after a deviation by player i , play forever a stage-game Nash equilibrium $\alpha(i)$ that is particularly bad for player i .

Theorem 0.0.21 (Folk theorem with Nash threats). *Consider an infinitely repeated game. Suppose there is a public correlation device and mixed stage-game actions are publicly observable. If v is feasible and strictly Nash-individually rational, then there exists $\underline{\delta} < 1$ such that, for every $\delta \in (\underline{\delta}, 1)$, there is a subgame-perfect equilibrium whose average utility profile is v .*

Why the theorem is stated in average-utility form. Feasibility is naturally defined in stage-payoff units. If a stage-game mixed action profile α is played forever, then player i 's discounted utility is

$$U_i(\alpha) = \frac{\pi_i(\alpha)}{1 - \delta},$$

while the associated average utility is

$$(1 - \delta)U_i(\alpha) = \pi_i(\alpha).$$

Thus the theorem is stated in terms of average utility because it is directly comparable to the feasible payoff set. In the proof, however, it is often more convenient to work with discounted continuation values.

Construction of the automaton. Fix a feasible payoff vector v . By feasibility, there exists a mixed action profile α^* such that

$$\pi(\alpha^*) = v.$$

For each player i , choose a stage-game Nash equilibrium $\alpha(i)$ such that

$$\pi_i(\alpha(i)) < v_i = \pi_i(\alpha^*).$$

Now define the repeated-game strategy profile as follows:

1. At the initial node, play α^* .
2. If no player deviates, continue to play α^* forever.
3. If player i deviates, switch permanently to $\alpha(i)$.

Because each $\alpha(i)$ is a stage-game Nash equilibrium, once the automaton enters a punishment state, repeating $\alpha(i)$ forever is sequentially rational in that continuation subgame. Hence the only remaining issue is to verify that no player wants to deviate on the equilibrium path.

On-path incentive constraint. Fix player i . If player i obeys at α^* , then the discounted utility is

$$U_i^{\text{eq}} = \frac{\pi_i(\alpha^*)}{1 - \delta} = \frac{v_i}{1 - \delta}.$$

If instead player i deviates once, let

$$B_i := \max_{a_i \in A_i} u_i(a_i, \alpha_{-i}^*)$$

denote the largest one-period payoff player i can obtain from a one-shot deviation while the other players follow α^* . After that deviation, the automaton moves forever to the punishment state $\alpha(i)$. Therefore the deviating payoff is bounded above by

$$U_i^{\text{dev}} \leq B_i + \frac{\delta \pi_i(\alpha(i))}{1 - \delta}.$$

Hence a deviation is unprofitable whenever

$$\frac{v_i}{1 - \delta} \geq B_i + \frac{\delta \pi_i(\alpha(i))}{1 - \delta}.$$

Multiplying both sides by $1 - \delta$, this is equivalent to

$$v_i \geq (1 - \delta)B_i + \delta \pi_i(\alpha(i)).$$

Rearranging,

$$\frac{\delta}{1 - \delta} (v_i - \pi_i(\alpha(i))) \geq B_i - v_i.$$

Since $v_i - \pi_i(\alpha(i)) > 0$, the left-hand side can be made arbitrarily large by taking δ sufficiently close to one. Therefore there exists $\underline{\delta}_i < 1$ such that, for all $\delta > \underline{\delta}_i$, player i does not want to deviate.

Doing this for every player and setting

$$\underline{\delta} := \max_i \underline{\delta}_i,$$

we conclude that for every $\delta \in (\underline{\delta}, 1)$, the proposed strategy profile is a subgame-perfect equilibrium.

Interpretation. The idea is simple. The gain from deviating is a one-period gain, while the loss from deviating is a permanent drop in continuation payoff. When players are sufficiently patient, the continuation loss dominates the short-run temptation.

A useful remark on units. The theorem is stated in average-utility form, but the proof is often written in discounted-utility form. These are equivalent because multiplying by $1 - \delta > 0$ does not change the incentive inequalities. In particular,

$$\frac{v_i}{1 - \delta} \geq B_i + \frac{\delta \pi_i(\alpha(i))}{1 - \delta} \iff v_i \geq (1 - \delta)B_i + \delta \pi_i(\alpha(i)).$$

So the apparent asymmetry in the proof is only a change of normalization.

Remark. The proof above uses a strict gap

$$v_i > \pi_i(\alpha(i)).$$

Thus the argument is most transparent under *strict* Nash individual rationality. If one writes the theorem with weak inequalities, then an additional approximation argument is needed.

Below-Nash punishments and why every deviation must be checked

We now study a punishment scheme that is sharper than the standard Nash-threat construction. In the usual approach, a deviation is punished by sending the deviator directly to a continuation payoff at his stage-game Nash equilibrium level. Here we consider a stricter alternative: after a deviation, the game first passes through a short transitional punishment phase and only then moves to a permanent punishment state. In this way, the deviator can be driven *below* his Nash equilibrium payoff.

The cooperative path we want to sustain is

$$TL, TL, TL, \dots$$

and the punishments are asymmetric:

- if player 1 deviates, the continuation is

$$MR, TC, TC, \dots;$$

- if player 2 deviates, the continuation is

$$BC, ML, ML, \dots$$

The basic idea is simple. A deviation from cooperation is met first with a one-period harsh response (*MR* or *BC*), and only afterwards with a permanent punishment state (*TC* or *ML*). **Because these continuation paths are not themselves stage-game Nash equilibria, their credibility is not automatic.** This is precisely what makes below-Nash punishments more delicate than standard Nash threats.

Remark. To prove that a repeated-game strategy profile is a subgame-perfect equilibrium, it is not enough to check incentives only on the intended path. One must verify sequential rationality after *every history*. In the present construction, this means checking (1) the cooperative state; (2) the transitional punishment states and; (3) the absorbing punishment states.²⁰

In discounted repeated games with perfect monitoring, the one-shot deviation principle simplifies the task. It is enough to check whether, after any history, any player can profit from deviating in a single period while the prescribed continuation is kept fixed thereafter. Thus the verification can be carried out state by state.

If the punishment state were already a stage-game Nash equilibrium, this credibility check would be immediate. Here, however, the punishment path passes through non-Nash action profiles, so each punishment state must be analyzed explicitly.

²⁰ *Intuition for future me:* A punishment can deter deviations on the equilibrium path only if the punishment itself is credible. In automaton language, the strategy must *close*: after any deviation, the prescribed continuation must itself remain optimal to follow.

An example of below-Nash punishments

Consider the stage game

	<i>L</i>	<i>C</i>	<i>R</i>
<i>T</i>	(10, 10)	(3, 15)	(0, 7)
<i>M</i>	(15, 3)	(7, 7)	(−4, 5)
<i>B</i>	(7, 0)	(5, −4)	(−15, −15)

We want to support the cooperative path

$$TL, TL, TL, \dots$$

using the asymmetric punishments

if player 1 deviates: *MR, TC, TC, ...*, if player 2 deviates: *BC, ML, ML, ...*

To obtain a complete repeated-game strategy profile, we must specify play after *every* deviation, not only after deviations on the equilibrium path. A convenient recursive description is the following:

1. Initially, play *TL*.
2. If no player deviates, continue playing *TL*.

3. After any one-shot deviation by player 1 from any state, play MR in the next period and then TC forever, unless another deviation occurs.
4. After any one-shot deviation by player 2 from any state, play BC in the next period and then ML forever, unless another deviation occurs.

Proposition 0.0.22. *The strategy profile above is a subgame-perfect equilibrium whenever*

$$\delta \geq \frac{4}{7}.$$

Hence the path $TL, TL, \dots$ can be supported by below-Nash punishments.

Proof. By the one-shot deviation principle, it is enough to check one-period deviations at every state of the automaton.

1. *The cooperative state TL .* If player 1 obeys forever, his continuation payoff is

$$V_1(TL) = \frac{10}{1-\delta}.$$

His best current deviation is to M , which yields 15 today and then triggers

$$MR, TC, TC, \dots,$$

so the deviation payoff is

$$15 + \delta(-4) + \delta^2 \frac{3}{1-\delta}.$$

Thus player 1 does not deviate if

$$\frac{10}{1-\delta} \geq 15 + \delta(-4) + \delta^2 \frac{3}{1-\delta},$$

which is equivalent to

$$7\delta^2 - 19\delta + 5 \leq 0.$$

In particular, this condition is satisfied for all

$$\delta \geq \frac{4}{7}.$$

By symmetry, the same is true for player 2.

2. *The transitional punishment state MR .* Suppose first that player 1 is the player currently being punished. If he obeys at MR , his continuation payoff is

$$V_1(MR) = -4 + \delta \frac{3}{1-\delta}.$$

If instead he deviates while player 2 plays R , his best deviation is to T , which yields

$$0 + \delta(-4) + \delta^2 \frac{3}{1-\delta},$$

because his deviation restarts the punishment path directed at player 1. Hence he does not deviate if

$$-4 + \delta \frac{3}{1-\delta} \geq \delta(-4) + \delta^2 \frac{3}{1-\delta},$$

or equivalently

$$\delta \geq \frac{4}{7}.$$

Now consider player 2 at the same state. If she obeys, she obtains

$$V_2(MR) = 5 + \delta \frac{15}{1-\delta}.$$

Her best deviation is to C , which yields 7 today and then triggers

$$BC, ML, ML, \dots,$$

so her deviation payoff is

$$7 + \delta(-4) + \delta^2 \frac{3}{1-\delta}.$$

Therefore player 2 does not deviate if

$$5 + \delta \frac{15}{1-\delta} \geq 7 + \delta(-4) + \delta^2 \frac{3}{1-\delta},$$

which is equivalent to

$$7\delta^2 - 21\delta + 2 \leq 0.$$

This inequality also holds for all

$$\delta \geq \frac{4}{7}.$$

3. *The absorbing punishment state TC.* If player 1 obeys forever at TC , his payoff is

$$V_1(TC) = \frac{3}{1-\delta}.$$

His best deviation is to M , which gives

$$7 + \delta(-4) + \delta^2 \frac{3}{1-\delta}.$$

Thus player 1 does not deviate if

$$\frac{3}{1-\delta} \geq 7 + \delta(-4) + \delta^2 \frac{3}{1-\delta},$$

that is,

$$7\delta^2 - 11\delta + 4 \leq 0.$$

Since

$$7\delta^2 - 11\delta + 4 = (7\delta - 4)(\delta - 1),$$

this reduces to

$$\delta \geq \frac{4}{7}.$$

Player 2 has no profitable deviation at *TC*, because she already receives the maximal stage payoff 15.

4. *The states BC and ML.* The same reasoning applies symmetrically.

At *BC*, the relevant constraint for player 2 is again

$$\delta \geq \frac{4}{7},$$

and player 1 faces the symmetric analogue of player 2's constraint at *MR*.

At *ML*, the relevant constraint for player 2 is again

$$\delta \geq \frac{4}{7},$$

while player 1 has no profitable deviation because he already receives the maximal stage payoff 15.

Conclusion. Every one-shot deviation at every state of the automaton is unprofitable whenever

$$\delta \geq \frac{4}{7}.$$

Hence, by the one-shot deviation principle, the strategy profile is a subgame-perfect equilibrium. Along the equilibrium path, play is

$$TL, TL, TL, \dots$$

as claimed. □

Remark. This example makes clear why every deviation must be checked. It is not enough that the punishment hurt the deviator on the equilibrium path; the punishment path must itself be sequentially rational after every possible history.

Remark. The punishment is *below-Nash*: the stage-game Nash equilibrium payoff is 7, whereas the punished player is driven first to -4 and then to a continuation payoff of only 3 per period.

The minimax automaton

We now strengthen the folk theorem by replacing Nash punishments with *minmax punishments*. The gain is that the deviator can be pushed all the way down to his minimax payoff. The cost is that minimax profiles need not be Nash equilibria of the stage game, so they are not automatically credible. The role of the automaton is precisely to make them credible.

Throughout, we work with normalized discounted payoffs:

$$U_i = (1 - \delta) \sum_{t=0}^{\infty} \delta^t u_i^t.$$

Hence, if a stage-game action profile α is repeated forever, the induced average utility profile is simply $\pi(\alpha)$.

Step 1: the geometric construction. Let $V \subset \mathbb{R}^2$ denote the feasible payoff set, and let

$$m_i := \min_{\alpha_{-i} \in \Delta A_{-i}} \max_{a_i \in A_i} \pi_i(a_i, \alpha_{-i})$$

be player i 's minimax payoff.²¹

Suppose

$$v = (v_1, v_2) \in \text{int}\left(V \cap \{x \in \mathbb{R}^2 : x_i > m_i, i = 1, 2\}\right).$$

Because v lies in the interior of the feasible and individually rational set, we can choose two nearby payoff vectors w^1, w^2 in the same set such that

$$w_1^1 < v_1, \quad w_2^1 > v_2,$$

and

$$w_1^2 > v_1, \quad w_2^2 < v_2.$$

Thus w^1 rewards player 2 and hurts player 1, while w^2 rewards player 1 and hurts player 2.

Since these vectors are feasible, there exist mixed action profiles

$$\alpha^*, \quad \alpha^*(1), \quad \alpha^*(2)$$

such that

$$\pi(\alpha^*) = v, \quad \pi(\alpha^*(1)) = w^1, \quad \pi(\alpha^*(2)) = w^2.$$

Step 2: the minimax profiles. For each player i , choose a mixed action profile $\alpha(i)$ such that player i 's opponents minimax him and player i is assigned a best reply to that minimaxing action. Hence

$$\pi_i(\alpha(i)) = m_i.$$

²¹ Fix a mixed action profile α_{-i} of the other players. Given that profile, player i chooses a best response and obtains $\max_{a_i \in A_i} \pi_i(a_i, \alpha_{-i})$. The minimax payoff m_i is the smallest value this best-response payoff can take as the opponents vary α_{-i} . Thus it is the lowest payoff that the others can force on player i , even when player i responds optimally.

Write

$$r_j^i := \pi_j(\alpha(i))$$

for the stage payoff obtained by player $j \neq i$ during the minimax phase against player i .

Step 3: the automaton. Fix an integer $L \geq 1$. The strategy profile is described by the following states.

- E : play α^* .
- $P_i(s)$, for $i \in \{1, 2\}$ and $s = 1, \dots, L$: play $\alpha(i)$.
- R_i , for $i \in \{1, 2\}$: play $\alpha^*(i)$.

The transition rule is:

1. Start in state E .
2. If nobody deviates in E , remain in E .
3. If player i deviates in any state, move next period to $P_i(L)$.
4. If nobody deviates in $P_i(s)$ and $s > 1$, move next period to $P_i(s - 1)$.
5. If nobody deviates in $P_i(1)$, move next period to R_i .
6. If nobody deviates in R_i , remain forever in R_i .

So, after a deviation by player i , the game spends L periods minimaxing player i , and then moves forever to a continuation w^i that rewards the other player.

Theorem 0.0.23 (Minimax folk theorem, two-player version). *Let v lie in the interior of the feasible and individually rational set. Then there exist $L \in \mathbb{N}$ and $\underline{\delta} < 1$ such that, for every $\delta \in (\underline{\delta}, 1)$, the automaton above defines a subgame-perfect equilibrium with average utility profile v .*

Proof. By the one-shot deviation principle, it is enough to check that no player has a profitable one-period deviation at any state of the automaton.

Let

$$B_i := \max_{a \in A} \pi_i(a)$$

be a uniform upper bound on player i 's one-period payoff.

1. *Incentives on the equilibrium path.* If player i obeys in state E , his continuation value is simply

$$V_i(E) = v_i.$$

If he deviates, he can get at most B_i today, and from next period on he is sent to $P_i(L)$. Therefore his deviation payoff is bounded above by

$$(1 - \delta)B_i + \delta V_i^i(L),$$

where

$$V_i^i(s) := (1 - \delta^s)m_i + \delta^s w_i^i$$

is player i 's continuation value when there are s minimax periods left before entering R_i .

Hence, at state E , player i does not deviate if

$$v_i \geq (1 - \delta)B_i + \delta[(1 - \delta^L)m_i + \delta^L w_i^i].$$

As $\delta \rightarrow 1$, the right-hand side converges to w_i^i , and by construction

$$w_i^i < v_i.$$

Therefore this inequality holds for all δ sufficiently close to one.

2. *Incentives of the punished player during the minimax phase.* Consider player i in state $P_i(s)$. If he obeys, his continuation value is

$$V_i^i(s) = (1 - \delta^s)m_i + \delta^s w_i^i.$$

If he deviates today, he cannot raise his current payoff above m_i , because $\alpha(i)$ minmaxes him. Moreover, the deviation restarts the punishment, so his continuation value is at most

$$(1 - \delta)m_i + \delta V_i^i(L) = V_i^i(L + 1).$$

Now $w_i^i > m_i$ because w^i is individually rational, so

$$V_i^i(s) = m_i + \delta^s(w_i^i - m_i)$$

is decreasing in s . Since $s \leq L$, we have

$$V_i^i(s) \geq V_i^i(L + 1).$$

Thus the punished player never wishes to deviate during the minimax phase.

3. *Incentives of the punisher during the minimax phase.* Now consider player $j \neq i$ in state $P_i(s)$. If he obeys, his continuation value is

$$V_j^i(s) := (1 - \delta^s)r_j^i + \delta^s w_j^i.$$

If instead he deviates today, he can get at most B_j this period, but from next period on he becomes the deviator and is sent to $P_j(L)$. Hence his deviation payoff is bounded above by

$$(1 - \delta)B_j + \delta V_j^j(L),$$

where

$$V_j^j(L) = (1 - \delta^L)r_j^j + \delta^L w_j^j.$$

As $\delta \rightarrow 1$,

$$V_j^i(s) \rightarrow w_j^i, \quad (1 - \delta)B_j + \delta V_j^j(L) \rightarrow w_j^j.$$

But by construction,

$$w_j^i > w_j^j.$$

Indeed, the continuation following a deviation by player i rewards player j , whereas the continuation following a deviation by player j hurts him. Therefore, for all δ sufficiently close to one, player j does not deviate in any punishment state $P_i(s)$.

4. *Incentives in the reward states R_i .* Consider first player $j \neq i$. If he obeys in R_i , he receives

$$w_j^i.$$

If he deviates, his payoff is again bounded above by

$$(1 - \delta)B_j + \delta V_j^j(L),$$

which converges to w_j^j as $\delta \rightarrow 1$. Since $w_j^i > w_j^j$, player j does not deviate for δ close enough to one.

Now consider player i in R_i . If he obeys, he receives

$$w_i^i.$$

If he deviates, he gets at most

$$(1 - \delta)B_i + \delta V_i^i(L).$$

Using the formula for $V_i^i(L)$, we obtain

$$w_i^i - \left[(1 - \delta)B_i + \delta V_i^i(L) \right] = (1 - \delta) \left[w_i^i - B_i + (\delta + \delta^2 + \dots + \delta^L)(w_i^i - m_i) \right].$$

Because $w_i^i > m_i$, we can first choose $\bar{\delta} < 1$ and then choose L large enough so that

$$w_i^i - B_i + (\bar{\delta} + \bar{\delta}^2 + \dots + \bar{\delta}^L)(w_i^i - m_i) > 0.$$

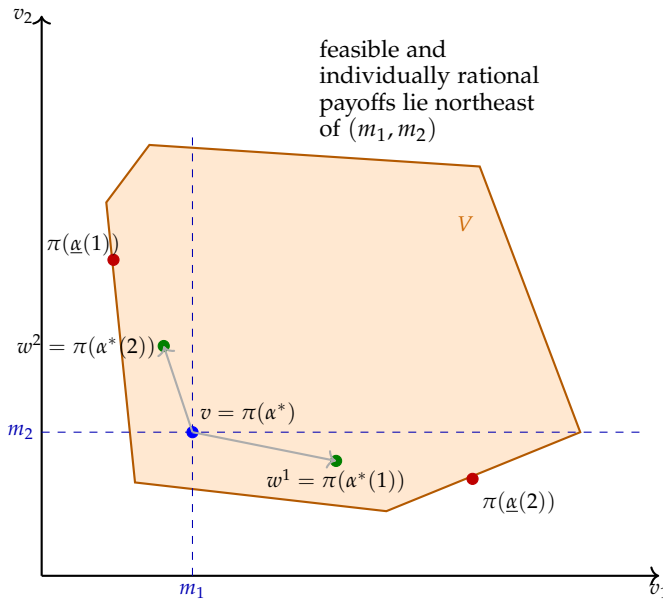
Since the sum $\delta + \dots + \delta^L$ is increasing in δ , the same L works for every $\delta \geq \bar{\delta}$. Hence player i does not deviate in R_i .

Conclusion. We have shown that, for a suitable integer L and for all δ sufficiently close to one, no player has a profitable one-shot deviation at any state of the automaton. By the one-shot deviation principle, the automaton defines a subgame-perfect equilibrium. Since the initial state is E , the equilibrium path plays α^* forever, and its average utility profile is

$$\pi(\alpha^*) = v.$$

□

Remark. The logic of the construction is now transparent. The minimax phase hurts the deviator, but may also be unpleasant for the punisher. This is why minimax is used only for a finite number of periods. The continuation w^i then compensates the punisher by giving him a payoff strictly above v_j , while the deviator receives a payoff strictly below v_i .



Geometry of the minimax construction. The target payoff v is an interior point of the feasible and individually rational set. The continuation payoffs w^1 and w^2 remain feasible and individually rational, but tilt rewards in favor of the punisher.

Appendix to Class 5: An Important Distinction Arising from Tit-for-Tat

Up to this point, we have not really thought of *strategy profiles themselves* as Nash equilibria, except through the simpler idea that equilibrium play in a repeated game might just consist of repeating a stage-game Nash equilibrium. The Tit-for-Tat exercise is useful because it forces us to make a more precise distinction: in a repeated game, equilibrium is a property of *entire contingent plans*, not only of the action profile played on the equilibrium path.

Strategies in a repeated game

Let the stage game be

$$G = (\{1, 2\}, \{C, D\}^2, u),$$

and let the game be repeated infinitely with common discount factor $\delta \in (0, 1)$.

A history at the beginning of period t is

$$h^t = (a^1, \dots, a^{t-1}) \in H^t,$$

where $a^\tau = (a_1^\tau, a_2^\tau) \in \{C, D\}^2$ is the action profile played in period τ , and $H^1 = \{\emptyset\}$. Let

$$H := \bigcup_{t \geq 1} H^t$$

denote the set of all finite histories.

A strategy for player i in the repeated game is a function

$$s_i : H \rightarrow \{C, D\}.$$

Thus, a strategy tells player i what to do after *every possible history*. A strategy profile

$$s = (s_1, s_2)$$

is therefore a *complete contingent plan*: it specifies what each player does after every history, both on and off the equilibrium path.

This is the key conceptual point. In a repeated game, an equilibrium is not simply a path of play such as “CC forever.” Rather, it is a profile of contingent plans capable of generating a path of play and also specifying what happens after any deviation.

The Tit-for-Tat exercise

Consider the infinitely repeated symmetric Prisoners’ Dilemma. Write the stage-game payoffs as

$$u_i(C, C) = R, \quad u_i(D, C) = T, \quad u_i(C, D) = S, \quad u_i(D, D) = P,$$

with

$$P = 0$$

and

$$2R > T + S.$$

As usual in a Prisoners' Dilemma, $T > R > P > S$.

Now consider the following strategy, called *Tit-for-Tat*:

Play C in the first period; from period 2 onward, play the action the opponent chose in the previous period.

Formally, for each player i ,

$$s_i^{TFT}(\emptyset) = C,$$

and for every history $h^t = (a^1, \dots, a^{t-1})$ with $t \geq 2$,

$$s_i^{TFT}(h^t) = a_{-i}^{t-1}.$$

The exercise asks:

- (a) Is the Tit-for-Tat strategy profile a Nash equilibrium of the repeated game for discount factors sufficiently close to 1?
- (b) Is it a subgame perfect equilibrium for discount factors sufficiently close to 1?

This exercise is important because it isolates a distinction that is easy to miss at first: a strategy profile may be a Nash equilibrium of the repeated game and yet fail to be subgame perfect.

Nash equilibrium versus subgame perfection

Recall the logical relation:

$$\text{SPE} \implies \text{Nash equilibrium},$$

but the converse is false.

The reason is that subgame perfection imposes a stronger requirement. A Nash equilibrium only requires that, starting from the initial node of the game, no player can profitably switch to another *entire strategy*. A subgame perfect equilibrium, by contrast, requires that the prescribed continuation be optimal *after every history*.

Equivalently:

- To show that a strategy profile is a Nash equilibrium, we fix the opponent's complete plan and ask whether there exists any alternative complete plan that yields a higher payoff from the beginning of the game.

- To show that a strategy profile is subgame perfect, we must check that after every history, the continuation strategy is still optimal. In other words, the profile must be sequentially rational in every subgame.

This is exactly where *non-credible threats* enter.

A Nash equilibrium may rely on punishments that are never actually optimal to carry out if the relevant off-path history is reached. Such punishments can still deter deviations *ex ante*, because when a player evaluates whether to deviate from the initial node, she takes the opponent's whole strategy as fixed. But a subgame perfect equilibrium cannot rely on such punishments, because once the off-path history is reached, the continuation must still be optimal for the player who is supposed to carry out the punishment.

So the slogan is:

A Nash equilibrium needs only to be stable from the initial node. A subgame perfect equilibrium must be stable after every history.

Part (a): Tit-for-Tat is a Nash equilibrium for δ sufficiently high

We now solve the first part carefully.

Step 1: What must be shown? As just discussed, to show that (s_1^{TFT}, s_2^{TFT}) is a Nash equilibrium, we must fix one player's strategy to be Tit-for-Tat and ask whether the other player can gain by switching to any other complete contingent plan.

So let player 2 play Tit-for-Tat, and consider an arbitrary unilateral deviation by player 1. We must compare:

- the payoff from also playing Tit-for-Tat, which yields CC forever and hence payoff R every period; and
- the payoff from any alternative complete strategy for player 1.

Since CC forever yields the constant discounted stream $R, R, R, \dots$, player 1's normalized discounted payoff on the equilibrium path is simply R .

Step 2: What can a unilateral deviation against Tit-for-Tat look like? This is the crucial observation. Because player 2 copies player 1's previous action, player 1's current action determines player 2's next-period action.

Suppose player 2 is about to play C. Then player 1 has two possibilities:

- If player 1 also plays C, the current outcome is CC, and player 2 will again play C next period.

- If player 1 plays D , the current outcome is DC , and player 2 will play D next period.

Suppose instead player 2 is about to play D . Again player 1 has two possibilities:

- If player 1 plays D , the current outcome is DD , and player 2 will again play D next period.
- If player 1 plays C , the current outcome is CD , and player 2 will play C next period.

Therefore, once player 1 deviates from perpetual cooperation, any induced deviation episode must have one of the following forms:

$$DC, \underbrace{DD, \dots, DD}_{k \text{ periods}}, CD \quad \text{for some } k \in \{0, 1, 2, \dots\},$$

or else

$$DC, DD, DD, \dots$$

forever.

Intuitively, a deviation episode begins with the one-shot temptation payoff T , may then generate some number of punishment periods DD , and if player 1 eventually tries to restore cooperation, it ends with the “sucker” payoff S . After that, either the play returns to CC , or a new such episode begins.

Step 3: Compare each deviation episode with cooperation If player 1 had simply continued cooperating, she would have received R in each of those periods. So the question is whether any of the deviation episodes above can beat a constant stream of R 's.

First consider the infinite episode

$$DC, DD, DD, \dots$$

Its normalized discounted payoff is

$$(1 - \delta)(T + \delta \cdot 0 + \delta^2 \cdot 0 + \dots) = (1 - \delta)T.$$

As $\delta \rightarrow 1$, this converges to 0, which is strictly less than R . Hence, for δ sufficiently close to 1, this infinite punishment path is strictly worse than perpetual cooperation.

Now consider a finite episode

$$DC, \underbrace{DD, \dots, DD}_{k \text{ periods}}, CD.$$

Its normalized discounted payoff over that block is

$$\bar{V}_k(\delta) = \frac{1-\delta}{1-\delta^{k+2}} (T + \delta^{k+1}S).$$

Taking the limit as $\delta \rightarrow 1$,

$$\bar{V}_k(\delta) \rightarrow \frac{T+S}{k+2}.$$

Since $k+2 \geq 2$, we have

$$\frac{T+S}{k+2} \leq \frac{T+S}{2}.$$

By assumption,

$$2R > T+S,$$

so

$$R > \frac{T+S}{2} \geq \frac{T+S}{k+2}.$$

Therefore, for every fixed k , and for δ sufficiently close to 1,

$$\bar{V}_k(\delta) < R.$$

So every possible deviation episode has discounted average payoff strictly below R when δ is high enough.

Step 4: Why does this imply Nash equilibrium? If player 1 also plays Tit-for-Tat, she gets R in every period. Any alternative complete strategy can only replace some stretches of CC's by deviation episodes of the forms studied above. But we have just shown that each such episode yields less than R in discounted average terms.

Hence no unilateral complete strategy can improve upon Tit-for-Tat when the opponent plays Tit-for-Tat. By symmetry, the same is true for player 2. Therefore, for discount factors sufficiently close to 1, the Tit-for-Tat strategy profile is a Nash equilibrium of the repeated game.

Part (b): Tit-for-Tat is not subgame perfect

We now turn to the second part.

Step 1: Why can we use a one-shot deviation argument here? Unlike part (a), here we are testing subgame perfection. So now we must check whether the continuation strategy is optimal after every history. In discounted repeated games with bounded payoffs, the one-shot deviation principle applies. Thus it is enough to find *one history* at which *one one-period deviation* is profitable.

Block normalization. Here $\bar{V}_k(\delta)$ is not the usual normalized infinite-horizon payoff. It is the discounted average payoff over the $k+2$ -period deviation block:

$$\bar{V}_k(\delta) = \frac{T + \delta^{k+1}S}{1 + \delta + \dots + \delta^{k+1}}.$$

Hence it can be compared directly with R , the discounted average from cooperating throughout the same block.

Step 2: Choose a history at which Tit-for-Tat prescribes alternation Consider a history after which Tit-for-Tat prescribes that player 1 play D and player 2 play C . Then, if both players continue following Tit-for-Tat, the induced continuation path is

$$DC, CD, DC, CD, \dots$$

forever.

From player 1's point of view, following Tit-for-Tat at such a history yields the discounted continuation value

$$V^{follow} = T + \delta S + \delta^2 T + \delta^3 S + \dots = \frac{T + \delta S}{1 - \delta^2}.$$

Step 3: A profitable one-shot deviation Now suppose that at this history player 1 deviates for one period only, playing C instead of the prescribed D . Then the current outcome is CC . Since the most recent observed action profile is now CC , Tit-for-Tat prescribes CC forever from the next period onward as well.

Thus the deviation yields continuation value

$$V^{dev} = R + \delta R + \delta^2 R + \dots = \frac{R}{1 - \delta}.$$

So the one-shot deviation is profitable whenever

$$\frac{R}{1 - \delta} > \frac{T + \delta S}{1 - \delta^2}.$$

Multiplying by $1 - \delta^2$, this is equivalent to

$$(1 + \delta)R > T + \delta S.$$

As $\delta \rightarrow 1$, the left-hand side converges to $2R$ and the right-hand side converges to $T + S$. Since the exercise assumes

$$2R > T + S,$$

the inequality holds for δ sufficiently close to 1.

Therefore, at that history, player 1 has a profitable one-shot deviation.

Step 4: Why does subgame perfection fail? Because there exists a history at which a player prefers to deviate from the continuation prescribed by Tit-for-Tat, the Tit-for-Tat strategy profile is not sequentially rational in every subgame. Hence it is *not* subgame perfect.

The intuition is transparent. Tit-for-Tat tries to “punish” a deviation by inducing alternation. But once such an off-path history is reached, the player who is supposed to continue the alternation would rather restore cooperation immediately. So the punishment is not credible. It may help deter deviations when evaluated from the initial node, which is enough for Nash equilibrium, but it cannot survive the stronger requirement of subgame perfection.

Takeaway

The Tit-for-Tat exercise is valuable because it forces us to distinguish between two different equilibrium requirements in repeated games.

- A Nash equilibrium asks whether any player can profit by replacing her entire contingent plan from the *beginning* of the game.
- A subgame perfect equilibrium asks whether the prescribed continuation is optimal *after every history*.

Thus:

Tit-for-Tat can be a Nash equilibrium because no global deviation is profitable from the initial node when δ is high enough. But it is not subgame perfect because, after certain off-path histories, a player has a profitable local deviation that restores cooperation.

This is one of the clearest illustrations of the fact that

Nash equilibrium $\nRightarrow$ subgame perfect equilibrium,

because Nash equilibrium may rely on off-path punishments that are not credible.

Class 6: Bayesian Games a primer approximation

Bayesian Games

So far we have studied games in which the players know the game being played. Bayesian games extend this framework to situations in which some relevant feature of the environment is privately known by some players but not by others.

Motivation: subjective uncertainty as objective chance

The key modeling device is the *Harsanyi transformation*. Suppose one player is uncertain about some feature of the environment—for example, the strength, cost, or type of another player. Even if this uncertainty is only subjective, we can model it *as if* Nature moved first, selected the relevant type according to a common prior, and revealed that realization only to the informed player.

This does *not* mean that the true state of the world is literally random. Rather, it is a convenient way of converting subjective uncertainty into a well-defined game of incomplete information.

Formal definition

A Bayesian game is a tuple

$$\langle N, A, u, \Theta, \mu \rangle,$$

where:

- $N = \{1, \dots, n\}$ is the set of players;
- $A = \prod_{i=1}^n A_i$ is the action space;
- $\Theta = \prod_{i=1}^n \Theta_i$ is the type space²²;
- μ is a common prior probability distribution over Θ ;
- $u_i : A \times \Theta \rightarrow \mathbb{R}$ is player i 's ex post payoff function.

The timing is:

²² For each player i , Θ_i denotes the set of possible types he may have, and a realization $\theta_i \in \Theta_i$ is the privately observed type of player i . Thus a profile $\theta = (\theta_1, \dots, \theta_n) \in \Theta$ is a complete specification of all players' types. The common prior $\mu \in \Delta(\Theta)$ assigns probabilities to type profiles, not merely to isolated individual types. That's why it lies on the simplex.

1. Nature draws a type vector

$$\theta = (\theta_1, \dots, \theta_n) \in \Theta$$

according to the common prior μ .

2. Each player i privately observes only his own type θ_i .
3. The players simultaneously choose actions $a_i \in A_i$.
4. Payoffs are determined by $u_i(a, \theta)$, where $a = (a_1, \dots, a_n)$.

Thus, a player's strategy in a Bayesian game is not just an action. It is a *contingent plan* that specifies what action he takes for each possible type he may observe.

The Strength Game

Consider the following situation: Westley knows his true strength, while Humperdinck does not. Objectively, Westley is weak. However, it is common knowledge that Humperdinck believes Westley is weak with probability $p < 1$. To model Humperdinck's uncertainty, we apply the Harsanyi transformation:

- Nature chooses Westley's type:

Weak with probability p , Strong with probability $1 - p$.

- Nature reveals the realization to Westley, but not to Humperdinck.
- Westley and Humperdinck then choose their actions simultaneously.

Strategies

Westley has one action when weak and one action when strong, so his pure strategy set is

$$S_W = \{(F, F), (F, G), (G, F), (G, G)\},$$

where the first component is the action chosen if Westley is weak and the second is the action chosen if Westley is strong.

Humperdinck does not observe either Westley's type or Westley's action before moving, so his pure strategy set is simply

$$S_H = \{F, G\}.$$

Normal-form representation

Using the payoffs from the tree, the Bayesian game can be written in normal form as follows:

	F	G
(F, F)	$(4 - 8p, 8p - 4)$	$(8, 0)$
(F, G)	$(-4p, 8 - 4p)$	$(4 + 4p, 4 - 4p)$
(G, F)	$(4 - 4p, 12p - 4)$	$(8 - 4p, 4p)$
(G, G)	$(0, 8)$	$(4, 4)$

Here the first entry in each cell is Westley's expected payoff and the second entry is Humperdinck's expected payoff.

Example. If Westley uses strategy (F, F) and Humperdinck chooses F , then

$$U_W((F, F), F) = p(-4) + (1 - p)(4) = 4 - 8p,$$

and

$$U_H((F, F), F) = p(4) + (1 - p)(-4) = 8p - 4.$$

Best responses by type

It is often cleaner to solve the Bayesian game directly from the type-contingent problem.

Let $r \in [0, 1]$ be the probability that Humperdinck chooses F .

Westley when strong. If Westley is strong, then:

$$u_W(F, F) = 4, \quad u_W(G, F) = 0,$$

if Humperdinck chooses F , and

$$u_W(F, G) = 8, \quad u_W(G, G) = 4,$$

if Humperdinck chooses G . Hence the strong type strictly prefers F regardless of Humperdinck's action. So for the strong type, F is a strictly dominant action.

Westley when weak. If Westley is weak, then:

$$u_W(F, F) = -4, \quad u_W(G, F) = 0,$$

if Humperdinck chooses F , and

$$u_W(F, G) = 8, \quad u_W(G, G) = 4,$$

if Humperdinck chooses G . Therefore the weak type's expected payoff from F is

$$r(-4) + (1 - r)8 = 8 - 12r,$$

while his expected payoff from G is

$$r(0) + (1 - r)4 = 4 - 4r.$$

So the weak type prefers F if and only if

$$8 - 12r \geq 4 - 4r \iff r \leq \frac{1}{2}.$$

Thus:

$$\text{Weak type chooses } \begin{cases} F & \text{if } r < \frac{1}{2}, \\ G & \text{if } r > \frac{1}{2}, \\ \text{either } F \text{ or } G & \text{if } r = \frac{1}{2}. \end{cases}$$

Humperdinck's best response

Let

$$x \equiv \Pr(a_W = F \mid \theta_W = \text{Weak}) = \sigma_W(F \mid \theta_W = \text{Weak}) \in [0, 1]$$

denote the probability that the weak type of Westley chooses F . Since the strong type chooses F with probability one,

$$\sigma_W(F \mid \theta_W = \text{Strong}) = 1.$$

Humperdinck's expected payoff from choosing F is

$$U_H(F) = p(4x + 8(1 - x)) + (1 - p)(-4),$$

which simplifies to

$$U_H(F) = 12p - 4px - 4.$$

His expected payoff from choosing G is

$$U_H(G) = p(0 \cdot x + 4(1 - x)) + (1 - p)0 = 4p - 4px.$$

Hence

$$U_H(F) - U_H(G) = 8p - 4.$$

Therefore:

$$\text{Humperdinck chooses } \begin{cases} F & \text{if } p > \frac{1}{2}, \\ G & \text{if } p < \frac{1}{2}, \\ \text{either } F \text{ or } G & \text{if } p = \frac{1}{2}. \end{cases}$$

Bayesian Nash equilibrium

Combining the best responses, we obtain:

Case 1: $p < \frac{1}{2}$. Humperdinck chooses G . Given that, the weak type of Westley chooses F , and the strong type also chooses F . So the unique pure Bayesian Nash equilibrium is

$$((F, F), G).$$

Case 2: $p > \frac{1}{2}$. Humperdinck chooses F . Given that, the weak type of Westley chooses G , while the strong type still chooses F . So the unique pure Bayesian Nash equilibrium is

$$((G, F), F).$$

Case 3: $p = \frac{1}{2}$. Humperdinck is indifferent between F and G . The strong type of Westley still chooses F , while the weak type is willing to choose either action when Humperdinck mixes with probability $1/2$ on F .

In pure strategies, there are two Bayesian Nash equilibria:

$$((F, F), G) \quad \text{and} \quad ((G, F), F).$$

Interpretation

This example is useful because it shows very clearly what changes in a Bayesian game.

The key novelty is that strategies are type-contingent plans. Westley does not choose a single action once and for all. Instead, he chooses what to do when weak and what to do when strong.

At the same time, Humperdinck must act without observing Westley's type, so his action is based on beliefs. The equilibrium therefore combines:

1. optimization by each type of each player, and
2. consistency of actions with the beliefs induced by the common prior.

In short, a Bayesian Nash equilibrium is just a Nash equilibrium in a game where players move under asymmetric information and therefore choose contingent plans rather than simple actions.

Strategies in a Bayesian game

In a Bayesian game, a player does not simply choose an action. Instead, each player chooses a contingent plan specifying what to do for each type he may observe.

Formally, player i 's strategy space is

$$S_i = \{ s_i : \Theta_i \rightarrow \Delta A_i \text{ measurable} \}.$$

Thus, a strategy for player i is a measurable function from the player's type space Θ_i into lotteries over actions in A_i . If we restrict attention to pure strategies, then $s_i : \Theta_i \rightarrow A_i$.

This is the key distinction from an ordinary normal-form game: in a Bayesian game, the strategic object is a *mapping from types to actions*.

Bayesian Nash equilibrium

Given a Bayesian game

$$\langle N, A, u, \Theta, \mu \rangle,$$

a strategy profile

$$s \in S = \prod_{i=1}^n S_i$$

is a *Bayesian Nash equilibrium* if for every player i and every alternative strategy $s'_i \in S_i$,

$$\mathbb{E}[u_i(s(\theta), \theta)] \geq \mathbb{E}[u_i(s'_i(\theta_i), s_{-i}(\theta_{-i}), \theta)].$$

That is, no player can profitably deviate by replacing his entire type-contingent plan with another type-contingent plan.

Remark. A Bayesian Nash equilibrium is simply a Nash equilibrium in the game

$$\langle N, S, \mathbb{E}(u) \rangle,$$

where the strategy spaces are the spaces of type-contingent plans and payoffs are expected payoffs computed under the common prior μ .

Type-by-type characterization

The previous definition is global: it compares a player's equilibrium strategy to every other possible contingent plan. But in practice, it is often more convenient to verify optimality *type by type*.

Theorem 0.0.24. In a Bayesian game $\langle N, A, u, \Theta, \mu \rangle$, a strategy profile

$$s \in \prod_{i=1}^n S_i$$

is a Bayesian Nash equilibrium if

$$\mathbb{E}[u_i(s(\theta), \theta) \mid \theta_i] \geq \mathbb{E}[u_i(a_i, s_{-i}(\theta_{-i}), \theta) \mid \theta_i]$$

for all $a_i \in A_i$, all $\theta_i \in \Theta_i$, and all $i \in N$.

This says that once player i observes his type θ_i , the action prescribed by the equilibrium strategy must be optimal given:

A caveat: There are three distinct objects. The strategy s_i is the player's complete type-contingent plan. Its value $s_i(\theta_i)$ is only the prescription for type θ_i : an action in A_i under pure strategies, or a lottery in $\Delta(A_i)$ under mixed strategies. Finally, $a_i \in A_i$ denotes a candidate pure action available to that same type. Thus, the type-by-type condition compares the action prescribed by $s_i(\theta_i)$ with the other actions that player i could choose after observing θ_i , holding fixed his beliefs and the other players' equilibrium strategies.

1. his posterior beliefs about the other players' types, and
2. the equilibrium strategies used by the other players.

So instead of checking deviations to arbitrary alternative *functions* s'_i , one can verify that, for each type separately, the prescribed action solves that type's conditional optimization problem.

Why this characterization works

The intuition is straightforward. Expected utility in a Bayesian game can be written by integrating over the player's own type. If for every possible type θ_i the prescribed action is conditionally optimal, then summing those conditional inequalities yields the global no-deviation condition.

Equivalently, if a deviation strategy s'_i changed the player's action only for some types, then it would improve payoffs only if it improved the player's conditional expected utility for those types. Hence, if no such type-wise improvement is possible, no overall profitable deviation is possible either.

A technical caveat

The theorem above is always sufficient, but the converse requires care.

Remark. The type-by-type condition is also necessary when the relevant types occur with positive probability—for example, when Θ is finite and the prior μ has full support.

The reason is subtle but important. Suppose there is a type θ_i that occurs with probability zero under μ . Then the player's action at that type does not affect unconditional expected payoff at all. So a strategy profile may satisfy the global Bayesian Nash equilibrium condition even if the prescribed action at that zero-probability type is not conditionally optimal.

Thus:

- the type-by-type inequalities always imply Bayesian Nash equilibrium;
- the reverse implication may fail at types that are assigned zero probability.

This is why, in finite type spaces with full support, the two formulations are equivalent, whereas without full support the global definition is weaker.

Interpretation

This characterization is especially useful computationally. In applications, one usually solves a Bayesian game by proceeding type by type:

1. Fix a player i and a type θ_i .
2. Compute the player's posterior beliefs about θ_{-i} .
3. Evaluate expected payoffs from each feasible action a_i .
4. Choose the action maximizing conditional expected payoff.

In this sense, a Bayesian Nash equilibrium is the natural extension of Nash equilibrium to environments with private information: each player's strategy must be optimal not just overall, but conditional on each type that the player may privately observe.

Appendix to Class 6: Eat Before Arguing

Caleb and Delilah are about to have an argument.

But first, Caleb is going to have breakfast. Caleb's personal taste for breakfast foods is given by $t \in [0, 1]$, which he privately knows. It is common knowledge that Delilah believes that t is distributed according to the uniform distribution on $[0, 1]$. At breakfast, Caleb chooses Beer, $B(t)$, or Quiche, $Q(t)$, as a function of t .

While Caleb is having breakfast, Delilah is deciding whether she will Fight, F , or Not Fight, NF . She does not know t , and she does not know what Caleb is having for breakfast when she makes her decision.

The payoffs are as follows. From breakfast, Caleb gets utility of t if he drinks beer and $1 - t$ if he eats quiche, and from the argument he gets additional utility of -2 if Delilah fights and 0 if Delilah does not fight. If Caleb drinks beer, he gets drunk, so Delilah gets utility 2 from fighting and 0 from not fighting. If Caleb eats quiche, he stays sober, so Delilah gets utility 0 from fighting and 1 from not fighting.

1. Draw an extensive form that represents the Harsanyi transformation of this game.
2. Find all Bayesian Nash equilibria.

We solve the game as a Bayesian game obtained from the Harsanyi transformation.

Environment. Nature draws Caleb's type $t \in [0, 1]$ from the uniform distribution on $[0, 1]$. Caleb privately observes t , while Delilah only knows the prior $t \sim U[0, 1]$.

Caleb chooses breakfast:

B (Beer) or Q (Quiche).

Delilah then chooses whether to

F (Fight) or NF (Not Fight).

A crucial feature of the problem is that when Delilah chooses, she observes neither Caleb's type nor Caleb's breakfast choice. Thus, in the extensive form, all of Delilah's decision nodes belong to a single information set. In particular, this is *not* a signaling game in which Delilah reacts to the observed breakfast.

Payoffs. Caleb's payoff is the sum of a breakfast payoff and an argument payoff:

$$u_C(B, F | t) = t - 2, \quad u_C(Q, F | t) = 1 - t - 2,$$

$$u_C(B, NF | t) = t, \quad u_C(Q, NF | t) = 1 - t.$$

Delilah's payoff depends on Caleb's breakfast:

$$u_D(B, F) = 2, \quad u_D(B, NF) = 0,$$

$$u_D(Q, F) = 0, \quad u_D(Q, NF) = 1.$$

Strategy spaces. A strategy for Caleb is a measurable function

$$s_C : [0, 1] \rightarrow \{B, Q\}.$$

A strategy for Delilah is an element of $\Delta\{F, NF\}$. Let

$$p := \Pr(F)$$

denote the probability with which Delilah fights.

Step 1: Caleb's optimal action for each type. Fix $p \in [0, 1]$. Caleb's expected payoff from each breakfast is

$$U_C(B | t, p) = p(t - 2) + (1 - p)t = t - 2p,$$

$$U_C(Q | t, p) = p(1 - t - 2) + (1 - p)(1 - t) = 1 - t - 2p.$$

Hence

$$U_C(B | t, p) - U_C(Q | t, p) = t - (1 - t) = 2t - 1.$$

Therefore Caleb's best response is

$$s_C^*(t) = \begin{cases} Q, & t < \frac{1}{2}, \\ B, & t > \frac{1}{2}, \\ \text{any mixture over } \{B, Q\}, & t = \frac{1}{2}. \end{cases}$$

Intuition. The term -2 from fighting is added to both breakfasts equally, so it cancels out when Caleb compares B and Q . Thus he simply chooses the breakfast he likes more: beer if t is high enough, quiche if t is low enough.

Step 2: The distribution of Caleb's breakfast. Since $t \sim U[0, 1]$, the cutoff rule above implies

$$\Pr(B) = \Pr\left(t > \frac{1}{2}\right) = \frac{1}{2}, \quad \Pr(Q) = \Pr\left(t < \frac{1}{2}\right) = \frac{1}{2}.$$

The type $t = \frac{1}{2}$ has probability zero, so Caleb's action at exactly $t = \frac{1}{2}$ is irrelevant for beliefs and payoffs.

Step 3: Delilah's optimal action. Given an arbitrary strategy s_C , let

$$q := \Pr(B).$$

Delilah's expected payoff from fighting is

$$U_D(F) = 2q + 0(1 - q) = 2q,$$

while her expected payoff from not fighting is

$$U_D(NF) = 0 \cdot q + 1 \cdot (1 - q) = 1 - q.$$

Thus Delilah prefers F whenever

$$2q \geq 1 - q \iff 3q \geq 1 \iff q \geq \frac{1}{3}.$$

Under Caleb's best response we have $q = \frac{1}{2}$, so

$$U_D(F) = 2 \cdot \frac{1}{2} = 1, \quad U_D(NF) = 1 - \frac{1}{2} = \frac{1}{2}.$$

Hence Delilah strictly prefers to fight:

F is her unique best response.

Step 4: Equilibrium. Since Caleb's best response does not depend on p , he best responds to F with the same cutoff strategy:

$$s_C^*(t) = \begin{cases} Q, & t < \frac{1}{2}, \\ B, & t > \frac{1}{2}, \\ \text{arbitrary at } t = \frac{1}{2}. \end{cases}$$

And given this strategy, Delilah's unique best response is F .

Therefore, all Bayesian Nash equilibria are of the form

$$(s_C^*, F),$$

where

$$s_C^*(t) = Q \text{ for } t < \frac{1}{2}, \quad s_C^*(t) = B \text{ for } t > \frac{1}{2},$$

and Caleb's action at $t = \frac{1}{2}$ may be any mixture over $\{B, Q\}$.

Conclusion. The equilibrium outcome is essentially unique:

Caleb chooses Quiche if $t < \frac{1}{2}$, Beer if $t > \frac{1}{2}$, and Delilah fights.

Why there are no other equilibria. If Delilah were to play NF , Caleb would still use the same cutoff rule, so $\Pr(B) = 1/2$. But then Delilah would strictly prefer F , so NF cannot be part of an equilibrium. Likewise, if Delilah mixed, Caleb's best response would still be the same cutoff rule, again implying $\Pr(B) = 1/2$, and therefore Delilah would again strictly prefer F . Hence no equilibrium with NF or mixing exists.

Class 7: Applications of Bayesian Games

Cournot duopoly with private costs

Consider a Bayesian Cournot duopoly with inverse demand

$$p(Q) = 12 - Q, \quad Q = q_1 + q_2.$$

The firms choose quantities simultaneously.

Firm 1 has zero costs. Firm 2 has private costs:

$$\theta_2 \in \{L, H\},$$

where

$$c_2(q_2, L) = 0, \quad c_2(q_2, H) = 2q_2.$$

Firm 1 believes that

$$\Pr(\theta_2 = L) = 1 - \alpha, \quad \Pr(\theta_2 = H) = \alpha.$$

Bayesian game

Firm 1 has no private information, so its strategy is just a single quantity q_1 . Firm 2 observes its own type, so its strategy is a type-contingent pair

$$(q_2(L), q_2(H)).$$

Payoffs are

$$u_1(q_1, q_2, \theta_2) = (12 - q_1 - q_2)q_1,$$

and

$$u_2(q_1, q_2, L) = (12 - q_1 - q_2)q_2, \quad u_2(q_1, q_2, H) = (10 - q_1 - q_2)q_2.$$

Type-by-type optimization

A Bayesian Nash equilibrium is a triple

$$(q_1^*, q_2^*(L), q_2^*(H))$$

such that each type solves its own optimization problem.

Firm 1 chooses q_1 to maximize expected profit:

$$\max_{q_1 \geq 0} \left[(1 - \alpha)(12 - q_1 - q_2(L))q_1 + \alpha(12 - q_1 - q_2(H))q_1 \right].$$

The first-order condition is

$$12 - 2q_1 - (1 - \alpha)q_2(L) - \alpha q_2(H) = 0,$$

so

$$q_1 = 6 - \frac{1}{2} \left((1 - \alpha)q_2(L) + \alpha q_2(H) \right).$$

If Firm 2 is low-cost, it solves

$$\max_{q_2 \geq 0} (12 - q_1 - q_2)q_2,$$

which yields

$$q_2(L) = \frac{12 - q_1}{2}.$$

If Firm 2 is high-cost, it solves

$$\max_{q_2 \geq 0} (10 - q_1 - q_2)q_2,$$

which yields

$$q_2(H) = \frac{10 - q_1}{2}.$$

Solving the system

Substituting the two type-contingent best responses of Firm 2 into Firm 1's condition gives

$$q_1 = 6 - \frac{1}{2} \left((1 - \alpha) \frac{12 - q_1}{2} + \alpha \frac{10 - q_1}{2} \right).$$

Since

$$(1 - \alpha)(12 - q_1) + \alpha(10 - q_1) = 12 - 2\alpha - q_1,$$

we obtain

$$q_1 = 6 - \frac{12 - 2\alpha - q_1}{4}.$$

Hence

$$4q_1 = 24 - (12 - 2\alpha - q_1) = 12 + 2\alpha + q_1,$$

so

$$3q_1 = 12 + 2\alpha \quad \implies \quad q_1^* = 4 + \frac{2\alpha}{3}.$$

Therefore

$$q_2^*(L) = \frac{12 - q_1^*}{2} = 4 - \frac{\alpha}{3},$$

and

$$q_2^*(H) = \frac{10 - q_1^*}{2} = 3 - \frac{\alpha}{3}.$$

Remark. As α increases, Firm 1 assigns more probability to Firm 2 being high-cost, so it behaves more aggressively:

$$q_1^* = 4 + \frac{2\alpha}{3}$$

is increasing in α . Because Firm 1 produces more, both types of Firm 2 optimally produce less.

Cournot duopoly with higher-order beliefs

Now enrich the previous example by allowing Firm 1 itself to have a type. The type of Firm 1 determines its belief about Firm 2's costs.

Environment

Again,

$$p(Q) = 12 - Q, \quad c_1(q_1) = 0.$$

Firm 2's type is

$$\theta_2 \in \{L, H\},$$

with

$$c_2(q_2, L) = 0, \quad c_2(q_2, H) = 2q_2.$$

Firm 1's type is

$$\theta_1 \in \{\ell, h\},$$

where

$$\Pr(\theta_1 = \ell) = 1 - \beta, \quad \Pr(\theta_1 = h) = \beta.$$

The interpretation is:

$$\theta_1 = \ell \Rightarrow \Pr(\theta_2 = H \mid \theta_1 = \ell) = \frac{1}{4},$$

while

$$\theta_1 = h \Rightarrow \Pr(\theta_2 = H \mid \theta_1 = h) = \frac{3}{4}.$$

So Nature first chooses Firm 1's type and then chooses Firm 2's type conditional on Firm 1's realized type.

Common prior

Let

$$\mu(\theta_1, \theta_2) := \Pr(\theta_1, \theta_2)$$

denote the common prior over the type profile

$$(\theta_1, \theta_2) \in \{\ell, h\} \times \{L, H\}.$$

Because the model specifies both the marginal distribution of θ_1 and the conditional distribution of θ_2 given θ_1 , the induced joint distribution is

$$\mu(\theta_1, \theta_2) = \Pr(\theta_1) \Pr(\theta_2 \mid \theta_1).$$

Therefore,

$$\begin{aligned}\mu(\ell, L) &= \Pr(\theta_1 = \ell, \theta_2 = L) = \Pr(\theta_1 = \ell) \Pr(\theta_2 = L \mid \theta_1 = \ell) = (1 - \beta) \frac{3}{4}, \\ \mu(\ell, H) &= \Pr(\theta_1 = \ell, \theta_2 = H) = \Pr(\theta_1 = \ell) \Pr(\theta_2 = H \mid \theta_1 = \ell) = (1 - \beta) \frac{1}{4}, \\ \mu(h, L) &= \Pr(\theta_1 = h, \theta_2 = L) = \Pr(\theta_1 = h) \Pr(\theta_2 = L \mid \theta_1 = h) = \beta \frac{1}{4}, \\ \mu(h, H) &= \Pr(\theta_1 = h, \theta_2 = H) = \Pr(\theta_1 = h) \Pr(\theta_2 = H \mid \theta_1 = h) = \beta \frac{3}{4}.\end{aligned}$$

A pure strategy profile is now

$$(q_1(\ell), q_1(h), q_2(L), q_2(H)).$$

Interpretation. The object μ is the *joint prior* over the pair of types. Later, when a player observes only his own type, posterior beliefs are obtained from μ by Bayes' rule. For example, after observing $\theta_2 = L$, Firm 2 forms beliefs about whether Firm 1 is type ℓ or type h using the conditional probabilities derived from this common prior.

Common prior

The common prior over (θ_1, θ_2) is therefore

$$\begin{aligned}\mu(\ell, L) &= \frac{3}{4}(1 - \beta), & \mu(\ell, H) &= \frac{1}{4}(1 - \beta), \\ \mu(h, L) &= \frac{1}{4}\beta, & \mu(h, H) &= \frac{3}{4}\beta.\end{aligned}$$

A pure strategy profile is now

$$(q_1(\ell), q_1(h), q_2(L), q_2(H)).$$

Posterior beliefs of Firm 2

Firm 2 observes only its own cost type, so it must infer Firm 1's type from Bayes' rule.

If $\theta_2 = L$, then

$$\Pr(\ell \mid L) = \frac{\mu(\ell, L)}{\mu(\ell, L) + \mu(h, L)} = \frac{\frac{3}{4}(1 - \beta)}{\frac{3}{4}(1 - \beta) + \frac{1}{4}\beta} = \frac{3(1 - \beta)}{3 - 2\beta},$$

and

$$\Pr(h \mid L) = \frac{\beta}{3 - 2\beta}.$$

If $\theta_2 = H$, then

$$\Pr(\ell \mid H) = \frac{\mu(\ell, H)}{\mu(\ell, H) + \mu(h, H)} = \frac{\frac{1}{4}(1 - \beta)}{\frac{1}{4}(1 - \beta) + \frac{3}{4}\beta} = \frac{1 - \beta}{1 + 2\beta},$$

and

$$\Pr(h \mid H) = \frac{3\beta}{1 + 2\beta}.$$

Type-by-type best responses

Each type of Firm 1 solves its own conditional Cournot problem.

Type ℓ of Firm 1 believes

$$\Pr(L \mid \ell) = \frac{3}{4}, \quad \Pr(H \mid \ell) = \frac{1}{4},$$

so it solves

$$\max_{q_1 \geq 0} \left[\frac{3}{4}(12 - q_1 - q_2(L))q_1 + \frac{1}{4}(12 - q_1 - q_2(H))q_1 \right].$$

The first-order condition gives

$$q_1(\ell) = 6 - \frac{1}{2} \left(\frac{3}{4}q_2(L) + \frac{1}{4}q_2(H) \right).$$

Type h of Firm 1 believes

$$\Pr(L \mid h) = \frac{1}{4}, \quad \Pr(H \mid h) = \frac{3}{4},$$

so it solves

$$\max_{q_1 \geq 0} \left[\frac{1}{4}(12 - q_1 - q_2(L))q_1 + \frac{3}{4}(12 - q_1 - q_2(H))q_1 \right].$$

The first-order condition gives

$$q_1(h) = 6 - \frac{1}{2} \left(\frac{1}{4}q_2(L) + \frac{3}{4}q_2(H) \right).$$

Type L of Firm 2 solves

$$\max_{q_2 \geq 0} \mathbb{E} \left[(12 - q_1(\theta_1) - q_2)q_2 \mid L \right],$$

hence

$$q_2(L) = 6 - \frac{1}{2} (\Pr(\ell \mid L) q_1(\ell) + \Pr(h \mid L) q_1(h)),$$

that is,

$$q_2(L) = 6 - \frac{1}{2} \left(\frac{3(1 - \beta)}{3 - 2\beta} q_1(\ell) + \frac{\beta}{3 - 2\beta} q_1(h) \right).$$

Type H of Firm 2 solves

$$\max_{q_2 \geq 0} \mathbb{E} \left[(10 - q_1(\theta_1) - q_2)q_2 \mid H \right],$$

so

$$q_2(H) = 5 - \frac{1}{2} (\Pr(\ell | H) q_1(\ell) + \Pr(h | H) q_1(h)),$$

that is,

$$q_2(H) = 5 - \frac{1}{2} \left(\frac{1-\beta}{1+2\beta} q_1(\ell) + \frac{3\beta}{1+2\beta} q_1(h) \right).$$

Equilibrium system

Therefore the Bayesian Nash equilibrium is the solution to the linear system

$$q_1(\ell) = 6 - \frac{1}{2} \left(\frac{3}{4} q_2(L) + \frac{1}{4} q_2(H) \right),$$

$$q_1(h) = 6 - \frac{1}{2} \left(\frac{1}{4} q_2(L) + \frac{3}{4} q_2(H) \right),$$

$$q_2(L) = 6 - \frac{1}{2} \left(\frac{3(1-\beta)}{3-2\beta} q_1(\ell) + \frac{\beta}{3-2\beta} q_1(h) \right),$$

$$q_2(H) = 5 - \frac{1}{2} \left(\frac{1-\beta}{1+2\beta} q_1(\ell) + \frac{3\beta}{1+2\beta} q_1(h) \right).$$

Solving yields

$$q_1(\ell) = \frac{50 + 51\beta - 50\beta^2}{12(1 + \beta - \beta^2)}, \quad q_1(h) = \frac{53 + 55\beta - 54\beta^2}{12(1 + \beta - \beta^2)},$$

$$q_2(L) = \frac{47 + 46\beta - 48\beta^2}{12(1 + \beta - \beta^2)}, \quad q_2(H) = \frac{35 + 30\beta - 32\beta^2}{12(1 + \beta - \beta^2)}.$$

Remark. The main lesson is conceptual. Firm 2's quantity depends not only on its own cost type, but also on its posterior beliefs about Firm 1's *beliefs*. So higher-order beliefs affect equilibrium behavior through the conditional expectations that each type uses when solving its own problem.

A lemons game

We now consider a simple lemons market.

Environment

The seller privately knows the quality of the good,

$$\theta \in [0, 1].$$

The buyer believes

$$\theta \sim U[0, 1].$$

The seller chooses a price

$$p \in [0, 1].$$

The buyer chooses a set

$$E \subset [0, 1]$$

of prices that she is willing to accept.

Trade occurs if and only if

$$p \in E.$$

Payoffs are:

$$u_S(p, E, \theta) = \begin{cases} p & \text{if } p \in E, \\ q\theta & \text{if } p \notin E, \end{cases}$$

and

$$u_B(p, E, \theta) = \begin{cases} \theta - p & \text{if } p \in E, \\ 0 & \text{if } p \notin E, \end{cases}$$

where

$$q \in \left(\frac{1}{2}, 1\right).$$

Remark. Since $q < 1$, trade is always ex post efficient: if trade occurs, total surplus is θ , whereas under no trade it is only $q\theta$. Thus

$$\theta - q\theta = (1 - q)\theta > 0$$

for every $\theta > 0$.

Step 1: no-trade equilibria

Suppose the buyer chooses

$$E^* = \emptyset.$$

Then no posted price is ever accepted, so for every seller type θ ,

$$u_S(p, E^*, \theta) = q\theta \quad \text{for all } p \in [0, 1].$$

Hence the seller is indifferent across all prices, and therefore any price choice is a best response for every type.²³

To obtain a Bayesian Nash equilibrium, however, we must also check that $E^* = \emptyset$ is a best response for the buyer against the seller's strategy p^* . A sufficient condition is

$$p^*(\theta) \geq \theta \quad \text{for all } \theta \in [0, 1],$$

because then trade at any posted price would give the buyer payoff

$$\theta - p^*(\theta) \leq 0.$$

Therefore rejecting every price is optimal.

²³ This does *not* create the same problem as in the ultimatum or finite-round bargaining examples. There, if the responder rejected the boundary offer, the proposer would face a maximization problem with a supremum that is not attained, so no best response would exist. Here, by contrast, when $E^* = \emptyset$, the seller's payoff is flat in p : every price yields exactly $q\theta$. Thus the best-response set is not empty; indeed, it is the whole interval $[0, 1]$ for each type. In this respect, the logic is closer to the simultaneous-demand bargaining example in which one player's action makes the other player indifferent across a whole set of responses.

Hence, for any seller strategy p^* such that $p^*(\theta) \geq \theta$ for all θ , the pair

$$(p^*, E^*) = (p^*, \emptyset)$$

is a pure-strategy Bayesian Nash equilibrium. Since there are infinitely many such seller strategies, there is a continuum of no-trade equilibria.

Remark. The condition $p^*(\theta) \geq \theta$ for all θ is sufficient, but not necessary. More generally, $E^* = \emptyset$ is a best response for the buyer whenever accepting any posted price yields weakly negative expected payoff given the seller types that choose that price. That is, for every price p posted with positive probability, it is enough that

$$\mathbb{E}[\theta \mid p(\theta) = p] \leq p.$$

Step 2: if $E^ \neq \emptyset$, let $e^* = \max E^*$*

Suppose now that $E^* \neq \emptyset$, and let

$$e^* = \max E^*.$$

For any seller type θ , compare two possibilities:

If the seller chooses some accepted price $p \in E^*$, his payoff is p , so among accepted prices the best one is e^* .

If the seller chooses some price $p \notin E^*$, there is no trade and his payoff is $q\theta$.

Thus the seller's best response to E^* is:

$$BR_S(E^* \mid \theta) = \begin{cases} e^* & \text{if } \theta < e^*/q, \\ \text{some } p \in (e^*, 1] & \text{if } \theta > e^*/q. \end{cases}$$

At the cutoff $\theta = e^*/q$, the seller is indifferent.

Step 3: show that $e^ = 0$*

Suppose, for contradiction, that

$$e^* > 0.$$

Then all seller types

$$\theta < e^*/q$$

choose the accepted price e^* , while types above the cutoff choose some price greater than e^* , which the buyer rejects.

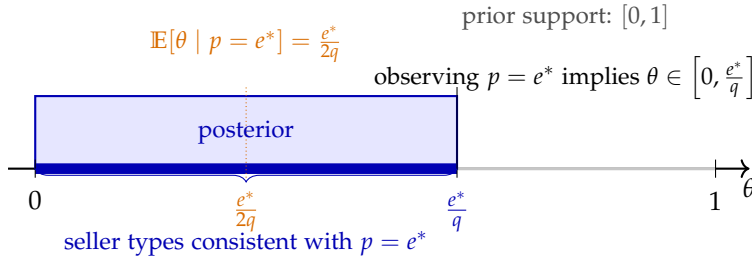
So if the buyer observes the price e^* , she infers that

$$\theta \in [0, e^*/q].$$

Since the prior is uniform, the posterior conditional on seeing e^* is uniform on that interval. Therefore the buyer's conditional expected quality is

$$\mathbb{E}[\theta \mid p = e^*] = \frac{1}{2} \cdot \frac{e^*}{q} = \frac{e^*}{2q}.$$

Figure 5 illustrates the logic. Observing the price e^* reveals that the seller's type must lie in the interval $[0, e^*/q]$. Since the prior is uniform, the posterior is uniform on that interval, and its conditional mean is simply the midpoint $e^*/(2q)$.



Observing the price e^* implies that the seller's type must lie in the interval $[0, \frac{e^*}{q}]$. Since the prior is uniform, the posterior is uniform on that interval, and its conditional mean is the midpoint $\frac{e^*}{2q}$.

Hence the buyer's expected payoff from accepting e^* is

$$\frac{e^*}{2q} - e^*.$$

Because $q > \frac{1}{2}$, we have

$$\frac{1}{2q} < 1,$$

so

$$\frac{e^*}{2q} - e^* < 0.$$

Thus the buyer strictly prefers to reject e^* , contradicting the assumption that $e^* \in E^*$.

Therefore any equilibrium with $E^* \neq \emptyset$ must satisfy

$$e^* = 0.$$

Step 4: almost-no-trade equilibria

If $E^* \neq \emptyset$, we have just shown that $\max E^* = 0$, so necessarily

$$E^* = \{0\}.$$

Given this acceptance rule, every seller type $\theta > 0$ strictly prefers no trade, since

$$q\theta > 0.$$

Hence these types choose some price $p(\theta) > 0$, which is rejected.

Type $\theta = 0$ is indifferent between trading at price 0 and not trading, since both give payoff 0. So it may choose $p(0) = 0$.

Thus there is a continuum of equilibria of the form:

$$E^* = \{0\}, \quad p(0) = 0, \quad p(\theta) > 0 \text{ for all } \theta > 0.$$

These are *almost-no-trade equilibria*: trade occurs only for the zero-quality seller, a type that has probability zero under the uniform prior.

Remark. The lemons example shows how severe adverse selection can be. Even though trade is ex post efficient for every $\theta > 0$, pure-strategy equilibrium can collapse to no trade or almost no trade. The basic force is that any positive accepted price attracts disproportionately low-quality sellers, making acceptance unprofitable for the buyer.

Appendix to Class 7: Bad mechanic

Quincy thinks there might be a problem with his car. It is common knowledge that with probability $\mu \in (0, 1)$ the problem is Serious (S), and with probability $1 - \mu$ the problem is Minor (M). Since Quincy does not know which is the case, he takes his car to Percy the mechanic.

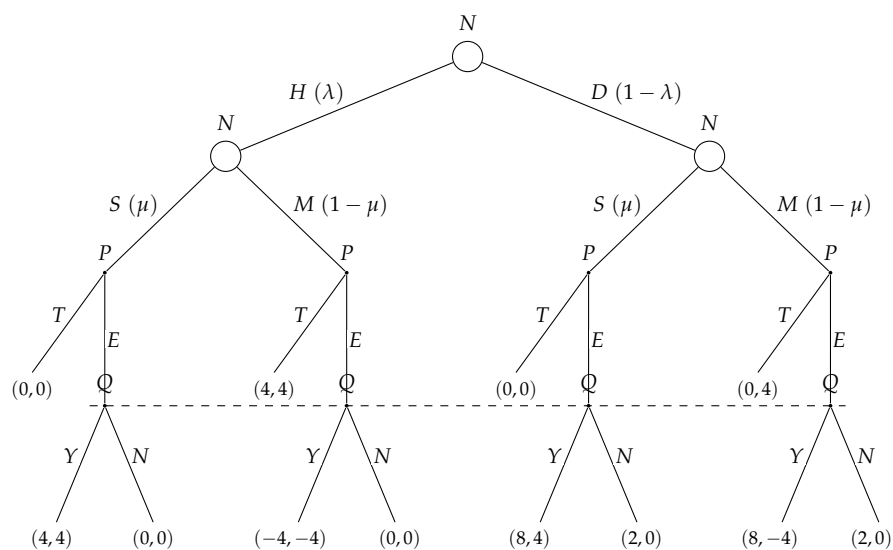
It is common knowledge that Quincy believes that with probability $\lambda \in (0, 1)$ Percy is Honest (H), and with probability $1 - \lambda$ she is Dishonest (D).

After figuring out what is wrong with Quincy's car, Percy can either perform a Tune-up (T) or suggest a new Engine (E). If Percy suggests a new Engine, then Quincy can say Yes (Y), repair it; or No (N), do not repair it.

Quincy's payoff is 4 if his car has a Minor problem and Percy gives it a Tune-up, or if his car has a Serious problem, Percy suggests a new Engine, and Quincy says Yes. Quincy's payoff is -4 if his car has a Minor problem, Percy suggests a new Engine, and Quincy says Yes. Otherwise Quincy's payoff is 0.

If Percy is Honest, her payoff is the same as Quincy's. If Percy is Dishonest, her payoff is 8 if she suggests a new Engine and Quincy says Yes, and 2 if she suggests a new Engine and Quincy says No. In all other cases Percy's payoff is 0.

The extensive form of the Harsanyi transformation of this game is shown below.



Extensive form of the Harsanyi transformation.

1. Write down each player's strategy space.
2. Which of Percy's strategies are dominated in the Bayesian normal

form?

3. Under what conditions on μ and λ , if any, does there exist a pure strategy Bayesian Nash equilibrium in which Percy's Honest type chooses a bad action (i.e., performs a Tune-up for a Serious problem, or suggests a new Engine for a Minor problem)?

We solve the Bayesian normal form of the game. Throughout, we order Percy's contingent actions as

$$(H, S), (H, M), (D, S), (D, M).$$

Thus a strategy such as *TTEE* means:

$$a_P(H, S) = T, \quad a_P(H, M) = T, \quad a_P(D, S) = E, \quad a_P(D, M) = E.$$

(a) *Strategy spaces.* Percy observes both her type (H or D) and the car problem (S or M), so she must specify an action in each of the four contingencies. Hence

$$S_P = \{T, E\}^4.$$

Equivalently,

$$S_P = \{TTTT, TTTE, TTET, TTEE, TETT, TETE, TEET, TEEE, \\ ET TT, ETTE, ETET, ETEE, EETT, EETE, EEET, EEEE\}.$$

Quincy moves only if Percy suggests a new engine, and all such decision nodes are in a single information set. Therefore Quincy has only two pure strategies:

$$S_Q = \{Y, N\}.$$

Remark (Information and contingent plans). The key distinction is that Percy observes both dimensions of uncertainty, whereas Quincy observes neither.

Percy observes both her personal type,

$$\theta_P \in \{H, D\},$$

and the condition of the car,

$$\omega \in \{S, M\}.$$

Therefore, Percy's effective type space is

$$\Theta_P = \{H, D\} \times \{S, M\}.$$

A pure strategy for Percy is a complete contingent plan assigning an action to each possible realization of this information:

$$s_P : \Theta_P \longrightarrow \{T, E\},$$

or equivalently,

$$s_P : \{H, D\} \times \{S, M\} \longrightarrow \{T, E\}.$$

Thus,

$$S_P = \{s_P : \{H, D\} \times \{S, M\} \rightarrow \{T, E\}\} = \{T, E\}^4.$$

There are four possible information realizations,

$$(H, S), \quad (H, M), \quad (D, S), \quad (D, M),$$

so each pure strategy specifies four actions, one for each contingency.

For example, fixing the order above, the strategy $TETE$ means

$$s_P(H, S) = T, \quad s_P(H, M) = E, \quad s_P(D, S) = T, \quad s_P(D, M) = E.$$

Since Percy has two actions at each of four contingencies, she has

$$2^4 = 16$$

pure strategies.

Quincy, by contrast, observes neither Percy's type nor the condition of the car. Formally, Quincy can be represented as having a singleton type space,

$$\Theta_Q = \{*\}.$$

Hence a pure strategy for Quincy is

$$s_Q : \{*\} \longrightarrow \{Y, N\},$$

so that

$$S_Q = \{s_Q : \{*\} \rightarrow \{Y, N\}\} \cong \{Y, N\}.$$

Thus, Quincy chooses only one action, Y or N , at her single information set. She must form beliefs about Percy's possible types and about the strategy used by each type in order to decide optimally, but those beliefs do not enlarge her strategy space. They affect her best response, not the set of actions she must specify in her contingent plan.

(b) *Dominated strategies of Percy in the Bayesian normal form.* A convenient way to proceed is to compare, for each type-contingency, Percy's payoff from T and E , given Quincy's two possible actions Y and N . Percy's payoff comparison:

Contingency	if Quincy plays Y	if Quincy plays N
$(H, S) : T \text{ vs } E$	0 vs 4	0 vs 0
$(H, M) : T \text{ vs } E$	4 vs -4	4 vs 0
$(D, S) : T \text{ vs } E$	0 vs 8	0 vs 2
$(D, M) : T \text{ vs } E$	0 vs 8	0 vs 2

From this table we obtain:

1. At (H, M) , T strictly dominates E .
2. At (D, S) , E strictly dominates T .
3. At (D, M) , E strictly dominates T .
4. At (H, S) , E weakly dominates T : if Quincy plays Y , then E is strictly better; if Quincy plays N , Percy is indifferent.

Because $\lambda, \mu \in (0, 1)$, each of the four type-state contingencies has strictly positive probability. Therefore any full strategy of Percy that uses a strictly dominated action in one of the contingencies above is itself strictly dominated by the otherwise identical strategy that replaces that action with the dominating one.

Hence:

- Every strategy with $a_P(H, M) = E$ is strictly dominated.
- Every strategy with $a_P(D, S) = T$ is strictly dominated.
- Every strategy with $a_P(D, M) = T$ is strictly dominated.

So the only *undominated candidates* that remain are

$$TTEE \quad \text{and} \quad ETEE.$$

Moreover, since the only difference between these two strategies is the action at (H, S) , and E weakly dominates T there, we also have:

$$ETEE \text{ weakly dominates } TTEE.$$

Conclusion for part (b). All strategies except $TTEE$ and $ETEE$ are strictly dominated. In addition, $TTEE$ is weakly dominated by $ETEE$.

(c) Existence of a pure-strategy BNE in which Honest chooses a “bad” action. An Honest mechanic’s actions are:

- at (H, S) : the “good” action is E , since with a serious problem the correct recommendation is a new engine;
- at (H, M) : the “good” action is T , since with a minor problem the correct action is a tune-up.

Thus an Honest type chooses a “bad” action if either:

$$a_P(H, S) = T \quad \text{or} \quad a_P(H, M) = E.$$

But from part (b), $a_P(H, M) = E$ is strictly dominated, so it can never be part of a Bayesian Nash equilibrium. Therefore the *only* possible

pure-strategy BNE with a bad action by the Honest type must have Percy play

$$TTEE.$$

So we now analyze whether $(TTEE, \sigma_Q)$, with $\sigma_Q \in \{Y, N\}$, can be a pure-strategy BNE.

Step 1: Quincy's best response to TTEE. Under $TTEE$, Percy recommends an engine only when she is dishonest:

$$(D, S) \quad \text{or} \quad (D, M).$$

Therefore, conditional on observing E , Quincy learns that Percy is dishonest, but he learns nothing new about whether the problem is serious or minor. Hence

$$\Pr(S \mid E) = \mu, \quad \Pr(M \mid E) = 1 - \mu.$$

If Quincy says Y , his expected payoff is

$$U_Q(Y \mid E) = 4\Pr(S \mid E) - 4\Pr(M \mid E) = 4\mu - 4(1 - \mu) = 4(2\mu - 1).$$

If he says N , his payoff is

$$U_Q(N \mid E) = 0.$$

Hence Quincy's best response is:

$$\begin{cases} Y, & \mu > \frac{1}{2}, \\ N, & \mu < \frac{1}{2}, \\ Y \text{ or } N, & \mu = \frac{1}{2}. \end{cases}$$

Step 2: Is TTEE a best response for Percy? If Quincy plays Y , then at contingency (H, S) an Honest Percy gets

$$u_P(T) = 0, \quad u_P(E) = 4.$$

So she would strictly deviate from T to E . Therefore $TTEE$ is *not* a best response to Y .

If Quincy plays N , then at (H, S) ,

$$u_P(T) = 0, \quad u_P(E) = 0,$$

so Honest Percy is indifferent between T and E . At the other three contingencies, $TTEE$ already uses the optimal action: T at (H, M) , and E at both dishonest contingencies.

Therefore $TTEE$ is a best response to N .

Conclusion for part (c). A pure-strategy Bayesian Nash equilibrium in which Honest chooses a bad action exists if and only if Quincy's best response to $TTEE$ is N , i.e.

$$\mu \leq \frac{1}{2}.$$

In that case, the equilibrium is

$$(TTEE, N).$$

This condition is *independent of* λ .

Final answer to part (c). Yes, such a pure-strategy Bayesian Nash equilibrium exists exactly when

$$\mu \leq \frac{1}{2}.$$

The bad action is that the Honest type performs a tune-up when the problem is serious. There is no equilibrium in which the Honest type suggests a new engine for a minor problem, because that action is strictly dominated.

Optional: full characterization of pure-strategy BNE. Since the only relevant candidate strategies for Percy are $TTEE$ and $ETEE$, we can characterize all pure Bayesian Nash equilibria.

1. *Equilibria with TTEE.* As shown above:

$$(TTEE, N) \text{ is a BNE iff } \mu \leq \frac{1}{2}.$$

2. *Equilibria with ETEE.* Under $ETEE$, Percy recommends an engine in contingencies

$$(H, S), (D, S), (D, M),$$

but not in (H, M) . Therefore

$$\Pr(E) = \lambda\mu + (1 - \lambda)\mu + (1 - \lambda)(1 - \mu) = 1 - \lambda(1 - \mu).$$

Among those who recommend E , the serious-problem contingencies are (H, S) and (D, S) , whose total probability is

$$\lambda\mu + (1 - \lambda)\mu = \mu.$$

Hence

$$\Pr(S | E) = \frac{\mu}{1 - \lambda(1 - \mu)}.$$

So Quincy prefers Y iff

$$\Pr(S | E) \geq \frac{1}{2} \iff \frac{\mu}{1 - \lambda(1 - \mu)} \geq \frac{1}{2} \iff \mu \geq \frac{1 - \lambda}{2 - \lambda}.$$

Thus:

$$(ETEE, Y) \text{ is a BNE iff } \mu \geq \frac{1-\lambda}{2-\lambda},$$

and

$$(ETEE, N) \text{ is a BNE iff } \mu \leq \frac{1-\lambda}{2-\lambda}.$$

Therefore the full set of pure-strategy BNE is:

$$\left\{ \begin{array}{ll} \{(TTEE, N), (ETEE, N)\}, & \mu < \frac{1-\lambda}{2-\lambda}, \\ \{(TTEE, N), (ETEE, N), (ETEE, Y)\}, & \mu = \frac{1-\lambda}{2-\lambda}, \\ \{(TTEE, N), (ETEE, Y)\}, & \frac{1-\lambda}{2-\lambda} < \mu \leq \frac{1}{2}, \\ \{(ETEE, Y)\}, & \mu > \frac{1}{2}. \end{array} \right.$$

Intuition. The only way an Honest mechanic can “mistreat” a serious problem in equilibrium is if Quincy is sufficiently pessimistic about the value of accepting an engine recommendation. Under $TTEE$, any engine recommendation reveals that Percy is dishonest, so Quincy accepts only if the serious problem is sufficiently likely. When $\mu \leq 1/2$, Quincy rejects, and once rejection is anticipated the Honest mechanic is indifferent between T and E in the serious state. That is exactly what allows the bad action T to survive in equilibrium.

Class 8: Auctions

Basic setup

We begin with the canonical auction environment with one indivisible object and n bidders.

- There is one prize and n bidders.
- Bidder i 's type is $\theta_i \in \Theta_i$.
- Each bidder simultaneously submits a bid

$$b_i \in [0, \infty).$$

- Bidder i 's payoff is

$$u_i(b, \theta) = x_i(b)v_i(\theta) - p_i(b).$$

Here:

- $x_i(b)$ is the probability that bidder i gets the prize,
- $v_i(\theta)$ is bidder i 's valuation for the prize,
- $p_i(b)$ is the payment made by bidder i .

This formulation is deliberately general. An auction mechanism is therefore summarized by two objects:

$$x_i(b) \quad \text{and} \quad p_i(b),$$

that is, by an allocation rule and a payment rule.

Second-price private values auction

We now specialize to the standard second-price private values environment.

- **Private values:**

$$\Theta_i = [0, \bar{\theta}] \quad \text{and} \quad v_i(\theta) = \theta_i.$$

Thus bidder i 's valuation depends only on her own type.

- **Second-price rule:** the highest bid wins, but pays the second-highest bid.

Hence, if bidder i wins, her payoff is

$$\theta_i - (\text{second-highest bid}),$$

and if she loses, her payoff is zero.

Theorem 0.0.25. *In the second-price private values auction, it is a weakly dominant strategy for each player to bid*

$$b_i(\theta_i) = \theta_i,$$

regardless of the distribution over Θ .

Proof. Fix bidder i , her type θ_i , and the bids of all other bidders b_{-i} . Let

$$m := \max_{j \neq i} b_j$$

denote the highest bid submitted by her opponents.

We compare truthful bidding, $b_i = \theta_i$, with any alternative bid $\hat{b}_i$.

Case 1: $m < \theta_i$.

If bidder i bids truthfully, then she wins for sure and pays m . Hence her payoff is

$$\theta_i - m > 0.$$

Now consider any deviation $\hat{b}_i$.

- If $\hat{b}_i > m$, she still wins and still pays m , so her payoff remains

$$\theta_i - m.$$

- If $\hat{b}_i = m$, she ties for the highest bid. Under any tie-breaking rule, her expected payoff is at most

$$\theta_i - m.$$

- If $\hat{b}_i < m$, she loses and gets payoff 0, which is strictly smaller than

$$\theta_i - m.$$

Thus truthful bidding is weakly optimal in this case.

Case 2: $m = \theta_i$.

If bidder i bids truthfully, then her payoff is 0: if she wins, she pays $m = \theta_i$, and if she loses, she also gets 0.

Now consider any deviation $\hat{b}_i$.

- If $\hat{b}_i > m$, she wins for sure and pays $m = \theta_i$, so her payoff is still 0.
- If $\hat{b}_i = m$, she ties for the highest bid, and her expected payoff is again 0.
- If $\hat{b}_i < m$, she loses and gets payoff 0.

Thus truthful bidding is weakly optimal in this case as well.

Case 3: $m > \theta_i$.

If bidder i bids truthfully, she loses and obtains payoff

$$0.$$

Now consider any deviation $\hat{b}_i$.

- If $\hat{b}_i < m$, she still loses and gets payoff 0.
- If $\hat{b}_i = m$, she ties for the highest bid. If she wins the tie, she pays $m > \theta_i$, so her payoff is negative. Hence her expected payoff is weakly smaller than 0.
- If $\hat{b}_i > m$, she wins and pays m , so her payoff is

$$\theta_i - m < 0.$$

Thus truthful bidding is weakly optimal in this case too.

Since for every realization of opponents' bids b_{-i} , bidding $b_i = \theta_i$ yields a payoff weakly greater than any alternative bid $\hat{b}_i$, truthful bidding is a weakly dominant strategy. $\square$

Remark. The logic of the result is simple: in a second-price auction, a bidder's own bid affects whether she wins, but it does not affect the price she pays conditional on winning. Therefore the bidder should use her bid only to decide whether obtaining the object is worthwhile, and the correct cutoff is exactly her valuation θ_i .

Remark. The statement that the result holds "regardless of the distribution over Θ " follows from the fact that the dominance argument is pointwise in opponents' bids. No belief about their types is needed to establish the result.

Remark (Another equilibrium). The truthful bidding profile is not the unique equilibrium of the second-price auction. Because truthful bidding is only a *weakly* dominant strategy, other Bayesian Nash equilibria may also exist.

For example, consider the strategy profile in which bidder 1 always bids

$$b_1(\theta_1) = \bar{\theta} \quad \text{with } \theta_i \in [0, \bar{\theta}],$$

while every other bidder $j \neq 1$ always bids

$$b_j(\theta_j) = 0.$$

Assume types lie in $[0, \bar{\theta}]$. This is also a Bayesian Nash equilibrium.

To see this, first consider bidder 1. Since every other bidder bids 0, any bid $b_1 > 0$ wins and pays the second-highest bid, which is 0. Hence under the proposed strategy bidder 1's payoff is

$$u_1 = \theta_1 \geq 0.$$

Any deviation to another bid $b'_1 > 0$ still wins and still yields payment 0, so it does not improve her payoff. A deviation to $b'_1 = 0$ only changes the tie-breaking outcome and therefore cannot yield a strictly higher payoff.

Now consider any bidder $j \neq 1$. If she deviates to some bid

$$b_j < \bar{\theta},$$

she still loses and gets payoff 0. If she deviates to

$$b_j = \bar{\theta},$$

she ties bidder 1; if she wins the tie, she pays $\bar{\theta}$, so her payoff is

$$\theta_j - \bar{\theta} \leq 0.$$

If she deviates to some bid

$$b_j > \bar{\theta},$$

she wins for sure and pays $\bar{\theta}$, so again her payoff is

$$\theta_j - \bar{\theta} \leq 0.$$

Thus no deviation is profitable.

Therefore this strategy profile is also a Bayesian Nash equilibrium.

Remark. This example shows that the second-price auction may have multiple equilibria, including inefficient ones. The special role of truthful bidding is not that it is the unique equilibrium, but that it is a weakly dominant strategy for every bidder and therefore yields a particularly robust equilibrium prediction.

First-price private values auction

We now turn to the first-price sealed-bid auction under independent private values.

- Each bidder i has type $\theta_i \in [0, \bar{\theta}]$.

- Types are i.i.d. with cumulative distribution function F and density f .
- The bidder with the highest bid wins.
- The winner pays her own bid.

In contrast with the second-price auction, truthful bidding is no longer optimal. A bidder now faces a trade-off: bidding more increases the probability of winning, but also increases the payment conditional on winning. Hence equilibrium bidding must involve *bid shading*.

Symmetric strictly increasing equilibrium

We look for a symmetric and strictly increasing bidding strategy

$$s : [0, \bar{\theta}] \rightarrow [0, \infty).$$

Because s is strictly increasing, it is invertible on its image. Thus if bidder i submits bid b_i , she wins if and only if every rival $j \neq i$ has type

$$\theta_j \leq s^{-1}(b_i).$$

Since types are i.i.d., the probability of winning is

$$\Pr(\text{win} \mid b_i) = (F(s^{-1}(b_i)))^{n-1}.$$

Therefore, if bidder i has type θ_i and submits bid b_i , her expected utility is

$$\bar{u}_i(b_i, \theta_i) = (F(s^{-1}(b_i)))^{n-1}(\theta_i - b_i).$$

If type θ_i follows the candidate equilibrium strategy, then her equilibrium utility is

$$U_i(\theta_i) = \bar{u}_i(b_i, \theta_i) \big|_{b_i=s(\theta_i)} = (F(\theta_i))^{n-1}(\theta_i - s(\theta_i)).$$

Proposition 0.0.26. *In a symmetric strictly increasing equilibrium of the first-price private values auction, the bidding strategy must satisfy*

$$s(\theta) = \theta - \int_0^\theta \left(\frac{F(z)}{F(\theta)} \right)^{n-1} dz.$$

Proof. Fix a bidder of type θ_i . Since $b_i = s(\theta_i)$ is optimal for that type, the first-order condition is

$$\frac{\partial \bar{u}_i(b_i, \theta_i)}{\partial b_i} \bigg|_{b_i=s(\theta_i)} = 0.$$

Now differentiate the equilibrium payoff $U_i(\theta_i)$ with respect to θ_i :

$$\frac{dU_i(\theta_i)}{d\theta_i} = \frac{\partial \bar{u}_i(b_i, \theta_i)}{\partial b_i} \bigg|_{b_i=s(\theta_i)} \frac{ds(\theta_i)}{d\theta_i} + \frac{\partial \bar{u}_i(b_i, \theta_i)}{\partial \theta_i} \bigg|_{b_i=s(\theta_i)}.$$

By the first-order condition, the first term vanishes, so

$$\frac{dU_i(\theta_i)}{d\theta_i} = \frac{\partial \bar{u}_i(b_i, \theta_i)}{\partial \theta_i} \Big|_{b_i=s(\theta_i)}.$$

Since

$$\bar{u}_i(b_i, \theta_i) = (F(s^{-1}(b_i)))^{n-1}(\theta_i - b_i),$$

we have

$$\frac{\partial \bar{u}_i(b_i, \theta_i)}{\partial \theta_i} = (F(s^{-1}(b_i)))^{n-1}.$$

Evaluating at $b_i = s(\theta_i)$, we obtain

$$\frac{dU_i(\theta_i)}{d\theta_i} = (F(\theta_i))^{n-1}.$$

Next, note that a type 0 bidder obtains zero utility in equilibrium:

$$U_i(0) = 0.$$

Hence, integrating from 0 to θ_i ,

$$U_i(\theta_i) = U_i(0) + \int_0^{\theta_i} \frac{dU_i(z)}{dz} dz = \int_0^{\theta_i} (F(z))^{n-1} dz.$$

But we also know that

$$U_i(\theta_i) = (F(\theta_i))^{n-1}(\theta_i - s(\theta_i)).$$

Equating the two expressions for $U_i(\theta_i)$,

$$\int_0^{\theta_i} (F(z))^{n-1} dz = (F(\theta_i))^{n-1}(\theta_i - s(\theta_i)).$$

Rearranging,

$$s(\theta_i) = \theta_i - \frac{\int_0^{\theta_i} (F(z))^{n-1} dz}{(F(\theta_i))^{n-1}} = \theta_i - \int_0^{\theta_i} \left(\frac{F(z)}{F(\theta_i)} \right)^{n-1} dz.$$

Renaming θ_i as θ yields the formula. $\square$

Remark. This formula shows explicitly why first-price bidding differs from second-price bidding. In a second-price auction, a bidder does not directly control the price paid conditional on winning, so truthful bidding is optimal. In a first-price auction, by contrast, increasing one's bid both raises the probability of winning and raises the payment upon winning. The equilibrium bid therefore lies strictly below the valuation: bidders optimally *shade* their bids.

Remark. The proof uses an envelope argument. The key insight is that when we differentiate the equilibrium value $U_i(\theta_i)$, the indirect effect through the optimal bid $s(\theta_i)$ disappears because the equilibrium bid solves the bidder's optimization problem. What remains is only the direct effect of a higher type, namely a higher surplus conditional on winning.

Second-Price Auctions with Common Values

We now consider a second-price sealed-bid auction in a *common values* environment. There are $n \geq 3$ bidders. Each bidder i observes a private signal

$$\theta_i \in [0, \bar{\theta}],$$

with signals drawn i.i.d. from distribution F with density f . The value of the object is common across bidders once all signals are taken into account, and is given by

$$V(\theta) = \frac{1}{n} \sum_{j=1}^n \theta_j.$$

Thus bidder i 's payoff is

$$u_i(b, \theta) = x_i(b)V(\theta) - p_i(b),$$

where $x_i(b) = 1$ if i wins and 0 otherwise, and in a second-price auction the winner pays the second-highest bid.

Why the private-values logic breaks down

Under private values, bidding one's valuation is weakly dominant because winning more often by raising one's bid is harmless: in the additional winning states, the object is still worth at least the price paid. Under common values, this logic fails. In fact, *winning itself is informative*. If you win, this means your signal was relatively high compared with the others', and that fact makes the object less valuable than your ex ante expectation based only on your own signal. This is the *winner's curse*.

The key economic idea is the following. In a second-price auction, a small change in your bid only matters in states in which you are *pivotal*: states in which your bid is approximately tied with the highest bid among the other players. Hence the relevant object for optimal bidding is not your expected value conditional only on θ_i , but rather your expected value conditional on θ_i and on being pivotal.

A natural but incorrect candidate strategy

A natural first guess is that bidder i should bid his interim expected value conditional only on his own signal:

$$s(\theta_i) = \mathbb{E}[V(\theta) \mid \theta_i].$$

Because signals are i.i.d., this equals

$$s(\theta_i) = \frac{1}{n}\theta_i + \frac{n-1}{n} \int_0^{\bar{\theta}} z f(z) dz.$$

This function is strictly increasing in θ_i , so if all bidders use it, the bidder with the highest signal wins.

Claim 1. The strategy profile in which each bidder bids

$$s(\theta_i) = \mathbb{E}[V(\theta) \mid \theta_i]$$

is *not* a Bayesian Nash equilibrium.

Proof. Fix bidder i , and suppose all other bidders use the strategy $s(\cdot)$. Since s is strictly increasing, bidder i wins if and only if his signal is the highest.

Now suppose bidder i lowers his bid by an arbitrarily small amount. This change has no effect except in states in which he is pivotal, namely those states in which his bid is just tied with the highest bid among the other bidders. In all non-pivotal states, either he still wins and pays the same second price, or still loses, so his payoff is unchanged.

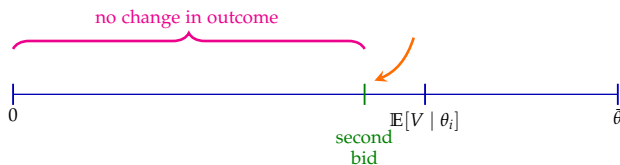
Consider therefore a pivotal state. Since bidding is strictly increasing, pivotality means that one rival's signal is approximately equal to θ_i , while all remaining rivals' signals are below θ_i . Thus, relative to the interim expectation conditional only on θ_i , the event of being pivotal reveals bad news about the value of the object: the remaining $n - 2$ signals are selected from below θ_i . Hence

$$\mathbb{E}[V(\theta) \mid \theta_i, \text{pivotal}] < \mathbb{E}[V(\theta) \mid \theta_i].$$

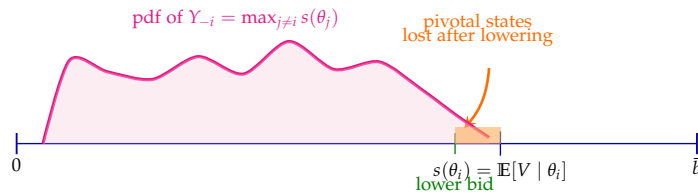
But under the candidate strategy, in a pivotal state bidder i 's price is approximately his own bid, which is precisely $\mathbb{E}[V(\theta) \mid \theta_i]$. Therefore his expected payoff from winning in those pivotal states is strictly negative:

$$\mathbb{E}[V(\theta) \mid \theta_i, \text{pivotal}] - s(\theta_i) < 0.$$

By lowering his bid slightly, bidder i avoids exactly those negative-surplus wins and changes nothing in all non-pivotal states. Hence the deviation is profitable. Therefore bidding the interim expected value cannot be a Bayesian Nash equilibrium. $\square$



The violet curve should be read heuristically as the conditional density of the highest rival bid. A small downward deviation only changes the outcome in the thin set of pivotal states where the second-highest bid lies just below the original bid.



The violet curve as the density of the highest rival bid. Lowering one's bid slightly only affects the thin pivotal region near the cutoff.

Intuition for the deviation

This argument is worth pausing over because it is the whole economics of the winner's curse.

If bidder i slightly lowers his bid, he is not changing the price he pays in the states in which he continues to win, because the price in a second-price auction is determined by the highest other bid. He is only removing himself from the most dangerous set of states: the marginal states in which he barely wins. Those are exactly the states in which winning is worst news about the object's value.

An analogy is useful. Imagine that several doctors observe noisy tests about the severity of a disease, and the treatment is assigned to the doctor reporting the highest severity. If your report barely beats the runner-up, then by construction the rest of the evidence is not especially strong. The event *that you are the winner* contains information; it is not innocuous. Common-value auctions work in exactly this way.

What should the equilibrium bid look like?

The previous claim shows that the equilibrium bid must be shaded below the interim expected value. The right way to think about the equilibrium bid is through the marginal, or pivotal, indifference condition.

Claim 2. In a symmetric strictly increasing equilibrium, bidder i 's bid must equal his expected value conditional on being pivotal, that is, conditional on tying with the highest bidder among the other players.

Proof. Let $b(\theta_i)$ denote a symmetric strictly increasing equilibrium bidding function. In a second-price auction, bidder i 's own bid affects only whether he wins, not the payment he makes in states in which he already wins. Therefore, if bidder i changes his bid by a small amount, the only affected states are the marginal states in which his bid is equal to the highest other bid.

If in those marginal states the conditional expected value of the object exceeded the price, then raising the bid slightly would be profitable, because bidder i would win additional states with strictly positive expected surplus. If instead the conditional expected value were below the price, then lowering the bid slightly would be profitable, because

bidder i would avoid additional states with strictly negative expected surplus.

Hence, at an optimal bid, bidder i must be indifferent in those marginal states:

$$b(\theta_i) = \mathbb{E}[V(\theta) \mid \theta_i, i \text{ is pivotal}].$$

Under symmetry and strict monotonicity, “ i is pivotal” is equivalent to the event that the highest signal among the other bidders is tied with θ_i . Thus the equilibrium bid equals the expected common value conditional on θ_i and on tying with the highest rival. $\square$

A useful heuristic formula

For intuition, it is convenient to write the pivotal-case expectation heuristically. In a continuous environment, if bidder i with signal θ_i is pivotal, then one rival has signal essentially equal to θ_i , while the remaining $n - 2$ rivals have signals drawn from the original distribution truncated below θ_i . Hence the equilibrium bid is heuristically

$$b(\theta_i) = \frac{1}{n} (2\theta_i + (n - 2)\mathbb{E}[Z \mid Z \leq \theta_i]),$$

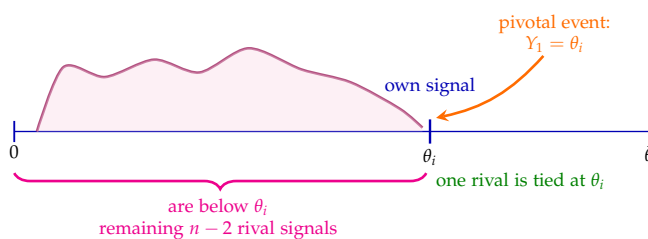
where Z is a generic signal drawn from F . Since

$$\mathbb{E}[Z \mid Z \leq \theta_i] < \mathbb{E}[Z]$$

whenever θ_i is not at the very top of the support, we obtain

$$b(\theta_i) < \mathbb{E}[V(\theta) \mid \theta_i].$$

So the winner’s curse pushes bids downward relative to the naive “bid your expected value” rule.



Heuristic structure of the pivotal event. Conditional on $Y_1 = \theta_i$, one rival signal is tied with bidder i 's signal, while the remaining $n - 2$ rivals have signals below θ_i . This motivates the decomposition $2\theta_i + (n - 2)\mathbb{E}[Z \mid Z \leq \theta_i]$.

Economic takeaway

The lesson is sharp:

In private-values second-price auctions, your own valuation is the right object to bid because winning conveys no additional information about value. In common-values second-price auctions, winning is bad news, so the relevant object is the value conditional on winning at the margin.

Equivalently, the bidder should not ask, “What is the object worth given my signal?” He should ask, “What is the object worth given my signal *and* given that I am just barely the winner?” That second question is the correct one in equilibrium.

Conclusion

We conclude that in second-price auctions with common values, bidding one’s interim expected value is not an equilibrium. The winner’s curse implies that the equilibrium bid must be lower: it must be the expected value of the object in the pivotal case, not the expected value based only on one’s own signal.

Appendix to Class 8: Simple Poker

Consider a simplified poker game with simultaneous bids. First, nature selects numbers x_1 and x_2 , drawn independently from the uniform distribution on $[0, 1]$. Each player i privately observes x_i . The players then simultaneously decide whether to FOLD or BID. If both FOLD, both earn payoffs of -1 . If only one player FOLDS, he obtains -1 while the other player gets 1. If both players BID, each player receives 2 if her number is at least as large as the other player's number; otherwise she gets -2 . Compute a Bayesian Nash equilibrium of this game. We solve the game by looking for a symmetric cutoff equilibrium.

1. Ex-post payoff matrices

Fix a pair of realized types (x_i, x_j) . Ex post, there are two possible cases.

If $x_i \geq x_j$			If $x_i < x_j$		
	F	B		F	B
F	$(-1, -1)$	$(-1, 1)$	F	$(-1, -1)$	$(-1, 1)$
B	$(1, -1)$	$(2, -2)$	B	$(1, -1)$	$(-2, 2)$

Thus, conditional on both players bidding, a higher card is strictly better.

2. Interim payoffs against an arbitrary opponent strategy

Fix player i and type $x_i \in [0, 1]$. Suppose player j uses an arbitrary strategy

$$s_j : [0, 1] \rightarrow \{F, B\}.$$

We now compute player i 's interim expected payoff from each action.

If player i chooses FOLD, then regardless of what player j does,

$$\mathbb{E}[u_i(F, s_j) \mid x_i] = -1.$$

If instead player i chooses BID, then there are three possibilities:

- player j folds, in which case player i gets 1;
- player j bids and has a lower type $x_j \leq x_i$, in which case player i gets 2;
- player j bids and has a higher type $x_j > x_i$, in which case player i gets -2 .

Hence

$$\mathbb{E}[u_i(B, s_j) \mid x_i] = 1 \cdot \Pr(s_j(x_j) = F) + 2 \cdot \Pr(x_j \leq x_i, s_j(x_j) = B) - 2 \cdot \Pr(x_j > x_i, s_j(x_j) = B).$$

Since $x_j \sim U[0, 1]$, this can also be written as

$$\mathbb{E}[u_i(B, s_j) \mid x_i] = \int_{\{x_j: s_j(x_j)=F\}} 1 dx_j + \int_{\{x_j \leq x_i: s_j(x_j)=B\}} 2 dx_j + \int_{\{x_j > x_i: s_j(x_j)=B\}} (-2) dx_j.$$

Key observation. As a function of x_i , the payoff from BID is weakly increasing. Indeed, the term $\Pr(s_j(x_j) = F)$ does not depend on x_i , while a larger x_i increases the set of opponent types against which player i wins when both bid, and decreases the set against which player i loses when both bid.

Therefore, against any strategy of the opponent, one can find a best response of *cutoff form*: there is some threshold $x^* \in [0, 1]$ such that it is optimal to FOLD below x^* and optimal to BID above x^* .

Intuition. This is the point that was being sketched informally in class. The graph was not meant to be an exact plot of the full payoff formula under an arbitrary strategy; rather, it was meant to show that the expected payoff from BID moves up with one's own type, while the payoff from FOLD stays flat at -1 . So the only sensible candidate equilibrium is a cutoff strategy.

3. Guess a symmetric cutoff equilibrium

Motivated by the previous observation, look for a symmetric Bayesian Nash equilibrium in which both players use the same cutoff rule:

$$s(x) = \begin{cases} B, & x \geq x^*, \\ F, & x < x^*. \end{cases}$$

So types below x^* fold, and types above x^* bid.

To verify that this is an equilibrium, we must check that:

1. types $x_i < x^*$ weakly prefer FOLD;
2. types $x_i \geq x^*$ weakly prefer BID.

4. Expected payoff from bidding when the opponent uses cutoff x^*

Assume player j uses the cutoff strategy above.

Case A: $x_i < x^$.* If player i bids, then whenever player j bids we must have $x_j \geq x^* > x_i$, so player i loses the showdown. Thus

$$\mathbb{E}[u_i(B, s) \mid x_i] = 1 \cdot \Pr(x_j < x^*) - 2 \cdot \Pr(x_j \geq x^*).$$

Since $x_j \sim U[0, 1]$,

$$\mathbb{E}[u_i(B, s) \mid x_i] = x^* - 2(1 - x^*) = 3x^* - 2.$$

Since folding gives -1 , low types prefer to fold provided

$$3x^* - 2 \leq -1 \quad \Longleftrightarrow \quad x^* \leq \frac{1}{3}.$$

Case B: $x_i \geq x^*$. If player i bids, there are now three regions for the opponent's type:

- if $x_j < x^*$, the opponent folds and player i gets 1;
- if $x^* \leq x_j \leq x_i$, the opponent bids and player i wins the showdown, obtaining 2;
- if $x_j > x_i$, the opponent bids and player i loses the showdown, obtaining -2 .

Therefore

$$\mathbb{E}[u_i(B, s) \mid x_i] = 1 \cdot \Pr(x_j < x^*) + 2 \cdot \Pr(x^* \leq x_j \leq x_i) - 2 \cdot \Pr(x_j > x_i).$$

Using uniformity,

$$\mathbb{E}[u_i(B, s) \mid x_i] = x^* + 2(x_i - x^*) - 2(1 - x_i) = 4x_i - x^* - 2.$$

Since folding gives -1 , high types prefer to bid provided

$$4x_i - x^* - 2 \geq -1 \quad \Longleftrightarrow \quad 4x_i - 1 \geq x^*.$$

Because the left-hand side is increasing in x_i , it is enough to check the lowest bidding type, namely $x_i = x^*$. This gives

$$4x^* - 1 \geq x^* \quad \Longleftrightarrow \quad 3x^* \geq 1 \quad \Longleftrightarrow \quad x^* \geq \frac{1}{3}.$$

Combining the two inequalities, we obtain

$$x^* = \frac{1}{3}.$$

5. Equilibrium

We conclude that a symmetric Bayesian Nash equilibrium is

$$s^*(x) = \begin{cases} F, & x < \frac{1}{3}, \\ B, & x \geq \frac{1}{3}. \end{cases}$$

Equivalently: each player bids if and only if her private draw is at least $1/3$.

Remark on the threshold type. At $x_i = \frac{1}{3}$, player i is indifferent between FOLD and BID. Thus the action at the cutoff can be assigned either way without affecting the Bayesian Nash equilibrium. Following the convention used in class, we let the threshold type bid.

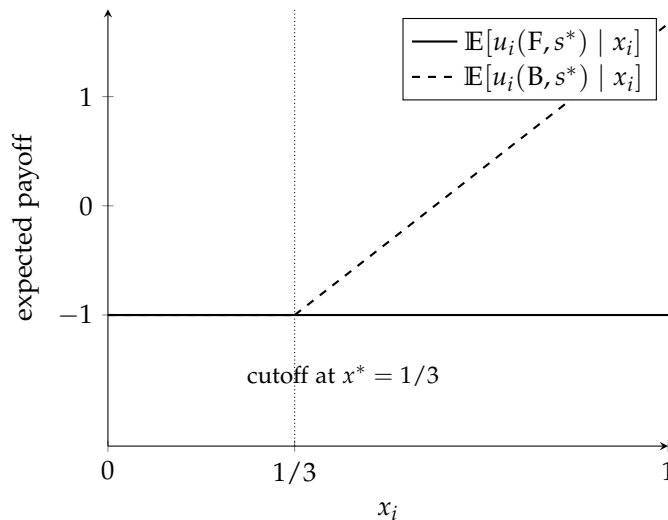
6. Graphical intuition

Compare this with Bruno's because it is different but I don't understand Bruno's one.

Under the equilibrium cutoff $x^* = \frac{1}{3}$, the payoff from folding is always -1 , while the payoff from bidding is

$$\mathbb{E}[u_i(B, s^*) \mid x_i] = \begin{cases} -1, & x_i < \frac{1}{3}, \\ 4x_i - \frac{7}{3}, & x_i \geq \frac{1}{3}. \end{cases}$$

So bidding yields exactly the same payoff as folding below the cutoff, and a strictly higher payoff above the cutoff.



Expected payoff from FOLD and BID when the opponent uses the equilibrium cutoff strategy.

Answer to the class comment. In the handwritten notes there was a question about why the graph seemed not to display the term $\Pr(F)$ inside the expected payoff from bidding. The answer is: that term *is* there. Under a cutoff strategy it is simply the constant x^* , so in the equilibrium calculation it gets absorbed into the linear expression above. The sketch in class was only trying to capture the shape of the payoff comparison, not every constant term separately.

Final intuition. The game has a threshold structure because folding always gives the same sure payoff, -1 , while the value of bidding rises with the strength of one's private signal. A better card never makes

bidding worse. Hence the natural equilibrium has a cutoff form, and the equilibrium cutoff is exactly $1/3$.

Class 9: Perfect Bayesian Equilibrium

Why Nash equilibrium and subgame perfection are not enough

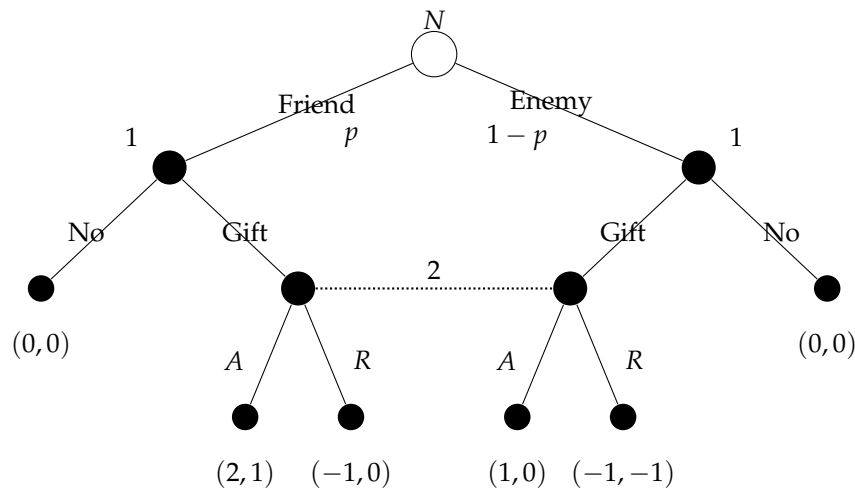
Nash equilibrium is a normal-form solution concept. Because of this, it places no discipline on what players plan to do at histories that are not reached in equilibrium. In dynamic environments, this is a serious limitation: an equilibrium may rely on behavior that is not credible once the relevant information set is actually reached.

Subgame perfection improves on Nash equilibrium by requiring equilibrium behavior in every subgame. But in games with private information, there are often very few proper subgames. The reason is conceptual and important: at the initial node of a subgame, the continuation game must be common knowledge, whereas private information precisely destroys that common knowledge requirement. Thus, in games with nontrivial information sets, subgame perfection often loses much of its force.

Perfect Bayesian equilibrium is designed to recover the core economic idea we still want to impose: at every information set, a player's behavior should be optimal *given her beliefs*, and those beliefs should be *correct whenever Bayes' rule can discipline them*.

The key pathology is immediate. If Player 1 chooses *No* regardless of type, Player 2's information set is never reached. Then any action by Player 2 is a best response in the normal-form sense, simply because it is never used. But if that information set were actually reached, Player 2 would strictly prefer to accept rather than reject. Thus Nash equilibrium, and even subgame perfection, may allow behavior that is not sequentially rational.

Remark. The hidden lesson is this: once information sets matter, we must evaluate actions *locally*, at the point where they are actually chosen. That is exactly what PBE is built to do.



Gift Game: a Nash equilibrium may fail sequential rationality.

Assessments

In games of imperfect information, a strategy profile alone is not enough to describe equilibrium behavior. At any information set, a player must choose an action without necessarily knowing which node in that information set has actually been reached. Thus, to evaluate whether her action is optimal, we must specify *not only what strategies are played*, but also *what she believes about where she is*.

Definition (Assessment). An *assessment* is a pair (β, μ) , where:

1. β is an independent behavior strategy profile;
2. μ is a belief system: for each information set I , $\mu(\cdot | I)$ is a probability measure over the nodes in I .

Given an information set I and a subset of nodes $x \subseteq I$, let:

- H_x denote the set of feasible paths that reach node x ;
- H_I denote the set of feasible paths that reach information set I .

The behavior strategy profile β induces a probability measure over feasible paths, denoted ϕ_β .

Definition (The path measure induced by β). Let Ω be the set of feasible paths of the game. Given an independent behavior strategy profile β , define

$$\phi_\beta : 2^\Omega \rightarrow [0, 1]$$

as the probability measure induced by β on feasible paths. For each path $h \in \Omega$, $\phi_\beta(\{h\})$ is the product of the probabilities of all actions

taken along h under β (and Nature's probabilities, if applicable). For any subset $H \subseteq \Omega$,

$$\phi_\beta(H) = \sum_{h \in H} \phi_\beta(\{h\}).$$

Hence:

$$\phi_\beta(H)$$

is the probability of a set of feasible paths H , and

$$\phi_\beta(H_x \mid H_I)$$

is the conditional probability that node x is reached, given that information set I is reached.

Remark. In a finite game, ϕ_β is simply the probability distribution over terminal histories generated by the behavior strategy profile β . Thus $\phi_\beta(H_x \mid H_I)$ is just the conditional probability that the true node is x , given that information set I has been reached.

Remark. This notation looks heavier than it really is. The object ϕ_β is simply the probability distribution over complete paths generated by the behavior strategy profile β . So when we write $\phi_\beta(H_x \mid H_I)$, we are just asking: conditional on having arrived at information set I , how likely is it that the true node is x ?

This definition is conceptually essential. A player's optimal action at an information set depends on two objects:

1. what she thinks has already happened inside that information set;
2. what she expects everyone else to do from that point onward.

The first object is encoded by μ ; the second by β .

Remark. An assessment is therefore the minimal object needed to speak meaningfully about equilibrium in dynamic games with imperfect information. Strategies alone tell us what players do. Beliefs tell us why those actions are rational from the player's point of view.

How this appears in the Gift Game

The notation becomes much simpler in the Gift Game. Player 2 has a single nontrivial information set after observing *Gift*. That information set (called I) contains exactly two nodes: one in which Player 1 is a Friend, and one in which Player 1 is an Enemy. Therefore, instead of writing the whole belief system abstractly, we can summarize Player 2's belief by a single number:

$$q \equiv \Pr(\text{Friend} \mid \text{Gift}).$$

Equivalently,

$$1 - q = \Pr(\text{Enemy} \mid \text{Gift}).$$

So in this example, the formal object $\mu(\cdot \mid I)$ is represented by the pair $(q, 1 - q)$. If x_F denotes the Friend node in Player 2's information set and x_E the Enemy node, then:

$$\mu(x_F \mid I) = \phi_\beta(H_{x_F} \mid H_I) = q, \quad \mu(x_E \mid I) = \phi_\beta(H_{x_E} \mid H_I) = 1 - q.$$

This is a useful bridge between the abstract notation and the concrete game. The abstract notation says that beliefs are probabilities over nodes in an information set. The Gift Game shows what that means in practice: Player 2 is simply asking herself,

conditional on having received a gift, how likely is it that I am facing a Friend rather than an Enemy?

Weak Perfect Bayesian Equilibrium

The idea is simple, even if the notation initially looks heavy: at each information set, beliefs must be correct whenever Bayes' rule applies, and actions must be optimal given those beliefs and the continuation strategies of the other players.

Definition (Weak Perfect Bayesian Equilibrium). An assessment (β, μ) is a *weak perfect Bayesian equilibrium* (wPBE) if, at each information set I , the following conditions hold:

1. **Bayesian updating on the equilibrium path.** If

$$\phi_\beta(H_I) > 0,$$

then for each node $x \in I$,

$$\mu(x \mid I) = \phi_\beta(H_x \mid H_I).$$

2. **Sequential rationality.** If player i moves at I , then

$$\beta_i(\cdot \mid H_I) \in \arg \max_{\hat{\beta}_i(\cdot \mid H_I)} \int_{x \in I} \int_{h \in H_x} u_i(h) d\phi_{\hat{\beta}_i, \beta_{-i}}(h \mid H_x) d\mu(x \mid I).$$

The first condition requires beliefs to be obtained from β by Bayes' rule whenever the information set is reached with positive probability under the equilibrium behavior strategy profile. The second condition requires the player moving at I to choose a continuation behavior strategy that maximizes expected continuation payoff, taking expectations both over the possible nodes in I and over the continuation play that may follow from each such node.

As we will see below, in Gift Game #1, this belief does not actually affect Player 2's optimal action, because she strictly prefers A to R for every value of q . In Gift Game #2, by contrast, the optimal action does depend on q , and that is exactly why beliefs become economically important there. So the notation (β, μ) is not decorative: it is what allows us to study situations in which optimal continuation behavior depends on what players infer from the path of play.

Remark. Sequential rationality involves two layers of uncertainty.

First, the player may be uncertain about which node in the information set has actually been reached. This is captured by the outer integral with respect to $\mu(\cdot | I)$.

Second, even conditional on a given node $x \in I$, continuation play may still be stochastic because behavior strategies may randomize. This is captured by the inner integral with respect to $\phi_{\hat{\beta}_i, \beta_{-i}}(\cdot | H_x)$.

In finite games, the same condition can be written more transparently in discrete form. If player i moves at I , sequential rationality is equivalent to

$$\beta_i(\cdot | H_I) \in \arg \max_{\hat{\beta}_i(\cdot | H_I)} \sum_{x \in I} \mu(x | I) V_i(\hat{\beta}_i, \beta_{-i}; x),$$

where

$$V_i(\hat{\beta}_i, \beta_{-i}; x) = \sum_{h \in H_x} u_i(h) \phi_{\hat{\beta}_i, \beta_{-i}}(h | H_x).$$

Remark. Thus sequential rationality says that, given her posterior belief over the nodes in I , the player chooses the continuation plan that maximizes expected continuation value:

$$\sum_{x \in I} \underbrace{\mu(x | I)}_{\text{belief that } x \text{ is the true node}} \cdot \underbrace{V_i(\hat{\beta}_i, \beta_{-i}; x)}_{\text{continuation value if } x \text{ is true}}.$$

Remark (Why the continuous and discrete formulations coincide). In a finite game, the conditional measure $\phi_{\hat{\beta}_i, \beta_{-i}}(\cdot | H_x)$ assigns probabilities to finitely many feasible continuation paths. Therefore the inner integral is simply a finite sum:

$$\int_{h \in H_x} u_i(h) d\phi_{\hat{\beta}_i, \beta_{-i}}(h | H_x) = \sum_{h \in H_x} u_i(h) \phi_{\hat{\beta}_i, \beta_{-i}}(h | H_x).$$

So the discrete formulation is not a different condition; it is exactly the same sequential rationality requirement written in finite-game notation.

How this works in the Gift Game

Player 2 has one nontrivial information set, reached after observing *Gift*. That information set contains two nodes: one in which Player 1 is a Friend and one in which Player 1 is an Enemy. Hence her belief system at that information set can be summarized by

$$q \equiv \Pr(\text{Friend} | \text{Gift}), \quad 1 - q \equiv \Pr(\text{Enemy} | \text{Gift}).$$

Formally, if we denote the two nodes in Player 2's information set by x_F and x_E , then

$$\mu(x_F | I) = q, \quad \mu(x_E | I) = 1 - q.$$

And when the information set is reached on the equilibrium path, wPBE requires these beliefs to satisfy

$$\mu(x_F | I) = \phi_\beta(H_{x_F} | H_I), \quad \mu(x_E | I) = \phi_\beta(H_{x_E} | H_I).$$

Remark. In the abstract definition, beliefs are probability measures over nodes in an information set. In the Gift Game, those nodes correspond to the two possible types of Player 1 conditional on the event that Player 2 received a gift. Thus the formal belief system μ becomes the scalar posterior q .

Why only on-path Bayes?

A crucial subtlety is that Bayes' rule only constrains beliefs at information sets reached with positive probability under β . If an information set is off the equilibrium path, then the conditioning event has probability zero, so Bayes' rule does not pin down a unique posterior.

That is precisely where the “weak” in weak perfect Bayesian equilibrium comes from. Off the equilibrium path, beliefs need only be well-defined enough to make sense of sequential rationality; they need not yet satisfy stronger consistency requirements. This flexibility is useful, but it also creates room for problematic off-path beliefs. That is why later refinements, such as sequential equilibrium, impose additional discipline on beliefs outside the equilibrium path.

Remark. A good mnemonic is:

- On path: Bayes + best responses.
- Off path: best responses, but Bayes may be silent.

It is informal, but it captures the concept very well.

Application I: The Gift Game

Let

$$q \equiv \Pr(\text{Friend} | \text{Gift})$$

denote Player 2's posterior belief at her information set.

Player 2's payoff from accepting is

$$U_2(A | q) = q \cdot 1 + (1 - q) \cdot 0 = q,$$

while her payoff from rejecting is

$$U_2(R | q) = q \cdot 0 + (1 - q) \cdot (-1) = -(1 - q).$$

Hence

$$U_2(A | q) - U_2(R | q) = q + (1 - q) = 1 > 0.$$

So Player 2 strictly prefers A for *every* belief $q \in [0, 1]$.

Proposition 0.0.27. *In the Gift Game, there is a unique wPBE. It is given by:*

$$\beta_1(\text{Friend}) = \text{Gift}, \quad \beta_1(\text{Enemy}) = \text{Gift}, \quad \beta_2 = A,$$

with belief

$$\mu(\text{Friend} \mid \text{Gift}) = p.$$

Proof. Since Player 2 strictly prefers A to R no matter what she believes, sequential rationality implies that she must choose A .

Given that Player 2 chooses A , the Friend type of Player 1 compares

$$u_1(\text{Gift} \mid \text{Friend}) = 2 \quad \text{to} \quad u_1(\text{No} \mid \text{Friend}) = 0,$$

so the Friend type strictly prefers Gift. Likewise, the Enemy type compares

$$u_1(\text{Gift} \mid \text{Enemy}) = 1 \quad \text{to} \quad u_1(\text{No} \mid \text{Enemy}) = 0,$$

so the Enemy type also strictly prefers Gift.

Thus both types choose Gift, and Player 2's information set is reached with probability one. Therefore Bayes' rule applies, and because both types pool on Gift, the action Gift conveys no information about Player 1's type. Hence Player 2's posterior equals the prior:

$$\Pr(\text{Friend} \mid \text{Gift}) = p.$$

This gives a wPBE, and uniqueness follows from the strict inequalities above. $\square$

Remark (The hidden detail that makes this example easy). The reason the example collapses so cleanly is that Player 2's decision is *belief-independent*: she wants to accept regardless of the posterior. Once one player's action is belief-independent, her behavior becomes pinned down immediately, and the rest of the equilibrium follows by simple best-response logic.

Application II: The Modified Gift Game

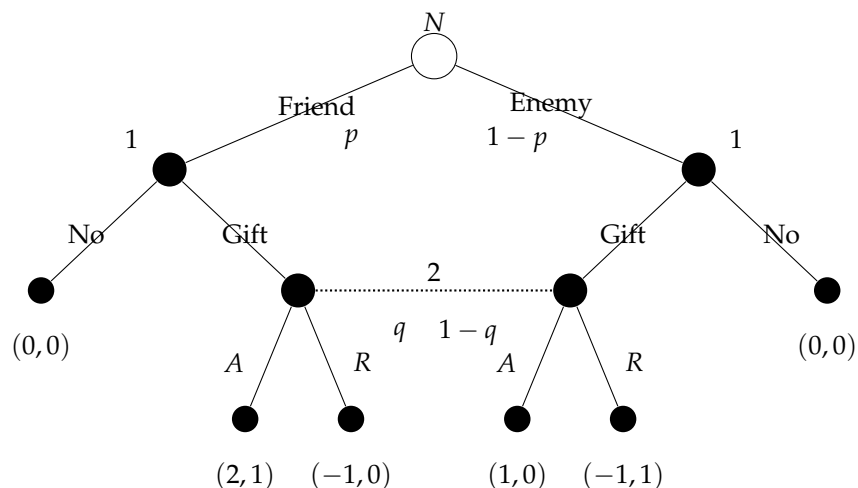
Now modify the game by changing Player 2's payoff from rejecting a gift from an Enemy from -1 to 1 .

Now Player 2's action depends on beliefs. If

$$q \equiv \Pr(\text{Friend} \mid \text{Gift}),$$

then

$$U_2(A \mid q) = q \cdot 1 + (1 - q) \cdot 0 = q, \quad U_2(R \mid q) = q \cdot 0 + (1 - q) \cdot 1 = 1 - q.$$



Modified Gift Game.

Therefore:

$$A \text{ is sequentially rational} \iff q \geq \frac{1}{2},$$

$$R \text{ is sequentially rational} \iff q \leq \frac{1}{2},$$

and mixing is possible only if

$$q = \frac{1}{2}.$$

Case 1: Player 2 accepts

Suppose Player 2 chooses A . Then both types of Player 1 prefer Gift to No, so both types pool on Gift. Since the information set is reached on path, Bayes' rule requires

$$q = p.$$

Hence acceptance is sequentially rational if and only if

$$p \geq \frac{1}{2}.$$

Case 2: Player 2 rejects

Suppose Player 2 chooses R . Then the Friend type gets -1 from Gift and 0 from No, while the Enemy type also gets -1 from Gift and 0 from No. Thus both types choose No. Player 2's information set is then off path, so Bayes' rule imposes no restriction there. Any belief satisfying

$$q \leq \frac{1}{2}$$

supports rejection as sequentially rational.

Case 3: Player 2 mixes

Suppose Player 2 rejects with probability r and accepts with probability $1 - r$. Then the expected payoff to the Friend type from Gift is

$$2(1 - r) + (-1)r = 2 - 3r,$$

and to the Enemy type it is

$$1(1 - r) + (-1)r = 1 - 2r.$$

Since No gives payoff 0, the incentives of the two types depend on r .

Proposition 0.0.28. *In the Modified Gift Game:*

1. *There is a pooling-on-Gift wPBE with Player 2 accepting if and only if*

$$p \geq \frac{1}{2}.$$

Proof. For this candidate equilibrium, let both types of Player 1 choose Gift and let Player 2 choose A. This is optimal for Player 1 because, if Player 2 accepts, the Friend type gets 2 from Gift instead of 0 from No, and the Enemy type gets 1 from Gift instead of 0 from No. Thus both types strictly prefer to give the gift, so pooling on Gift is indeed optimal.

Since Gift is observed on the equilibrium path, Bayes' rule pins down Player 2's belief at

$$q = \Pr(\text{Friend} \mid \text{Gift}) = p.$$

Player 2 accepts if and only if acceptance is sequentially rational, that is, if and only if $q \geq \frac{1}{2}$. Hence this pooling-on-Gift profile is a wPBE if and only if

$$p \geq \frac{1}{2}.$$

Intuitively, once both types want to give the gift, the only remaining issue is whether receiving a gift makes Player 2 sufficiently optimistic that the sender is a Friend rather than an Enemy; that requires exactly $p \geq \frac{1}{2}$. $\square$

2. *There is always a pooling-on-No wPBE with Player 2 rejecting, supported by any off-path belief satisfying*

$$q \leq \frac{1}{2}.$$

Proof. Consider the candidate profile in which both types of Player 1 choose No and Player 2 chooses R if reached. Given rejection, choosing Gift yields payoff -1 for both the Friend and the Enemy type, whereas choosing No yields 0. Hence both types strictly prefer No, so pooling on No is optimal for Player 1.

Because Gift is never observed on the equilibrium path, Player 2's information set is off path, so Bayes' rule does not restrict the belief $q = \Pr(\text{Friend} \mid \text{Gift})$. Sequential rationality only requires that, at that off-path information set, rejection be optimal. Since

$$U_2(R \mid q) = 1 - q \quad \text{and} \quad U_2(A \mid q) = q,$$

Player 2 prefers to reject if and only if $q \leq \frac{1}{2}$. Therefore, for any off-path belief satisfying

$$q \leq \frac{1}{2},$$

there is a pooling-on-No wPBE with Player 2 rejecting. Intuitively, once rejection makes a gift unprofitable for both types, no one gives; and because no gift is ever observed, beliefs after a gift can be chosen freely as long as they make rejection optimal. $\square$

3. If Player 2 mixes, then:

- (a) if $r \in [\frac{2}{3}, 1)$, both types of Player 1 can optimally choose No, and because the information set is off path Player 2 may hold the belief $q = \frac{1}{2}$ and mix;
- (b) if $r \in (\frac{1}{2}, \frac{2}{3}]$, no wPBE exists of this form, because only the Friend type is willing to give a gift, implying $q = 1$, which contradicts the indifference condition $q = \frac{1}{2}$;
- (c) if $r = \frac{1}{2}$, the Friend type strictly prefers Gift, the Enemy type is indifferent, and a mixed-type wPBE exists when the Enemy type gives a gift with probability

$$g = \frac{p}{1-p},$$

which requires

$$p \leq \frac{1}{2};$$

- (d) if $r < \frac{1}{2}$, both types strictly prefer Gift, so Bayes' rule gives $q = p$ and Player 2 can mix only if

$$p = \frac{1}{2}.$$

Proof. If Player 2 mixes, she must be indifferent between accepting and rejecting, so necessarily

$$q = \frac{1}{2}.$$

Given that Player 2 rejects with probability r , the expected payoff from Gift is

$$u_1^F(\text{Gift}) = 2(1-r) + (-1)r = 2-3r$$

for the Friend type, and

$$u_1^E(\text{Gift}) = 1(1-r) + (-1)r = 1-2r$$

for the Enemy type. In contrast, No always yields payoff 0. Thus the analysis reduces to comparing $2-3r$ and $1-2r$ with zero.

If $r \in [\frac{2}{3}, 1)$, then $2-3r \leq 0$ and $1-2r \leq 0$, so both types can optimally choose No. Hence Gift is off path, Bayes' rule imposes no restriction on q , and we may assign the off-path belief $q = \frac{1}{2}$, which makes Player 2 indifferent and therefore willing to mix. If instead $r \in (\frac{1}{2}, \frac{2}{3})$, then $2-3r > 0$ while $1-2r < 0$: the Friend type strictly prefers Gift, but the Enemy type strictly prefers No. Therefore only the Friend type gives a gift on path, so Bayes' rule implies $q = 1$, contradicting the indifference condition $q = \frac{1}{2}$. Hence no wPBE exists in this region.

If $r = \frac{1}{2}$, then

$$u_1^F(\text{Gift}) = \frac{1}{2} > 0, \quad u_1^E(\text{Gift}) = 0,$$

so the Friend type strictly prefers Gift and the Enemy type is indifferent. Let the Enemy type give a gift with probability g . Then along the equilibrium path Bayes' rule gives

$$q = \Pr(\text{Friend} \mid \text{Gift}) = \frac{p}{p + (1-p)g}.$$

For Player 2 to mix we need $q = \frac{1}{2}$, so

$$\frac{p}{p + (1-p)g} = \frac{1}{2} \implies g = \frac{p}{1-p}.$$

This is feasible if and only if $g \leq 1$, that is, if and only if

$$p \leq \frac{1}{2}.$$

Finally, if $r < \frac{1}{2}$, then $2 - 3r > 0$ and $1 - 2r > 0$, so both types strictly prefer Gift. Thus both types give the gift on path, and Bayes' rule implies

$$q = \Pr(\text{Friend} \mid \text{Gift}) = p.$$

Since Player 2 mixes only if $q = \frac{1}{2}$, we must have

$$p = \frac{1}{2}.$$

This proves all four cases. $\square$

Remark (Why this example is much richer). The original Gift Game was easy because Player 2's action did not depend on beliefs. Here beliefs matter directly. Once that happens, equilibrium analysis becomes a feedback loop:

beliefs $\rightarrow$ optimal action $\rightarrow$ type incentives $\rightarrow$ on-path behavior $\rightarrow$
Bayes' rule $\rightarrow$ beliefs.

That loop is the whole logic of PBE.

Remark (A very useful exam heuristic). Whenever you solve a PBE problem, force yourself to separate three steps:

1. **Fix beliefs** and solve for sequentially rational actions.
2. **Given those actions**, solve the incentive problem of earlier movers or types.
3. **Only then** impose Bayes' rule wherever the information set is reached on path.

Students often try to do all three at once. That is exactly when mistakes appear.

Why PBE is still not the final word

wPBE is stronger than Nash equilibrium, but it is still not strong enough. The core problem is off-path beliefs.

A first repair, often used informally in class, is to define *perfect Bayesian equilibrium* as a refinement of wPBE that also imposes equilibrium discipline in every proper subgame. This fixes some pathologies, but not all of them.

Two important limitations remain:

1. **Incorrect off-path beliefs.** A player may assign high probability to nodes that should be essentially impossible once one uses the known strategies of the other players.
2. **Violations of strategic independence.** A player may hold beliefs that effectively attribute to others a correlation in their deviations that the extensive form does not justify.

The economic message is subtle but important. Bayes' rule alone does not tell us how to think about zero-probability events. But equilibrium analysis still wants players to share some coherent theory of how they got there. PBE does not always impose enough coherence.

Class 10: Applications of $wPBE$

Applications of weak Perfect Bayesian Equilibrium

Having defined weak Perfect Bayesian Equilibrium, we now study two canonical applications. The first is a signaling problem, where an informed player moves first and an uninformed player updates beliefs from the observed action. The second is a simple reputation problem, where a privately informed long-run player may mimic a “good” type in order to affect the future behavior of an uninformed opponent.

In both examples the discipline is the same:

1. specify an *assessment*, i.e. a strategy profile together with beliefs at each information set;
2. impose *sequential rationality* at every information set;
3. impose *Bayesian updating* whenever the information set is reached with positive probability;
4. check whether any player has a profitable deviation.

The examples are useful because they show two different roles for beliefs. In the signaling example, off-path beliefs are crucial and generate multiplicity. In the reputation example, the relevant information set is reached on the equilibrium path, so Bayes’ rule bites sharply and pins down the posterior belief that supports the equilibrium.

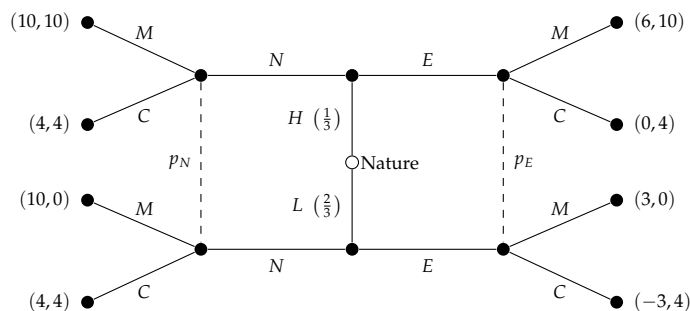
Simple signaling

The game

Nature draws the sender’s type:

$$\Pr(H) = \frac{1}{3}, \quad \Pr(L) = \frac{2}{3}.$$

The sender then chooses a signal, either N or E . The receiver observes the signal but not the type, and then chooses between actions M and C .



Simple signaling. The first payoff is the sender's, the second is the receiver's.

Step 1: the receiver's problem

Let $p_N = \Pr(H \mid N)$ and $p_E = \Pr(H \mid E)$. After observing either signal, the receiver compares M and C.

After N, the receiver's expected payoff is

$$u_2(M \mid N) = 10p_N, \quad u_2(C \mid N) = 4.$$

Hence,

$$M \text{ is optimal after } N \iff 10p_N \geq 4 \iff p_N \geq \frac{2}{5}.$$

After E, the receiver's expected payoff is

$$u_2(M \mid E) = 10p_E, \quad u_2(C \mid E) = 4,$$

so

$$M \text{ is optimal after } E \iff p_E \geq \frac{2}{5}.$$

This is already a useful observation. The receiver uses the *same cutoff* after both signals. Therefore the signals matter only through the beliefs they induce.

Step 2: sender incentives

Now fix the receiver's continuation actions and look at the sender's incentives.

Type H.

	M	C
N	10	4
E	6	0

Type H prefers N whenever the receiver reacts in the same way after both signals, but if the receiver chooses C after N and M after E, then type H prefers E because $6 > 4$.

Type L.

	M	C
N	10	4
E	3	-3

Type L has a stronger preference for N. In particular, if the receiver chooses C after N and M after E, then type L still prefers N, because $4 > 3$.

This is the key asymmetry in the model: signal E is *less costly* for the high type than for the low type. That is what makes separation possible.

Separating equilibrium

Consider the candidate assessment

$$H \mapsto E, \quad L \mapsto N,$$

with receiver actions

$$C \text{ after } N, \quad M \text{ after } E.$$

If this strategy profile is played, then Bayes' rule gives

$$p_N = \Pr(H \mid N) = 0, \quad p_E = \Pr(H \mid E) = 1.$$

Given these beliefs, the receiver's actions are sequentially rational:

$$p_N = 0 < \frac{2}{5} \implies C \text{ after } N, \quad p_E = 1 > \frac{2}{5} \implies M \text{ after } E.$$

Now check sender deviations:

$$u_1(H; E, M) = 6 > 4 = u_1(H; N, C),$$

so type H does not want to deviate, and

$$u_1(L; N, C) = 4 > 3 = u_1(L; E, M),$$

so type L does not want to deviate either.

Therefore there is a separating wPBE:

$H \rightarrow E, \quad L \rightarrow N; \quad C \text{ after } N, \quad M \text{ after } E; \quad p_N = 0, \quad p_E = 1.$

Pooling on N

Now consider pooling on N: both types choose N.

Then the receiver's posterior after N is just the prior:

$$p_N = \Pr(H \mid N) = \frac{1}{3}.$$

Since

$$\frac{1}{3} < \frac{2}{5},$$

the receiver chooses C after N .

So along the equilibrium path, both types of sender get payoff 4.

What about the off-path signal E ? Since E is never observed in this candidate equilibrium, wPBE imposes no Bayesian restriction on p_E . Thus, if we choose off-path beliefs satisfying

$$p_E \leq \frac{2}{5},$$

the receiver optimally chooses C after E as well.

Given that continuation, neither type wants to deviate:

$$u_1(H; N, C) = 4 > 0 = u_1(H; E, C),$$

$$u_1(L; N, C) = 4 > -3 = u_1(L; E, C).$$

Therefore there is also a pooling wPBE on N :

$H \rightarrow N, \quad L \rightarrow N; \quad C \text{ after } N, \quad C \text{ after } E; \quad p_N = \frac{1}{3}, \quad p_E \leq \frac{2}{5}.$
--

Pooling on E is impossible

Suppose both types choose E . Then

$$p_E = \frac{1}{3} < \frac{2}{5},$$

so the receiver must choose C after E . Then sender payoffs on the equilibrium path are

$$u_1(H) = 0, \quad u_1(L) = -3.$$

But then type H would deviate to N if N induced C , because $4 > 0$, and also if N induced M , because $10 > 0$. Type L would deviate as well since both 4 and 10 exceed -3 . Hence pooling on E cannot be supported.

What this example teaches

This example is extremely instructive.

First, wPBE does not necessarily select a unique outcome. Here we have both a separating equilibrium and a pooling equilibrium on N . The multiplicity comes entirely from the freedom to assign off-path beliefs after E .

Second, the separating equilibrium is sustained by a *differential cost of signaling*. Signal E is not attractive for the low type when the receiver rewards it with M , but it is attractive for the high type.

Third, the pooling equilibrium on N reminds us that wPBE is a weak refinement. It rules out sequentially irrational behavior, but it still permits pessimistic off-path beliefs that can block deviation and sustain pooling.

Class 11: Reputation Game and Perfect Bayesian Equilibrium (PBE)

Simple reputation

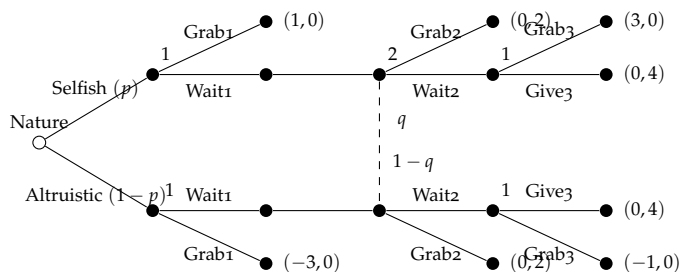
The game

Player 1 may be one of two types:

$$\Pr(\text{Selfish}) = p, \quad \Pr(\text{Altruistic}) = 1 - p, \quad p > \frac{1}{2}.$$

Nature selects the type and reveals it to Player 1, but not to Player 2.

Player 1 moves first. If she waits, then Player 2 moves at an information set where he knows only that Player 1 waited, not her type. If Player 2 also waits, then the game continues to a final move by Player 1. At that last move, the selfish type wants to grab, whereas the altruistic type wants to give.



Simple reputation with $p > \frac{1}{2}$. Here $q = \Pr(\text{Selfish} \mid \text{Player 2 moves})$.

Step 1: solve the last move

At the last move, the selfish type compares

$$3 \text{ vs. } 0,$$

so she chooses Grab3.

The altruistic type compares

$$0 \text{ vs. } -1,$$

so she chooses Give₃.

Thus the continuation after Wait₂ is fully pinned down:

$$\text{Selfish} \Rightarrow \text{Grab}_3, \quad \text{Altruistic} \Rightarrow \text{Give}_3.$$

Step 2: Player 2's problem

Let

$$q = \Pr(\text{Selfish} \mid \text{Player 2 moves}).$$

If Player 2 chooses Grab₂, his payoff is always 2.

If instead he chooses Wait₂, then he gets

$$0 \quad \text{if type is Selfish}, \quad 4 \quad \text{if type is Altruistic}.$$

Hence

$$u_2(\text{Wait}_2) = 4(1 - q).$$

Therefore Player 2 chooses:

$$\text{Wait}_2 \iff 4(1 - q) \geq 2 \iff q \leq \frac{1}{2},$$

$$\text{Grab}_2 \iff q \geq \frac{1}{2},$$

and he is indifferent exactly when $q = \frac{1}{2}$.

So the reputation problem is already transparent: Player 2 waits only if, conditional on reaching his information set, he thinks it is sufficiently likely that he is facing the altruistic type.

Step 3: Player 1's first move

The altruistic type prefers Wait₁ to Grab₁ no matter what:

$$0 > -3.$$

So the altruistic type never grabs at the first opportunity.

The selfish type compares:

$$u_1(\text{Grab}_1) = 1$$

with the payoff from waiting.

If Player 2 grabs after observing Wait₁, then waiting yields 0 to the selfish type.

If Player 2 waits, then the selfish type grabs at the last move and gets 3.

Thus, if Player 2 chooses Grab₂ with probability λ , the selfish type's expected payoff from Wait₁ is

$$u_1(\text{Wait}_1 \mid \text{Selfish}) = 0 \cdot \lambda + 3(1 - \lambda) = 3(1 - \lambda).$$

So the selfish type prefers to wait iff

$$3(1 - \lambda) \geq 1 \iff \lambda \leq \frac{2}{3}.$$

No pure equilibrium

It is useful to see first why no pure wPBE exists.

Case 1: suppose the selfish type always waits. Then both types wait, so when Player 2 moves his posterior is just the prior:

$$q = p > \frac{1}{2}.$$

Hence Player 2 chooses Grab2. But then the selfish type would prefer Grab1, since $1 > 0$. Contradiction.

Case 2: suppose the selfish type always grabs immediately. Then whenever Player 2 gets to move, he knows the type must be altruistic, so

$$q = 0.$$

Hence Player 2 chooses Wait2. But then the selfish type would like to deviate from Grab1 to Wait1, because waiting would lead to payoff 3, which is strictly better than 1. Contradiction.

So the equilibrium must be *semi-separating*: the altruistic type waits for sure, while the selfish type mixes between grabbing immediately and waiting.

Semi-separating equilibrium

Let

$$\sigma = \Pr(\text{Selfish waits at the first move}).$$

Since the altruistic type waits with probability 1, Bayes' rule implies that, conditional on Player 1 waiting,

$$q = \Pr(\text{Selfish} \mid \text{Player 1 wait}) = \frac{p\sigma}{p\sigma + (1-p)}.$$

For Player 2 to mix, he must be indifferent:

$$q = \frac{1}{2}.$$

Hence

$$\frac{p\sigma}{p\sigma + (1-p)} = \frac{1}{2} \implies 2p\sigma = p\sigma + (1-p) \implies p\sigma = 1-p,$$

so

$$\boxed{\sigma = \frac{1-p}{p}}.$$

Since $p > \frac{1}{2}$, this indeed lies in $(0, 1)$.

Now let

$$\lambda = \Pr(\text{Player 2 chooses Grab2}).$$

For the selfish type to mix, he must be indifferent between Grab₁ and Wait₁:

$$1 = 3(1 - \lambda).$$

Therefore

$$\lambda = \frac{2}{3}.$$

Putting everything together, we obtain the equilibrium assessment:

$$\begin{aligned} \Pr(\text{Selfish chooses Wait}_1) &= \frac{1-p}{p}, \\ \Pr(\text{Selfish chooses Grab}_1) &= 1 - \frac{1-p}{p} = \frac{2p-1}{p}, \\ \Pr(\text{Altruistic chooses Wait}_1) &= 1, \\ \Pr(\text{Player 2 chooses Grab}_2) &= \frac{2}{3}, \quad \Pr(\text{Player 2 chooses Wait}_2) = \frac{1}{3}, \\ \Pr(\text{Selfish} \mid \text{Player 1 wait}) &= \frac{1}{2}, \\ \text{if the final move is reached: Selfish grabs, Altruistic gives.} \end{aligned}$$

This is a wPBE.

Intuition

The logic of the model is beautiful.

The altruistic type waits because she truly is willing to pass up current opportunism.

The selfish type may nevertheless imitate the altruistic type. Why? Because by waiting, she may induce Player 2 to wait as well, which then allows her to grab a larger amount later on.

But imitation cannot be too frequent. If the selfish type waited too often, then once Player 2 observed waiting he would infer that the selfish type was too likely, so he would simply grab in the middle stage. That would destroy the value of building a reputation.

Thus the equilibrium is pinned down by two indifference conditions:

- Player 2 must be indifferent between taking 2 now and waiting for a potentially altruistic continuation;
- the selfish type must be indifferent between grabbing 1 immediately and waiting for a chance to grab 3 later.

This is the central economic idea of reputation models: a strategically opportunistic type may optimally mimic a more cooperative type in order to manipulate future behavior.

Takeaway from the two examples

These two applications highlight two distinct uses of wPBE.

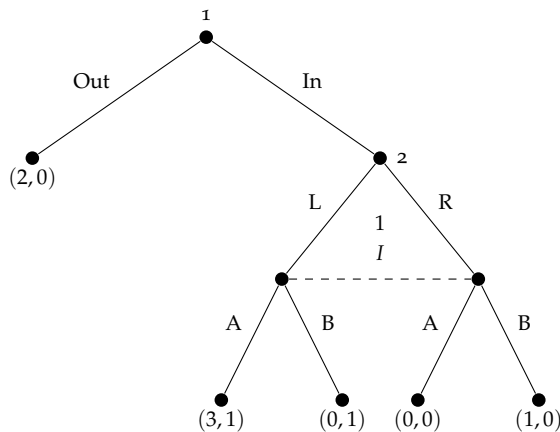
In the signaling game, the main issue is how observed actions shape beliefs and how off-path beliefs can sustain either separation or pooling. The signal matters because it changes the receiver's posterior, and the equilibrium set is large because off-path beliefs are weakly disciplined.

In the reputation game, the main issue is dynamic manipulation of beliefs. A player may be willing to take a costly short-run action in order to affect what an opponent infers and thus how that opponent behaves in the future. Here Bayes' rule on the equilibrium path is crucial: because the key information set is reached with positive probability, the posterior must be correct, and this sharply constrains equilibrium behavior.

So the broader lesson is the following: *wPBE is the equilibrium concept that lets us talk seriously about strategic behavior when actions reveal information, and when beliefs feed back into continuation play.* That is exactly why it is the natural bridge from static Bayesian games to signaling, reputation, and more sophisticated sequential equilibrium refinements.

Limitations of weak Perfect Bayesian Equilibrium

Weak Perfect Bayesian Equilibrium is often too permissive off the equilibrium path. The problem is that sequential rationality is imposed *given beliefs*, but at information sets that are not reached in equilibrium those beliefs are largely unrestricted. As a result, arbitrary off-path beliefs can rationalize continuation behavior that would not survive stronger equilibrium requirements.



A game in which a wPBE need not be subgame perfect.

Consider the following assessment:

$$\beta_1 = (\text{Out at the root, } B \text{ at } I), \quad \beta_2 = L,$$

and let Player 1's belief at the off-path information set I satisfy

$$\mu(\text{right node} \mid I) = 1, \quad \mu(\text{left node} \mid I) = 0.$$

Why this is a wPBE. Since I is not reached in equilibrium, weak PBE imposes no Bayesian restriction on μ . Given belief μ , Player 1's expected payoff at I from actions A and B is

$$U_1(A \mid I, \mu) = 3(1 - \mu), \quad U_1(B \mid I, \mu) = \mu,$$

where $\mu := \mu(\text{right node} \mid I)$. Hence B is optimal whenever

$$\mu \geq \frac{3}{4}.$$

In particular, under $\mu = 1$, action B is sequentially rational.

Given that Player 1 would choose B at I , Player 2 compares

$$u_2(L, B) = 1, \quad u_2(R, B) = 0,$$

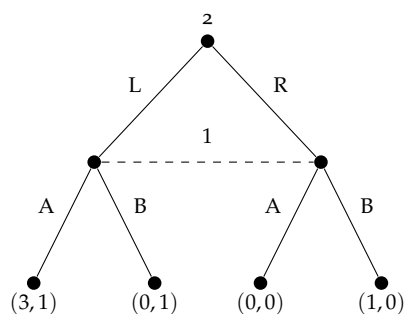
so L is optimal for Player 2.

Finally, at the root Player 1 compares

$$u_1(\text{Out}) = 2 \quad \text{and} \quad u_1(\text{In}, L, B) = 0,$$

so choosing Out is optimal. Therefore the assessment is a weak Perfect Bayesian Equilibrium.

Why this is not subgame perfect. Now look at the proper subgame that starts after action In.



The proper subgame following In.

In this subgame, the strategy profile induced by the candidate equilibrium is (L, B) . But (L, B) is *not* a Nash equilibrium of the subgame, because given L , Player 1 strictly prefers A to B :

$$u_1(A, L) = 3 > 0 = u_1(B, L).$$

Hence the continuation play after In is not an equilibrium of that subgame. Therefore the candidate assessment is not subgame perfect.

The underlying intuition. The problem is not that action B is always unreasonable. In fact, B is optimal for Player 1 whenever her belief assigns sufficiently high probability to the right node. The real problem is that weak PBE allows those beliefs to be chosen arbitrarily whenever the information set is off the equilibrium path.

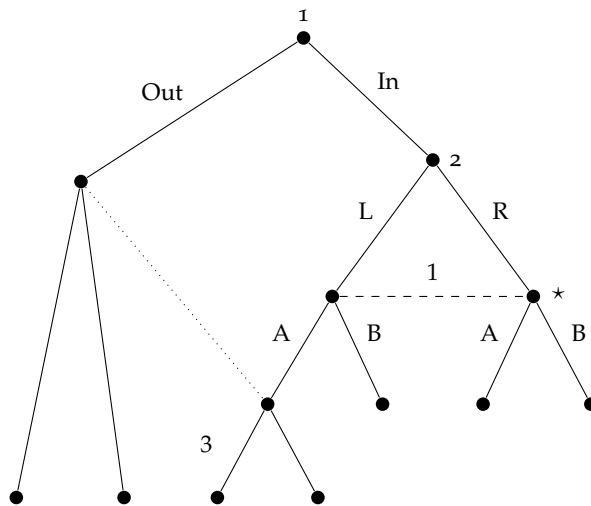
So wPBE may validate the following kind of reasoning:

“If this information set were ever reached, I would believe I am at the node that makes my prescribed action optimal.”

But such beliefs may have no strategic justification at all. In the example above, they support continuation play that would immediately unravel if the subgame were actually played.

PBE does not imply correct beliefs

The next problem is that PBE may allow beliefs that are *sequentially useful* but not really *correct* in any global sense. The issue is again off-path information sets: once an information set is not reached in equilibrium, PBE gives us substantial freedom in assigning beliefs there.



PBE may allow off-path beliefs that are not globally well-founded.

A typical assessment here is to prescribe

(Out, B)

and then assign belief 1 to the starred node at Player 1's off-path information set. With that belief, action B can be made sequentially rational, even though the information set is never reached on the equilibrium path.

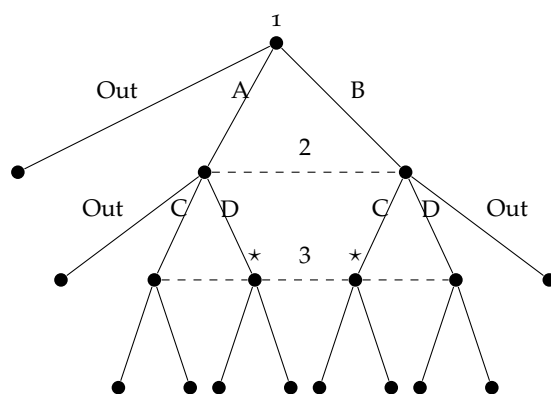
The basic problem is the following: PBE only asks that beliefs be updated correctly *where Bayes' rule is actually triggered on the equilibrium path*. Off the path, it does not require that beliefs come from a single

coherent probabilistic story about trembles. So the belief system may be locally useful, but not genuinely “correct.”

Remark. Thus, PBE can rationalize continuation play by assigning convenient beliefs at unreached information sets, even when those beliefs are not supported by any common perturbation of the equilibrium strategies.

PBE does not imply strategic independence

A related weakness is that PBE does not force beliefs at different off-path information sets to be linked by the same underlying strategy profile. In that sense, it may violate *strategic independence*.



PBE may fail to impose a common probabilistic discipline across off-path information sets.

Here the underlying issue is simple. Player 2 moves at a single information set, so his strategy should generate a single probability distribution over *Out*, *C*, and *D*. That same strategy should then help determine the relative likelihood of the nodes in Player 3's later information set.

But PBE may treat these beliefs too separately. One can choose beliefs at Player 2's information set to justify one action, and then choose beliefs at Player 3's information set to justify a continuation that is not actually induced by the same strategy choices. So the whole assessment may fit together only *locally*, not globally.

Remark. This is why PBE can be too weak: it may fail to impose a single coherent tremble structure linking all off-path beliefs. Sequential equilibrium is designed precisely to rule out this kind of inconsistency.

Bottom line. The two basic weaknesses are:

1. PBE may allow off-path beliefs that are not genuinely well founded;
2. PBE may allow different off-path beliefs to be chosen independently, even when they should be linked by the same strategic perturbation.

For this reason, after PBE the next step is to impose a stronger notion of consistency, leading to *sequential equilibrium*.

Sequential Equilibrium I

The weaknesses of weak Perfect Bayesian Equilibrium come from the fact that, off the equilibrium path, beliefs may be too weakly disciplined. Sequential equilibrium strengthens the belief requirement.

Definition (Consistent assessments). In a countable game, an assessment (β, μ) is *consistent* if there exists a sequence of assessments $\{(\beta^k, \mu^k)\}_{k=1}^{\infty}$ such that:

1. each behavior strategy profile β^k is *completely mixed*, i.e. every action at every information set is chosen with strictly positive probability;
2. each belief system μ^k is derived from β^k by Bayes' rule;
3. $(\beta^k, \mu^k) \rightarrow (\beta, \mu)$ uniformly.

Definition (Sequential equilibrium). In a countable game, an assessment is a *sequential equilibrium* if it is

1. sequentially rational, and
2. consistent.

Basic idea. A consistent assessment must be approximable by a sequence of fully mixed strategy profiles. Since every action is played with positive probability in β^k , Bayes' rule determines beliefs at *every* information set. The limiting belief system μ must therefore come from a coherent perturbation of the strategy profile.

Why this helps. This requirement rules out the most arbitrary off-path beliefs allowed by wPBE. In particular, it imposes two basic forms of discipline:

1. beliefs must be compatible with *some* sequence of small trembles;
2. beliefs at different information sets must fit together with a single underlying perturbation.

Interpretation. Sequential equilibrium can be read as:

sequential equilibrium = sequential rationality + consistent beliefs.

So the solution concept keeps the optimality requirement of PBE, but strengthens the notion of which beliefs are admissible.

Remark. The key improvement over weak PBE is that off-path beliefs are no longer chosen freely. They must arise as limits of Bayesian updating from completely mixed assessments.

Appendix to Class 11: A wrapper of notions under incomplete information

This appendix is meant to organize, in one place, the main equilibrium notions that appear once we move from standard dynamic games to environments with private information. The goal is not to give the most abstract or most general treatment possible, but rather to clarify the logic behind the concepts we have seen in class: Bayesian Nash Equilibrium (BNE), weak Perfect Bayesian Equilibrium (wPBE), and Perfect Bayesian Equilibrium (PBE).

A useful way to think about the progression is the following. In static games of complete information, Nash equilibrium is the natural baseline concept. In dynamic games of complete and perfect information, subgame perfect equilibrium refines Nash by forcing players to behave optimally not only overall, but also after every possible history. Once we introduce private information, however, this standard logic becomes more delicate. The reason is that players may no longer know exactly where they are in the game tree when they move, so we need equilibrium notions that keep track not only of strategies, but also of beliefs.

From incomplete information to Bayesian normal form

The first object that naturally appears under incomplete information is the BNE. Here the primitive idea is still NE. The difference is that the game now includes uncertainty about payoff-relevant characteristics, typically represented as *types*. At least one player may have several possible types, and the other players do not know with certainty which type has been realized.

Following the Harsanyi transformation, we represent this uncertainty by introducing an initial move of Nature. Nature draws a type profile according to a commonly known probability distribution. Each player observes her own type, but not necessarily the types of the others. In this way, a situation of incomplete information is transformed into a well-defined game in which Nature moves first.

Once this is done, the relevant equilibrium concept is a NE of the associated *Bayesian normal form game*. That is exactly what a BNE is. Formally, if player i has type θ_i , then her strategy may depend on θ_i , and equilibrium requires that each type of each player chooses a best response given:

1. her beliefs about the other players' **types**²⁴, and
2. the type-contingent strategies played by the other players.

So the basic logic is still Nash: each player is optimizing given

²⁴ I emphasize *types* here because, as we will see in a moment, in dynamic games we must also keep track of beliefs about which node within an information set a player may be at.

what she expects the others to do. The only difference is that now the optimization is conditional on the player's own type. In that sense, solving for a BNE amounts to solving the optimization problem of each type separately.

This is already a major step forward, because it gives us a coherent way to analyze strategic interaction under incomplete information. Still, BNE remains, in an important sense, a fairly coarse notion. It works very naturally in static Bayesian games, but in dynamic games it may fail to impose enough discipline on what players would do at information sets that are reached only after some particular history of play.

Why BNE is not enough in dynamic games

The problem is that BNE works at the level of a complete contingent plan. It tells us whether a player's overall strategy is optimal, but it does not explicitly require us to ask what a player would want to do if she were to arrive at a particular information set in the course of the game.

This is the key tension. In a dynamic game with private information, a player may move at an information set containing several nodes. She does not know exactly which node she is at, so in order to decide optimally she must form beliefs about where she is. But standard BNE does not explicitly build these beliefs into the equilibrium object.

A useful way to phrase the issue is this: BNE asks whether the player's full plan is globally optimal, while a more refined dynamic notion should ask something stronger, namely:

if the player actually reached this information set, would she still want to do what her strategy prescribes, given her beliefs there?

That second question is the essence of sequential rationality.

Assessments and weak Perfect Bayesian Equilibrium

This is where wPBE enters. In a wPBE, the equilibrium object is no longer just a strategy profile. Instead, it is an *assessment*,

$$(\beta, \mu),$$

where:

- β is a profile of behavioral strategies, and
- μ is a system of beliefs assigning to each information set a probability distribution over the nodes in that information set.

So the idea is very simple, even if the formalism is not: once we are in a dynamic game with private information, strategies alone are not enough. We also need to specify what players believe when they move.

A wPBE requires two ingredients. First, *sequential rationality*: at every information set, each player's continuation strategy must be optimal given her beliefs at that information set and given the continuation strategies of the other players.

Second, *Bayesian updating on the equilibrium path*: whenever an information set is reached with positive probability under the equilibrium strategies, the beliefs at that information set must be obtained from Bayes' Rule. This means that on the equilibrium path, beliefs are pinned down by observed actions and the equilibrium strategies that generated those actions. In particular, beliefs are not updated from "observed payoffs," but rather from the relative likelihood that different types or histories would have generated the action that was actually observed.

At this point the equilibrium logic becomes a little circular, but that is not a bug. It is exactly what one should expect from a fixed-point notion. Strategies must be optimal given beliefs, while beliefs must be consistent with strategies, at least on the equilibrium path.

A simple intuition

Consider a simple signaling environment with two players. Player 1 has private information, say different possible types, and chooses an action first. Player 2 observes the action but not Player 1's type, and then moves. When Player 2 acts, she may be at an information set containing several nodes, one for each possible type of Player 1 that could have generated the observed action.

At that point, Player 2 cannot choose optimally unless she has some belief about which type is more likely. A wPBE says that these beliefs must be reasonable in the following sense:

1. if the information set is reached on the equilibrium path, then beliefs must come from Bayes' Rule;
2. given those beliefs, Player 2 must indeed be willing to choose the action prescribed by equilibrium.

So one can think of wPBE as the equilibrium notion that asks for local optimality, node by node, rather than only global optimality of an entire plan.

What happens off the equilibrium path?

This brings us to the main limitation of wPBE. If an information set is never reached in equilibrium, then Bayes' Rule does not pin down

beliefs there. As a result, off-path beliefs may be chosen rather freely, as long as they support sequential rationality.

This is why wPBE can still be too permissive. In a simple example, Player 1 may choose an equilibrium strategy that makes it impossible to reach Player 2's information set. Then, since that information set is off the equilibrium path, Player 2's beliefs there can be specified almost "by hand." If those beliefs are chosen appropriately, they may justify some continuation behavior and hence support a wPBE, even if the underlying interpretation feels somewhat artificial.

In short, wPBE imposes discipline on beliefs on the path of play, but much less discipline off the path.

Perfect Bayesian Equilibrium

Perfect Bayesian Equilibrium is meant to go one step further. In the version emphasized in these notes, the idea is that we do not want only the game as a whole to satisfy the weak perfect Bayesian conditions. We also want those conditions to hold *within each subgame*, so as to recover the spirit of subgame perfection.

The motivation is clear. In dynamic games of complete information, SPE improves on Nash by requiring equilibrium behavior after every possible history that defines a subgame. Under incomplete information, true subgames may be scarce because information sets often link together multiple histories. Even so, whenever a subgame exists, it is natural to require equilibrium-like behavior there as well.

So, in the interpretation adopted here, PBE strengthens wPBE by asking for the same type of discipline not only globally, but also subgame by subgame. Still, even this may not be enough to capture all the implications one might want from "correct beliefs" and from independence restrictions across histories. In particular, PBE can remain somewhat loose in the way it treats off-path beliefs. This is one reason why even stronger refinements, such as sequential equilibrium, were later introduced.

Class 12: Introduction to Mechanism Design

The quasilinear environment

We won't work in the abstract social choice environment as in most books. Instead of that, we will go straight to the quasilinear environment, in which transfers can be used to affect agents' incentives directly.

There are n agents, indexed by

$$N = \{1, \dots, n\}.$$

There is a space X of feasible alternatives or outcomes, and for each agent i there is a type space Θ_i of private information. The overall type space is

$$\Theta = \Theta_1 \times \dots \times \Theta_n.$$

The agents share a common prior ϕ over Θ .

Agent i 's utility takes the quasilinear form

$$u_i(\chi, \theta, \tau) = v_i(\chi, \theta) + \tau_i,$$

where:

- $v_i(\chi, \theta)$ is agent i 's valuation of alternative χ given the type profile θ ,
- $\tau = (\tau_1, \dots, \tau_n) \in \mathbb{R}^n$ is the vector of monetary transfers.

The quasilinear structure is central because it allows us to separate the utility generated by the chosen allocation from the utility generated by transfers. In that sense, money acts as a numeraire for utility.

Mechanisms

A mechanism specifies a game to be played by the agents. Formally, a mechanism is a triple

$$\langle M, x, t \rangle,$$

where:

- $M = M_1 \times \cdots \times M_n$ is the message space,
- $x : M \rightarrow X$ is the outcome rule,
- $t : M \rightarrow \mathbb{R}^n$ is the transfer rule.

The timing is simple:

1. agents simultaneously send messages

$$m = (m_1, \dots, m_n) \in M,$$

2. the mechanism selects the outcome $x(m) \in X$,
3. each agent i receives transfer $t_i(m)$.

Thus, the designer does not choose the final outcome directly. Instead, the designer chooses the game form, and the final outcome is generated by equilibrium behavior inside that game.

Mechanism design objectives

The designer typically has some objective function to maximize. Common examples are:

- **Social welfare:**

$$\mathbb{E} \left[\sum_i v_i(x(m(\theta)), \theta) \right].$$

- **Revenue:**

$$- \mathbb{E} \left[\sum_i t_i(m(\theta)) \right].$$

- **Weighted sum of agents' utilities:**

$$\mathbb{E} \left[\sum_i \lambda_i \left(v_i(x(m(\theta)), \theta) + t_i(m(\theta)) \right) \right].$$

Hence mechanism design is an optimization problem subject to incentive and feasibility constraints.

Three categories of constraints

There are three broad classes of constraints.

Incentive compatibility

This is the equilibrium concept imposed on the mechanism. The idea is that agents should find it optimal to send the equilibrium messages prescribed by the mechanism.

The main notions are:

- interim incentive compatibility (Bayesian IC),
- ex post incentive compatibility (the one we will use),
- dominant strategy incentive compatibility.

These differ in how strong the required notion of optimality is.

Individual rationality

Agents must be willing to participate in the mechanism. Depending on the timing of participation, one can impose:

- ex ante IR,
- interim IR,
- ex post IR.

Budget balance

Transfers must satisfy some feasibility requirement. Common notions are:

- ex ante budget balance,
- ex post no subsidy,
- ex post budget balance.

Interim incentive compatibility (Bayesian IC)

Definition (Bayesian Nash equilibrium of a mechanism). The strategy profile m^* is a *Bayesian Nash equilibrium* of the mechanism $\langle M, x, t \rangle$ if

$$\mathbb{E}[v_i(x(m^*(\theta)), \theta) + t_i(m^*(\theta)) \mid \theta_i] \geq \mathbb{E}[v_i(x(m_i, m_{-i}^*(\theta_{-i})), \theta) + t_i(m_i, m_{-i}^*(\theta_{-i})) \mid \theta_i]$$

for all $m_i \in M_i$, for all $\theta_i \in \Theta_i$, and for all i .

Definition (Implementation in Bayesian Nash equilibrium). The mechanism $\langle M, x, t \rangle$ *implements* the outcome rule

$$\hat{x} : \Theta \rightarrow X$$

in Bayesian Nash equilibrium if there exists a Bayesian Nash equilibrium m^* of $\langle M, x, t \rangle$ such that

$$x(m^*(\theta)) = \hat{x}(\theta) \quad \text{for all } \theta.$$

Such a mechanism is called *interim incentive compatible*.

The logic here is that each agent conditions only on her own type. She chooses her message optimally given her posterior beliefs about other agents' types and given the equilibrium behavior of the others.

Ex post incentive compatibility

Definition (Ex post equilibrium of a mechanism). The strategy profile m^* is an *ex post equilibrium* of the mechanism $\langle M, x, t \rangle$ if

$$v_i(x(m^*(\theta)), \theta) + t_i(m^*(\theta)) \geq v_i(x(m_i, m_{-i}^*(\theta_{-i})), \theta) + t_i(m_i, m_{-i}^*(\theta_{-i}))$$

for all $m_i \in M_i$, for all $\theta \in \Theta$, and for all i .

Definition (Implementation in ex post equilibrium). The mechanism $\langle M, x, t \rangle$ implements the outcome rule

$$\hat{x} : \Theta \rightarrow X$$

in ex post equilibrium if there exists an ex post equilibrium m^* of $\langle M, x, t \rangle$ such that

$$x(m^*(\theta)) = \hat{x}(\theta) \quad \text{for all } \theta.$$

Such a mechanism is called *ex post incentive compatible* (EPIC).

Ex post IC is stronger than Bayesian IC. It requires that, after all types are realized, no agent would regret her equilibrium message even if she were somehow told the realized types of the others.

Dominant strategy incentive compatibility

Definition (Dominant strategy equilibrium of a mechanism). The strategy profile m^* is a *dominant strategy equilibrium* of the mechanism $\langle M, x, t \rangle$ if

$$v_i(x(m_i^*(\theta_i), m_{-i}), \theta) + t_i(m_i^*(\theta_i), m_{-i}) \geq v_i(x(m), \theta) + t_i(m)$$

for all $m \in M$, for all $\theta \in \Theta$, and for all i .

Definition (Implementation in dominant strategies). The mechanism $\langle M, x, t \rangle$ implements the outcome rule

$$\hat{x} : \Theta \rightarrow X$$

in *dominant strategy equilibrium* if there exists a dominant strategy equilibrium m^* of $\langle M, x, t \rangle$ such that

$$x(m^*(\theta)) = \hat{x}(\theta) \quad \text{for all } \theta.$$

Such a mechanism is called *dominant strategy incentive compatible* (DSIC).

This is the strongest notion among the three. Unlike ex post incentive compatibility, which requires each agent's prescribed message to be optimal against the other agents' equilibrium messages (and, in a direct truthful mechanism, against their truthful reports), dominant-strategy incentive compatibility requires it to be optimal for every possible message profile of the others.

DSIC versus EPIC

It is useful to compare dominant-strategy and ex post implementation.

Definition (Private values). The environment satisfies *private values* if

$$v_i(\chi, \theta) = v_i(\chi, \theta_i, \theta'_{-i}) \quad \text{for all } \theta'_{-i}, \text{ all } \theta, \text{ and all } i.$$

Under private values, agent i 's valuation depends only on her own type, not on the other agents' true types. In a direct mechanism (x, t) , this implies that the incentive constraints can be written pointwise in the opponents' reports. In particular, truth-telling is *ex post incentive compatible* if

$$v_i(x(\theta_i, \theta_{-i}), \theta_i) + t_i(\theta_i, \theta_{-i}) \geq v_i(x(\hat{\theta}_i, \theta_{-i}), \theta_i) + t_i(\hat{\theta}_i, \theta_{-i})$$

for all i , all $\theta_i, \hat{\theta}_i \in \Theta_i$, and all $\theta_{-i} \in \Theta_{-i}$.

By contrast, truth-telling is *dominant-strategy incentive compatible* if

$$v_i(x(\theta_i, m_{-i}), \theta_i) + t_i(\theta_i, m_{-i}) \geq v_i(x(\hat{\theta}_i, m_{-i}), \theta_i) + t_i(\hat{\theta}_i, m_{-i})$$

for all i , all $\theta_i, \hat{\theta}_i \in \Theta_i$, and all $m_{-i} \in M_{-i}$.

In a direct mechanism, $M_{-i} = \Theta_{-i}$. Therefore, under private values, the two conditions coincide:

$$\text{private values} \implies \text{DSIC} = \text{EPIC}.$$

By contrast, under *interdependent values*, agent i 's valuation depends on the other agents' types as well, so the two notions need not coincide. In that case, DSIC is generally more demanding than EPIC and may rule out many interesting or efficient mechanisms.

EPIC versus BIC/IIC

The distinction between ex post and Bayesian notions is also conceptually important.

Interim/Bayesian IC

Bayesian IC:

- relies on the details of the common prior ϕ ,
- is not robust to higher-order beliefs,
- is not robust to information leakage.

Ex post IC

Ex post IC:

- does not depend on ϕ at all,
- is robust to higher-order beliefs,
- is robust to information leakage,
- is often simpler to work with.

The reason is straightforward. Bayesian IC is an *interim* notion: it depends on what agents believe, conditional on their information. Ex post IC instead asks whether truthful equilibrium behavior remains optimal after the uncertainty over types is fully resolved. Because of this, ex post IC is much less sensitive to the informational details of the environment.

Takeaway

Mechanism design in the quasilinear environment begins with three basic ingredients:

1. a space of alternatives,
2. private information,
3. transfers that enter utility linearly.

A mechanism is a message game $\langle M, x, t \rangle$, and the core problem is to design this game so that the desired outcome rule is induced in equilibrium. The central constraints are incentive compatibility, individual rationality, and budget balance. Among incentive notions, Bayesian IC is the weakest, ex post IC is stronger, and DSIC is strongest. In private value environments, ex post and dominant-strategy implementation coincide, while in interdependent value environments dominant strategies are often too restrictive.

Class 13: Revelation Principle and Groves Mechanisms

Direct mechanisms and the revelation principle

A *direct mechanism* is one in which each agent simply reports a type. Thus,

$$M_i = \Theta_i \quad \text{for all } i,$$

so the message space is the type space itself. The mechanism is then described by an outcome rule

$$\hat{x} : \Theta \rightarrow X$$

and a transfer rule

$$\hat{t} : \Theta \rightarrow \mathbb{R}^n.$$

For direct mechanisms, we usually write simply

$$\langle \hat{x}, \hat{t} \rangle$$

instead of $\langle \Theta, \hat{x}, \hat{t} \rangle$.

The key result is the revelation principle.

Theorem 0.0.29 (The Revelation Principle). *If there exists a mechanism $\langle M, x, t \rangle$ that implements $\hat{x}$ in a Bayesian, ex post, or dominant strategy equilibrium m^* , then there exists a direct mechanism*

$$\langle \Theta, \hat{x}, \hat{t} \rangle,$$

where

$$\hat{t}(\hat{\theta}) = t(m^*(\hat{\theta})) \quad \text{for all } \hat{\theta},$$

with an equilibrium

$$\hat{m}_i(\theta_i) = \theta_i \quad \text{for all } \theta_i \text{ and all } i.$$

That is, the direct mechanism truthfully implements $\hat{x}$.

Interpretation. The revelation principle does *not* say that every mechanism is direct. Rather, it says that for the purpose of characterizing what can be implemented, there is no loss in restricting attention to direct and truthful mechanisms.

This is enormously useful. Instead of searching over all possible message games, extensive forms, and equilibrium strategies, we may restrict attention to the much simpler problem: can we find a direct mechanism in which truthful reporting is an equilibrium?²⁵

²⁵ From this point on, when studying implementability, we may focus on direct truthful mechanisms.

Intuition. Suppose some complicated mechanism $\langle M, x, t \rangle$ implements the desired rule $\hat{x}$, and in equilibrium agents use the message profile $m^*(\theta)$. Then we can construct a new mechanism that asks each agent to report her type $\hat{\theta}_i$, internally simulates the old equilibrium message $m_i^*(\hat{\theta}_i)$, and then chooses exactly the same outcome and transfers that the old mechanism would have produced. In that new direct mechanism, reporting truthfully is optimal because a truthful report reproduces precisely the equilibrium play of the original mechanism.

Remark. The Revelation Principle does not require the equilibrium message map $\theta_i \mapsto m_i^*(\theta_i)$ to be injective. Different types may induce the same equilibrium message in the original mechanism, and hence the same allocation and transfer in the associated direct mechanism. Truthful implementation only requires that truthful reporting be an equilibrium, not that it be uniquely optimal or strictly optimal.

Remark (Pooling and weak truthfulness under the Revelation Principle). Let $\langle M, x, t \rangle$ be a mechanism and let m^* be an equilibrium that implements the outcome rule $\hat{x}$. The associated direct mechanism is $\langle \Theta, \hat{x}, \hat{t} \rangle$, where

$$\hat{x}(\hat{\theta}) = x(m^*(\hat{\theta})), \quad \hat{t}(\hat{\theta}) = t(m^*(\hat{\theta})).$$

The Revelation Principle does *not* require the equilibrium message map

$$\theta_i \mapsto m_i^*(\theta_i)$$

to be injective. In particular, it may happen that for some agent i and some distinct types $\theta_i \neq \theta'_i$,

$$m_i^*(\theta_i) = m_i^*(\theta'_i).$$

If so, then for every $\theta_{-i} \in \Theta_{-i}$,

$$m^*(\theta_i, \theta_{-i}) = (m_i^*(\theta_i), m_{-i}^*(\theta_{-i})) = (m_i^*(\theta'_i), m_{-i}^*(\theta_{-i})) = m^*(\theta'_i, \theta_{-i}),$$

and therefore

$$\hat{x}(\theta_i, \theta_{-i}) = \hat{x}(\theta'_i, \theta_{-i}), \quad \hat{t}(\theta_i, \theta_{-i}) = \hat{t}(\theta'_i, \theta_{-i}).$$

Thus, different types may be *pooled*: they may induce exactly the same allocation and the same transfer in the direct mechanism. The Revelation Principle does not require “separation” of types; it only requires that the direct mechanism replicate the equilibrium outcome rule of the original mechanism.

This also means that truthful reporting need not be *strictly* optimal. Truthful implementation only requires that truthful reporting be an equilibrium, not that it be the unique optimal action. Since incentive compatibility is stated with weak inequalities, an agent may be indifferent between reporting truthfully and misreporting. In particular, if two reports induce the same allocation and the same transfer for that agent, then both reports yield the same utility. This does not violate truthfulness.

Groves mechanisms

We now restrict attention to the *private values* case. This means that each agent’s valuation depends only on the chosen outcome and on her own type, not on the types of others. In that case, we may write

$$v_i(\chi, \theta) = v_i(\chi, \theta_i).$$

The main theorem is the following.

Theorem 0.0.30 (Groves Mechanisms). *Assume private values. If $x^*(\theta)$ maximizes social welfare for each θ , then x^* is truthfully implemented in ex post equilibrium by any direct mechanism $\langle x^*, t \rangle$ satisfying*

$$t_i(\theta) = \sum_{j \neq i} v_j(x^*(\theta), \theta_j) + h_i(\theta_{-i}),$$

where

$$h_i : \Theta_{-i} \rightarrow \mathbb{R}$$

is an arbitrary function, for all θ and all i .

These are also called *VCG mechanisms* (Vickrey–Clarke–Groves).

Why this works. Fix agent i , her true type θ_i , and the reports of the others θ_{-i} . If she reports $\hat{\theta}_i$, her utility is

$$v_i(x^*(\hat{\theta}_i, \theta_{-i}), \theta_i) + \sum_{j \neq i} v_j(x^*(\hat{\theta}_i, \theta_{-i}), \theta_j) + h_i(\theta_{-i}).$$

The crucial point is that $h_i(\theta_{-i})$ does not depend on i ’s report. Therefore agent i chooses her report only to maximize

$$\sum_j v_j(x^*(\hat{\theta}_i, \theta_{-i}), \theta_j),$$

that is, total social welfare. Since x^* is defined to maximize social welfare at the true type profile, truthful reporting is optimal.

Proof. Suppose, toward a contradiction, that the Groves mechanism $\langle x^*, t \rangle$ is not truthfully implementable in ex post equilibrium. Then there exist i , $\theta = (\theta_i, \theta_{-i})$, and $\hat{\theta}_i \neq \theta_i$ such that

$$v_i(x^*(\hat{\theta}_i, \theta_{-i}), \theta_i) + t_i(\hat{\theta}_i, \theta_{-i}) > v_i(x^*(\theta_i, \theta_{-i}), \theta_i) + t_i(\theta_i, \theta_{-i}).$$

Using the Groves form of transfers,

$$t_i(\theta) = \sum_{j \neq i} v_j(x^*(\theta), \theta_j) + h_i(\theta_{-i}),$$

and since $h_i(\theta_{-i})$ does not depend on i 's report, we can add the valuations of the other agents to both sides and cancel $h_i(\theta_{-i})$, obtaining

$$\sum_{j=1}^n v_j(x^*(\hat{\theta}_i, \theta_{-i}), \theta_j) > \sum_{j=1}^n v_j(x^*(\theta_i, \theta_{-i}), \theta_j).$$

But this contradicts the assumption that $x^*(\theta_i, \theta_{-i})$ maximizes social welfare at the true type profile θ . Therefore no such profitable deviation exists, and truthful reporting is an ex post equilibrium. $\square$

Economic intuition. A Groves mechanism makes each agent internalize the effect of her report on total welfare, not just on her own payoff. The transfer adds to agent i 's utility the welfare of everyone else, up to a term $h_i(\theta_{-i})$ that is irrelevant for incentives. Thus each agent behaves as if she were a social planner choosing the report that maximizes aggregate surplus.

Remark (The role of $h_i(\theta_{-i})$ and indifferent misreports). In a Groves mechanism,

$$t_i(\theta) = \sum_{j \neq i} v_j(x^*(\theta), \theta_j) + h_i(\theta_{-i}),$$

the term $h_i(\theta_{-i})$ depends only on the types of the other agents. Hence, from the viewpoint of agent i , and holding θ_{-i} fixed, $h_i(\theta_{-i})$ is a constant with respect to i 's own report. It can affect the *level* of agent i 's utility, but it does not affect which report maximizes that utility.

A natural question is whether some other agent j could misreport—for instance, if she is indifferent between truth-telling and lying—in a way that changes agent i 's transfer, perhaps making agent i pay more. In principle, yes: a misreport by j may affect the transfer formula faced by other agents, even in cases in which the chosen allocation x^* remains unchanged.

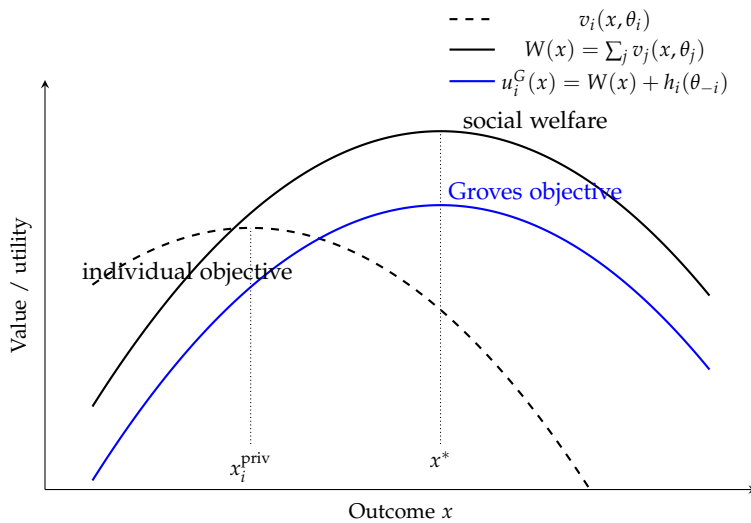
However, agent j has no strict strategic reason to do this merely to affect agent i 's payment. Under a Groves mechanism, agent j 's utility

can be written as

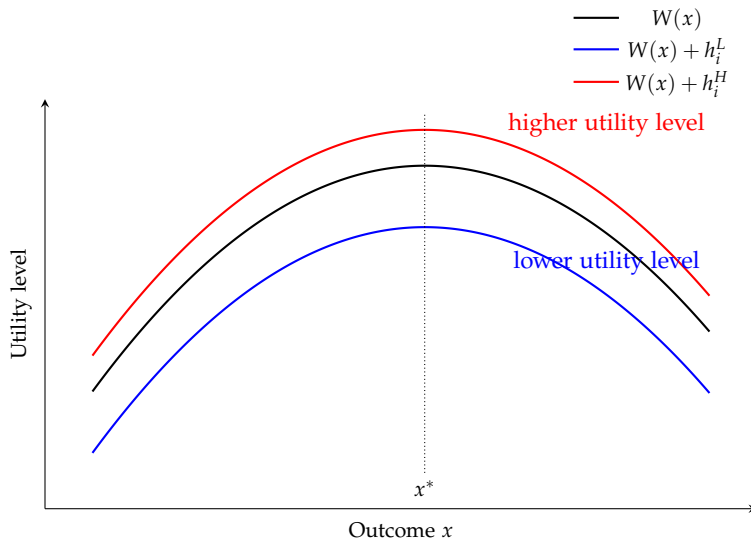
$$u_j(\hat{\theta}_j, \theta_{-j}; \theta_j) = \sum_{k=1}^n v_k(x^*(\hat{\theta}_j, \theta_{-j}), \theta_k) + h_j(\theta_{-j}).$$

Therefore, if two different reports by j induce the same allocation x^* , then they also induce the same utility for agent j , since $h_j(\theta_{-j})$ does not depend on j 's own report. So such a lie is, at most, payoff-equivalent for agent j , not a strictly profitable deviation.

This is again compatible with truthful implementation. Truth-telling only has to be a weak best response; it need not be the unique best response.



Without Groves, agent i would like to maximize $v_i(x, \theta_i)$, whose maximizer is x_i^{priv} . Under Groves transfers, agent i 's relevant objective becomes $W(x) + h_i(\theta_{-i})$, so the maximizer is x^* , the welfare-maximizing outcome.



Changing $h_i(\theta_{-i})$ shifts agent i 's utility up or down without changing its shape. Hence h_i affects the utility level, but not the welfare-maximizing choice x^* .

Remark (Graphical intuition for Groves mechanisms). A useful way to think about a Groves mechanism is that it transforms agent i 's private optimization problem into the problem of maximizing total social welfare. Indeed, once the transfer is chosen as

$$t_i(\theta) = \sum_{j \neq i} v_j(x^*(\theta), \theta_j) + h_i(\theta_{-i}),$$

agent i 's utility from inducing outcome x becomes

$$u_i(x) = v_i(x, \theta_i) + t_i = \sum_{j=1}^n v_j(x, \theta_j) + h_i(\theta_{-i}) = W(x) + h_i(\theta_{-i}).$$

Thus, as a function of the report of agent i , her objective has the same shape as social welfare. The term $h_i(\theta_{-i})$ is just a vertical shift: it changes the level of utility, but not the maximizing report. This is why Groves mechanisms preserve incentives while leaving room to adjust transfers for other purposes.

A warning. The theorem guarantees truthful *ex post* implementation, but it says nothing by itself about:

- budget balance,
- no-subsidy,
- individual rationality.

These constraints must be checked separately.

Remark (Why Groves does not automatically imply individual rationality). The Groves theorem guarantees truthful *ex post* incentive compatibility, but it does not by itself guarantee individual rationality.

The reason is that incentive compatibility and individual rationality are different requirements:

- Incentive compatibility asks whether, *conditional on participating in the mechanism*, truthful reporting maximizes an agent's utility relative to other reports.
- Individual rationality asks whether participation in the mechanism gives the agent at least as much utility as her outside option.

For example, if the outside option is normalized to zero, then *ex post* individual rationality requires

$$v_i(x^*(\theta), \theta_i) + t_i(\theta) \geq 0 \quad \text{for all } i \text{ and all } \theta.$$

Now, in a Groves mechanism,

$$t_i(\theta) = \sum_{j \neq i} v_j(x^*(\theta), \theta_j) + h_i(\theta_{-i}),$$

and $h_i(\theta_{-i})$ is arbitrary. Since h_i does not affect incentives, one may replace it by

$$\tilde{h}_i(\theta_{-i}) = h_i(\theta_{-i}) - C$$

for some large constant $C > 0$. This preserves truthful incentive compatibility, but lowers agent i 's utility by C in every state. If C is large enough, the agent's utility may fall below her outside option, so individual rationality fails.

Thus, truthful implementation does not imply that agents are willing to participate. Individual rationality must be checked separately.

In some important special cases, such as the VCG mechanism with the Clarke pivot rule, one can obtain ex post individual rationality under additional assumptions (for example, an allocation corresponding to non-participation is feasible, and valuations are normalized so that an agent who is not served gets utility zero). But this is an additional result, not a consequence of the Groves theorem alone.

Example: A public project

Consider the following environment.

- $X = [0, 1]$, where $\chi \in X$ is the probability that the project is built.
- $\Theta_i = [-1, 1]$, where $\theta_i \in \Theta_i$ is agent i 's value for the project.
- Payoffs are

$$v_i(\chi, \theta) = \theta_i \chi.$$

The objective is to maximize social welfare:

$$\chi \sum_i \theta_i.$$

Hence the efficient allocation rule is

$$x^*(\theta) = \begin{cases} 1 & \text{if } \sum_i \theta_i > 0, \\ 0 & \text{otherwise.} \end{cases}$$

We want a mechanism that satisfies:

- ex post incentive compatibility,
- no subsidy:

$$-\sum_i t_i(\theta) \geq 0 \quad \text{for all } \theta.$$

Groves transfers

In this environment, the Groves formula becomes

$$t_i(\theta) = x^*(\theta) \left(\sum_{j \neq i} \theta_j \right) + h_i(\theta_{-i}).$$

A convenient choice that ensures no subsidy is

$$h_i(\theta_{-i}) = \min \left\{ 0, - \sum_{j \neq i} \theta_j \right\}.$$

Let

$$S_{-i} = \sum_{j \neq i} \theta_j.$$

Then

$$t_i(\theta) = x^*(\theta) S_{-i} + \min\{0, -S_{-i}\},$$

which can be written as

$$t_i(\theta) = \begin{cases} 0 & \text{if } x^*(\theta) = 1 \text{ and } S_{-i} > 0, \\ 0 & \text{if } x^*(\theta) = 0 \text{ and } S_{-i} \leq 0, \\ S_{-i} \leq 0 & \text{if } x^*(\theta) = 1 \text{ and } S_{-i} \leq 0, \\ -S_{-i} \leq 0 & \text{if } x^*(\theta) = 0 \text{ and } S_{-i} > 0. \end{cases}$$

Since the notation here uses t_i as a transfer *to* the agent, a negative t_i means that agent i pays $-t_i$ to the mechanism.

Intuition

This example is worth unpacking carefully.

Step 1: What is efficient? The project should be built if and only if total value is positive. If

$$\sum_i \theta_i > 0,$$

then society as a whole gains from building. If the sum is nonpositive, it should not be built.

Step 2: What is the incentive problem? Each individual may want to exaggerate her value in order to push the decision in her preferred direction:

- an agent who likes the project may overreport to make sure it is built,
- an agent who dislikes the project may underreport to make sure it is rejected.

So the mechanism must make such manipulations unattractive.

Step 3: Who pays? The transfer depends on whether agent i is *pivotal*. The quantity S_{-i} measures what the others alone want.

- If $S_{-i} > 0$, the others already want the project. Then if the project is built, agent i is not pivotal and pays nothing.
- If $S_{-i} \leq 0$, the others alone do not want the project. Then if the project is nevertheless built, it must be because agent i 's report tipped the decision toward building. In that case she pays.
- Symmetrically, if $S_{-i} > 0$ but the project is not built, then agent i must have tipped the decision toward rejection, and again she pays.

Step 4: What is the payment measuring? The pivotal agent pays the externality imposed on the others. If agent i causes the project to be built when the others alone would reject it, then the others' total valuation is $S_{-i} \leq 0$, so the agent pays

$$-t_i(\theta) = -S_{-i} \geq 0.$$

This is exactly the magnitude of the loss inflicted on the rest of society by switching the decision from 0 to 1.

Likewise, if agent i causes the project to be rejected when the others alone would approve it, then she pays

$$-t_i(\theta) = S_{-i} > 0,$$

which equals the gain the others would have enjoyed from building.

Step 5: Why does this create truth-telling? Because whenever an agent manipulates the outcome in her preferred direction, she must pay for the harm imposed on everyone else. She therefore faces the correct social tradeoff:

private gain from changing the decision versus loss imposed on others.

That is precisely why the efficient rule can be sustained truthfully.

Step 6: Why no subsidy? In every case above, $t_i(\theta) \leq 0$. Therefore every transfer is weakly negative, meaning that the mechanism never pays money out to the agents. Thus

$$-\sum_i t_i(\theta) \geq 0$$

for every θ .

Example: Externalities

Now consider a different environment.

- $X = \mathbb{R}_+^n$, where $x_i \in \mathbb{R}_+$ is the pollution generated by agent i .
- $\Theta_i = \mathbb{R}_+$, where $\theta_i \in \Theta_i$ is agent i 's value for generating pollution.
- Payoff functions are

$$v_i(x, \theta) = \theta_i x_i - \sum_j x_j^2.$$

Thus each agent benefits from her own pollution x_i , but everyone suffers from total pollution through the quadratic damage term.

The planner's objective is to maximize total welfare:

$$\sum_i (\theta_i x_i - n x_i^2).$$

Therefore the efficient choice solves, for each i ,

$$\max_{x_i \geq 0} \theta_i x_i - n x_i^2,$$

which yields

$$x_i^*(\theta) = \frac{\theta_i}{2n}.$$

The constraints are:

- EPIC,
- t_i measurable with respect to θ_i .

Groves transfers

Applying the Groves formula,

$$t_i(\theta) = \sum_{j \neq i} \left(\theta_j x_j^*(\theta) - \sum_k (x_k^*(\theta))^2 \right) + h_i(\theta_{-i}).$$

Substituting

$$x_j^*(\theta) = \frac{\theta_j}{2n} \quad \text{and} \quad \sum_k (x_k^*(\theta))^2 = \sum_k \frac{\theta_k^2}{4n^2},$$

we obtain

$$t_i(\theta) = \sum_{j \neq i} \left(\frac{\theta_j^2}{2n} - \sum_k \frac{\theta_k^2}{4n^2} \right) + h_i(\theta_{-i}).$$

We now choose $h_i(\theta_{-i})$ to eliminate dependence on θ_j for $j \neq i$:

$$h_i(\theta_{-i}) = - \sum_{j \neq i} \left(\frac{\theta_j^2}{2n} - \sum_{k \neq i} \frac{\theta_k^2}{4n^2} \right).$$

Then the transfer simplifies to

$$t_i(\theta) = -\frac{(n-1)\theta_i^2}{4n^2}.$$

Since

$$x_i^*(\theta) = \frac{\theta_i}{2n},$$

we can rewrite this as

$$t_i(\theta) = -(n-1)(x_i^*(\theta))^2.$$

Economic intuition

This example is especially instructive, because the transfer can be interpreted as a Pigouvian correction.

Step 1: What would agent i do on her own? Ignoring transfers, agent i cares about

$$\theta_i x_i - x_i^2$$

from her own choice x_i , because she internalizes the benefit of her pollution and only the damage she herself suffers. The first-order condition would be

$$\theta_i - 2x_i = 0 \quad \Rightarrow \quad x_i = \frac{\theta_i}{2}.$$

This is too high.

Step 2: Why is it too high? Because when agent i increases x_i , the damage term x_i^2 is borne by *everyone*, not just by her. In social welfare, the total cost of x_i is

$$nx_i^2,$$

not merely x_i^2 . Therefore the efficient choice solves

$$\max_{x_i \geq 0} \theta_i x_i - nx_i^2,$$

leading to

$$x_i^* = \frac{\theta_i}{2n},$$

which is much smaller.

Step 3: What does the Groves transfer do? Since

$$t_i(\theta) = -(n-1)(x_i^*(\theta))^2,$$

agent i is effectively charged for the external harm imposed on the other $n-1$ agents.

If we define the payment by

$$p_i(\theta) = -t_i(\theta),$$

then

$$p_i(\theta) = (n-1)(x_i^*(\theta))^2.$$

This is precisely the missing part of the social cost: agent i already internalizes one copy of x_i^2 , and the transfer makes her internalize the remaining $n-1$ copies.

Step 4: Why is this a particularly nice example? Because the efficient choice $x_i^*(\theta) = \theta_i/(2n)$ depends only on agent i 's own type. Once the transfer is designed properly, we do not even need a direct revelation mechanism.

Indirect implementation

Instead of asking agent i to report θ_i , let her directly choose x_i , and then charge her

$$(n-1)x_i^2.$$

Her objective becomes

$$\theta_i x_i - x_i^2 - (n-1)x_i^2 = \theta_i x_i - n x_i^2.$$

Hence she solves exactly the social problem for her own pollution level, and her optimal choice is

$$x_i = \frac{\theta_i}{2n} = x_i^*(\theta).$$

Interpretation. This is a clean example of *indirect implementation*. Agents do not report types at all. Instead, they choose the relevant action directly, and the pricing rule is designed so that their privately optimal action coincides with the socially efficient one.

Why this is useful. In many applications, asking agents to report preferences is artificial or informationally demanding. A well-designed tax or fee schedule can sometimes achieve the same allocation more naturally by decentralizing the decision problem.

Summary

The revelation principle tells us that implementability can be studied through direct truthful mechanisms. Groves mechanisms then provide a powerful constructive result: in private value environments, any socially efficient allocation rule can be truthfully implemented in ex

post equilibrium by assigning each agent the welfare of the other agents, up to a term independent of her own report.

The examples illustrate two different economic logics:

- in the public project problem, the pivotal agent pays the externality imposed on the rest of society;
- in the pollution problem, the transfer works like a Pigouvian charge that makes each agent internalize the external cost of her action.

These examples are especially important because they show that mechanism design is not just about abstract reporting games. It is often about constructing the right prices, taxes, or transfer rules so that decentralized incentives replicate the socially desired outcome.